Sorting Out Secure Analytics with Thousands of Users

October 31, 2025

Sam Buxbaum

Roadmap

1. Deployment of MPC for Political Polling

2. Oblivious Sorting

3. ORQ

Privacy-Preserving Machine Learning on Web Browsing for Public Opinion

To appear at CSCML 25!

Paper

GitHub

Privacy-Preserving Machine Learning on Web Browsing for Public Opinion

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[4] Washington University dinopc@wustl.edu

Abstract. We present a real-world deployment of secure multiparty computation to predict political preference from private web browsing data. To estimate aggregate preferences for the 2024 U.S. presidential election candidates, we collect and analyze secret-shared data from nearly 8000 users from August 2024 through February 2025, with over 2000

Plan

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1. Background on Political Polling

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2. Our Approach and Contributions

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3. Our System

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7. Next Steps and Future Work

Traditional Political Polling

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- Data collection takes time

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- Data collection is human-intensive

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West Virginia 2024 Presidential Election Polls

Harris vs. Trump

Source	Date	Sample	Harris	Trump	Other
Research America	8/30/2024	400 LV ±4.9%	34%	61%	5%

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Michigan 2024 Presidential Election Polls

Instantly compare a poll to prior one by same pollster

Harris vs. Trump

Source	Date	Sample	Harris	Trump	Other
Average of 23 Polls†			**48.6%**	**46.8%**	-
FAU / Mainstreet	11/04/2024	713 LV	49%	47%	4%
Emerson College	11/04/2024	790 LV ±3.4%	50%	48%	2%
Research Co.	11/04/2024	450 LV ±4.6%	49%	47%	4%
InsiderAdvantage	11/03/2024	800 LV ±3.7%	47%	47%	6%
Trafalgar Group	11/03/2024	1,079 LV ±2.9%	47%	48%	5%
MIRS / Mich. News Source	11/03/2024	585 LV ±4%	50%	48%	2%
NY Times / Siena College	11/03/2024	998 LV ±3.7%	47%	47%	6%
Morning Consult	11/03/2024	1,108 LV ±3%	49%	48%	3%
AtlasIntel	11/02/2024	1,198 LV ±3%	48%	50%	2%
Redfield & Wilton	11/01/2024	1,731 LV ±2.2%	47%	47%	6%
The Times (UK) / YouGov	11/01/2024	942 LV ±3.9%	48%	45%	7%
EPIC-MRA	11/01/2024	600 LV ±4%	48%	45%	7%
Marist Poll	11/01/2024	1,214 LV ±3.5%	51%	48%	1%
AtlasIntel	10/31/2024	1,136 LV ±3%	49%	49%	2%
Echelon Insights	10/31/2024	600 LV ±4.4%	48%	48%	4%
MIRS / Mich. News Source	10/31/2024	1,117 LV ±2.5%	47%	49%	4%
UMass Lowell	10/31/2024	600 LV ±4.5%	49%	45%	6%
Washington Post	10/31/2024	1,003 LV ±3.7%	47%	46%	7%
Fox News	10/30/2024	988 LV ±3%	49%	49%	2%
CNN	10/30/2024	726 LV ±4.7%	48%	43%	9%
Suffolk University	10/30/2024	500 LV ±4.4%	47%	47%	6%

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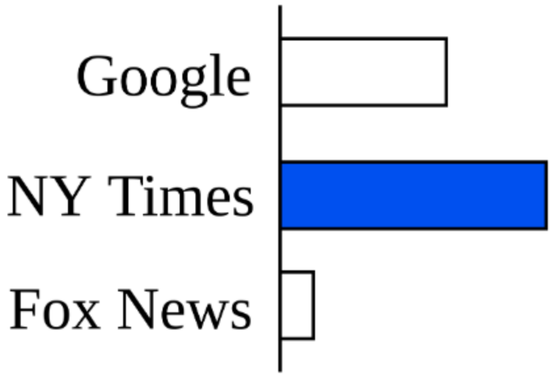

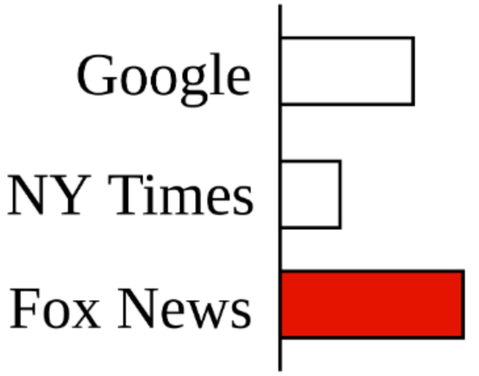

Web Browsing for Political Polling

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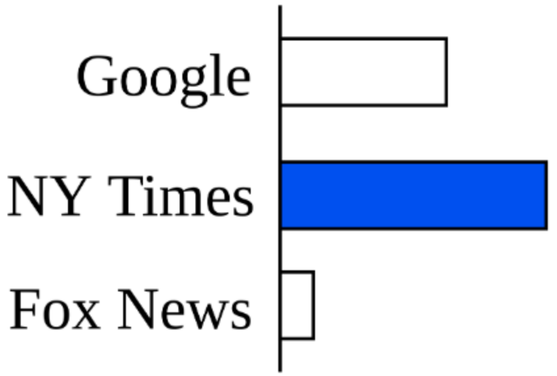

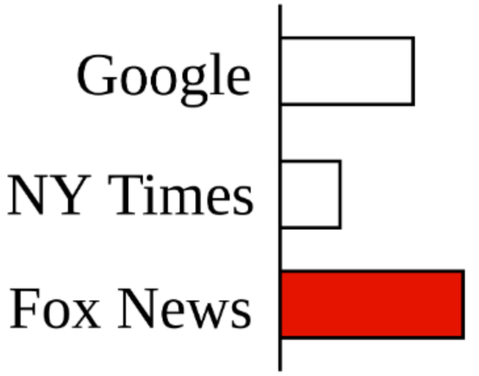

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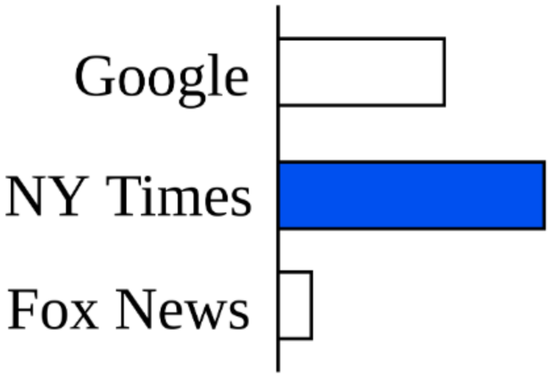
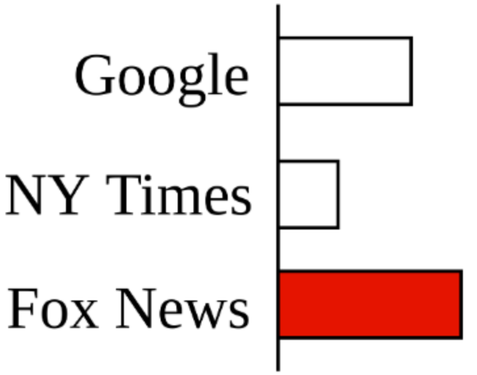
- Example - news websites
- More data
- Fully automated





Prior Work

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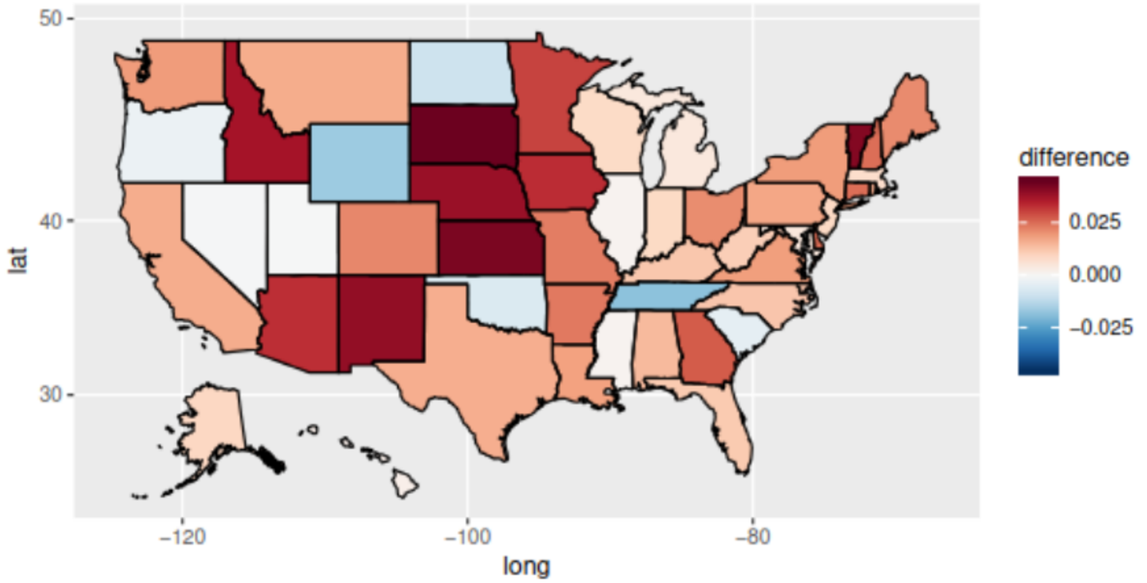

Figure 8: Impact of the 'Comey letter' at the state level.

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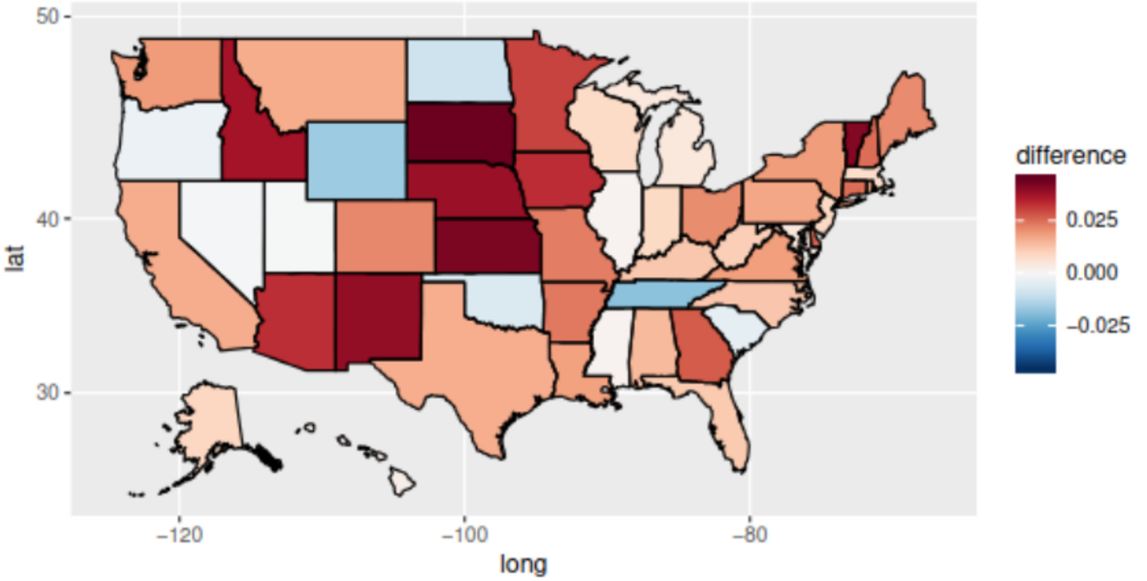

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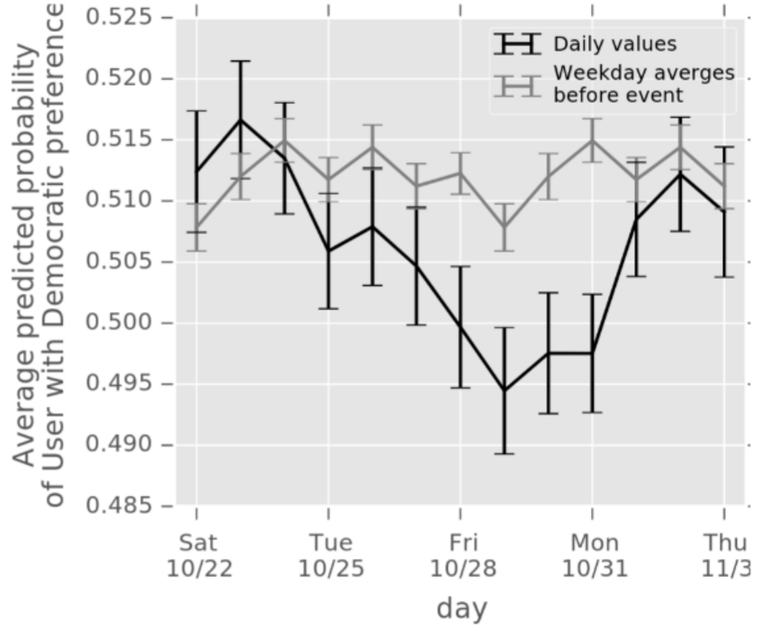

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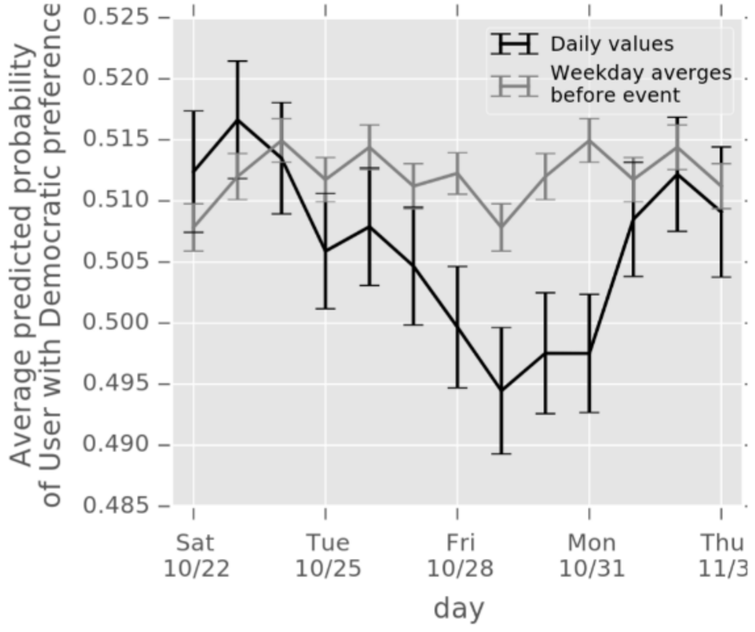

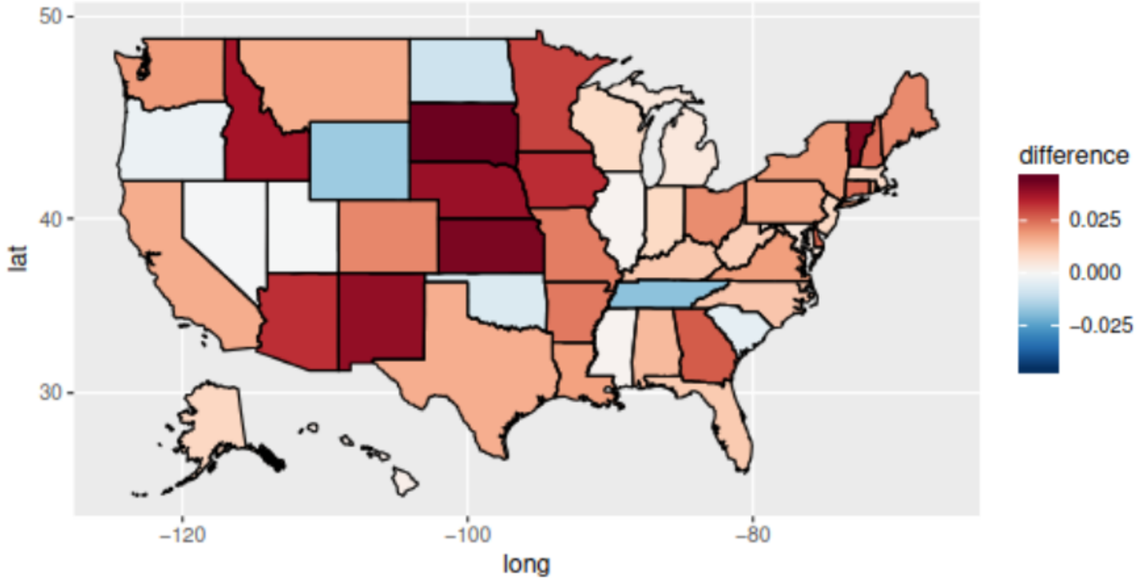

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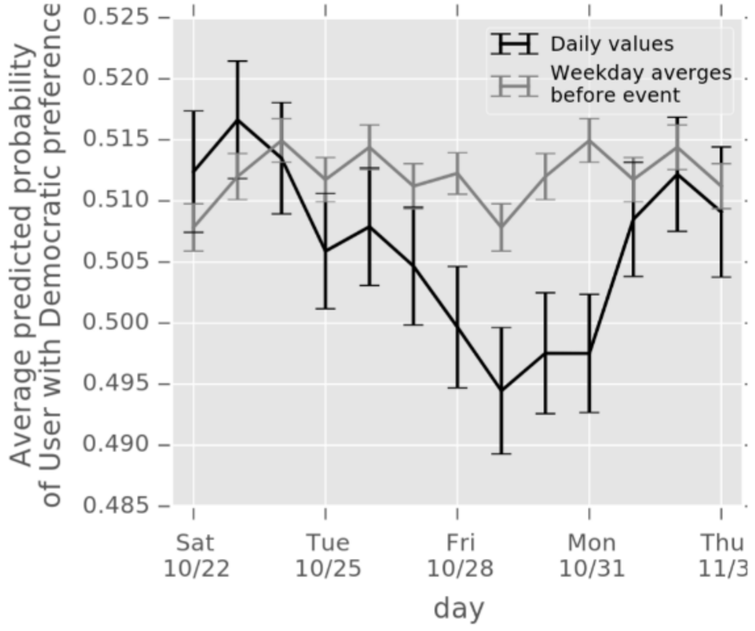
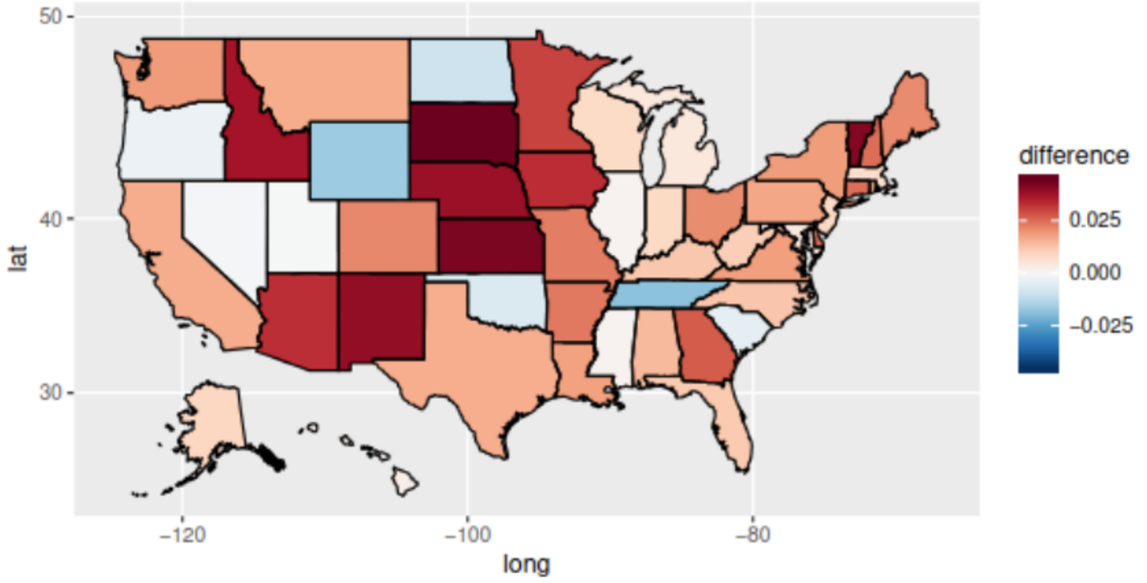
Prior Work

- Web browsing behavior can predict voting results
- Quantifying the 'Comey letter' (Comarela et al.)
- Social media referrals are the best signal
- Used large plaintext dataset





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Two Approaches to Political Polling

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Traditional Polling

- Slow
- Expensive
- Coarse-grained insights

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VS

Web Behavior Analysis

- Immediate
- Cheap
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What about privacy?

Our Contributions

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- We collected and analyzed data from almost 8000 unique users

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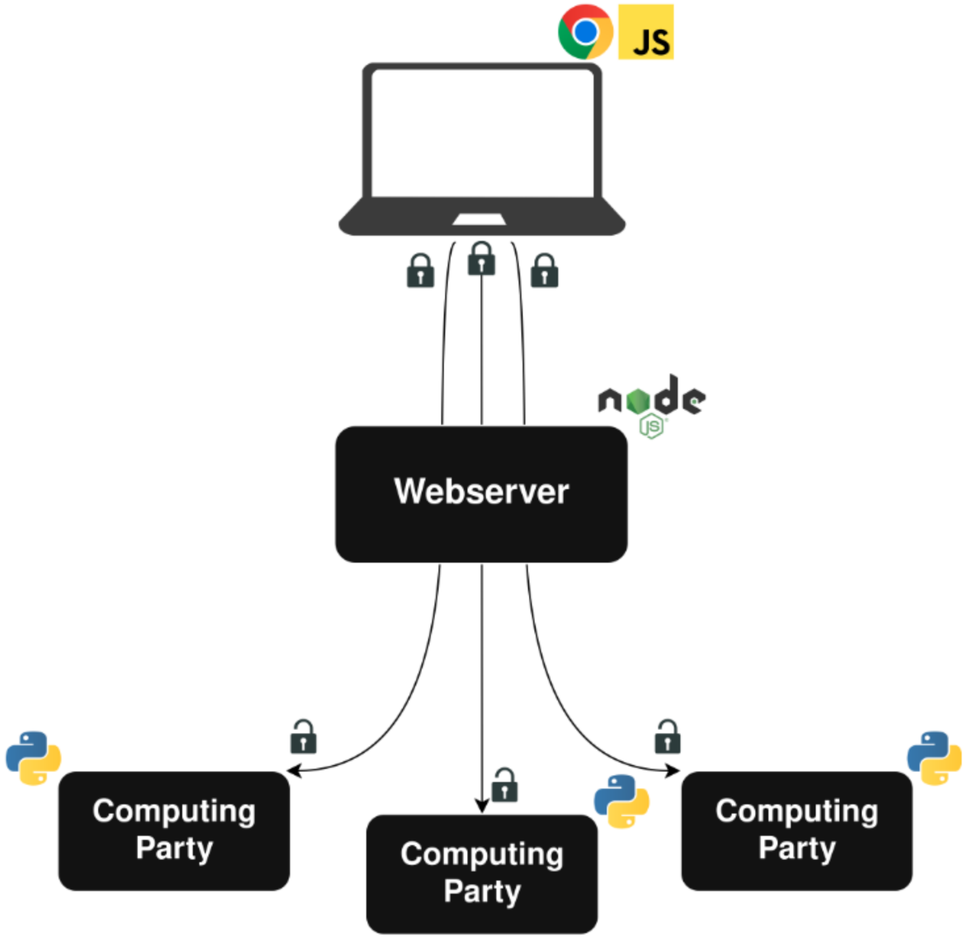
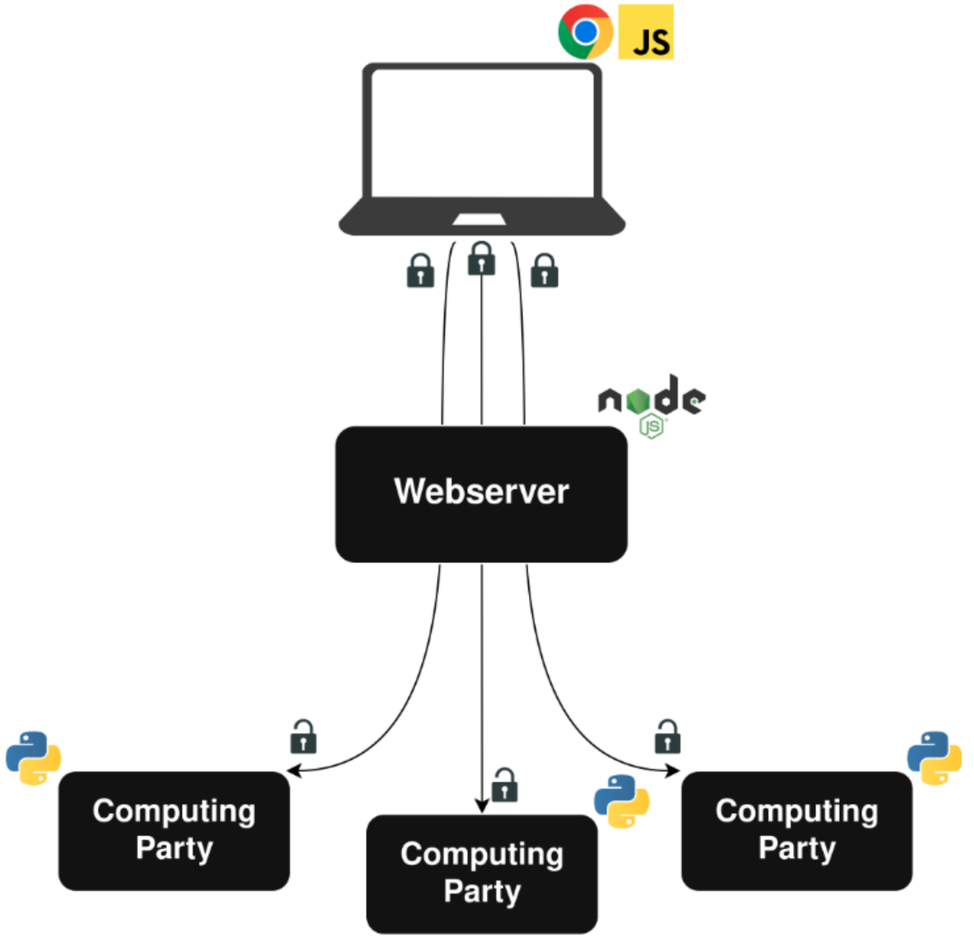
- We built a system for securely predicting political preferences from web browsing data
- We collected and analyzed data from almost 8000 unique users
- All analysis took place under MPC

System Design



System Design

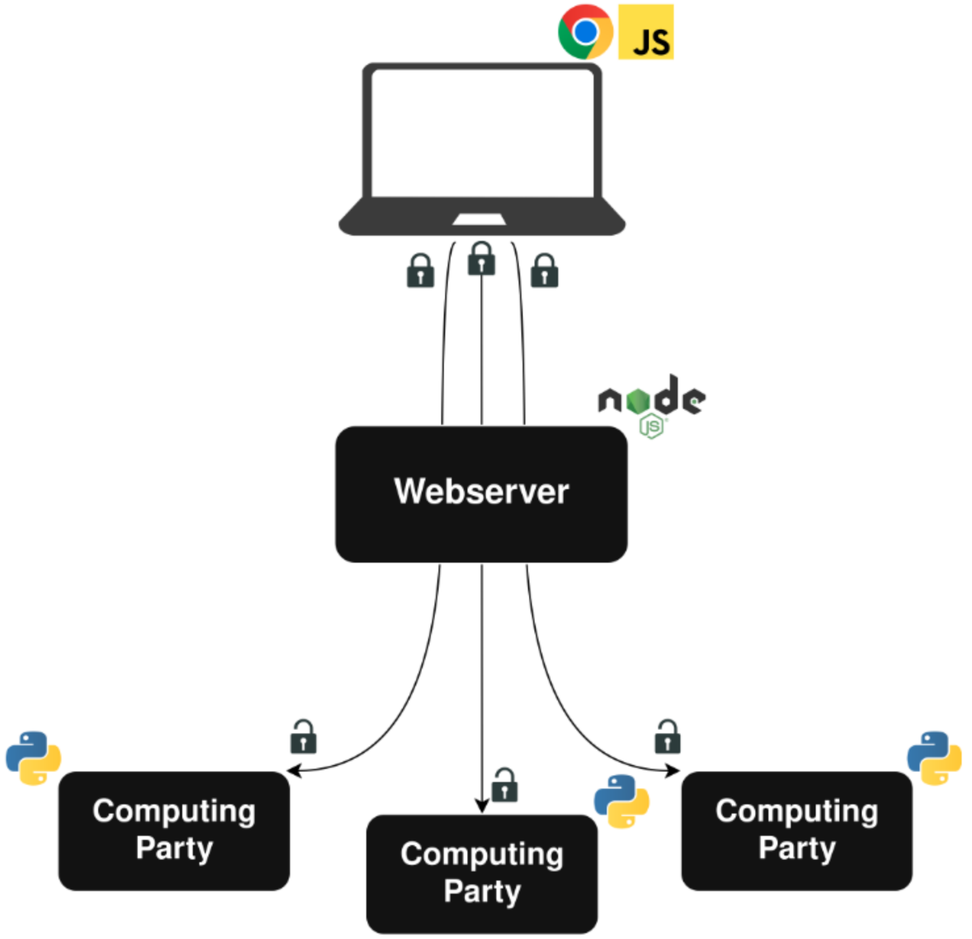
Client Plugin



System Design

Client Plugin

Webserver



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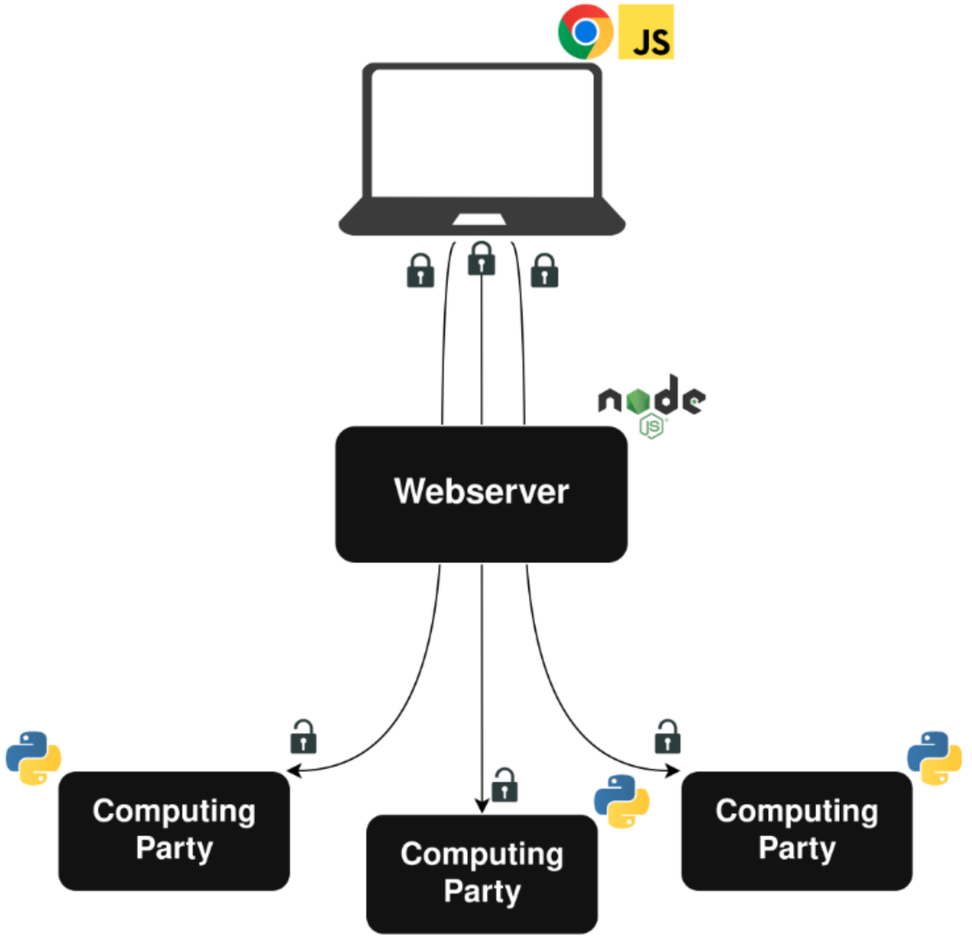

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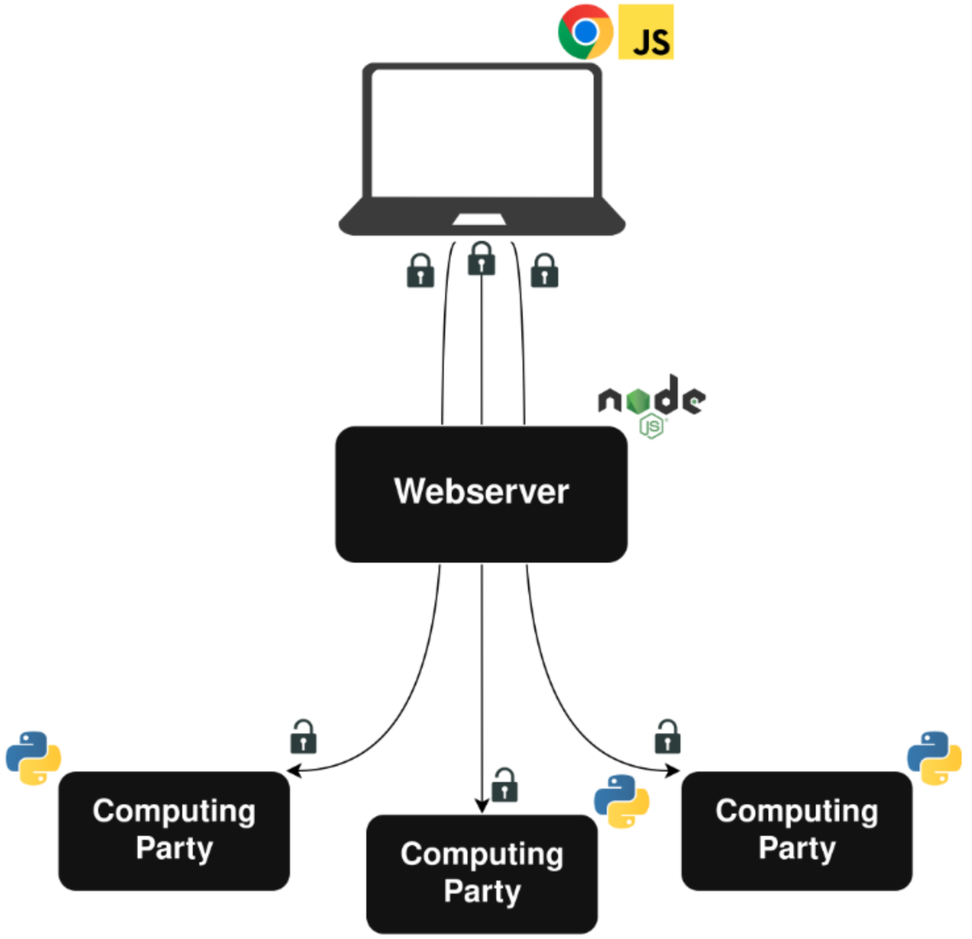

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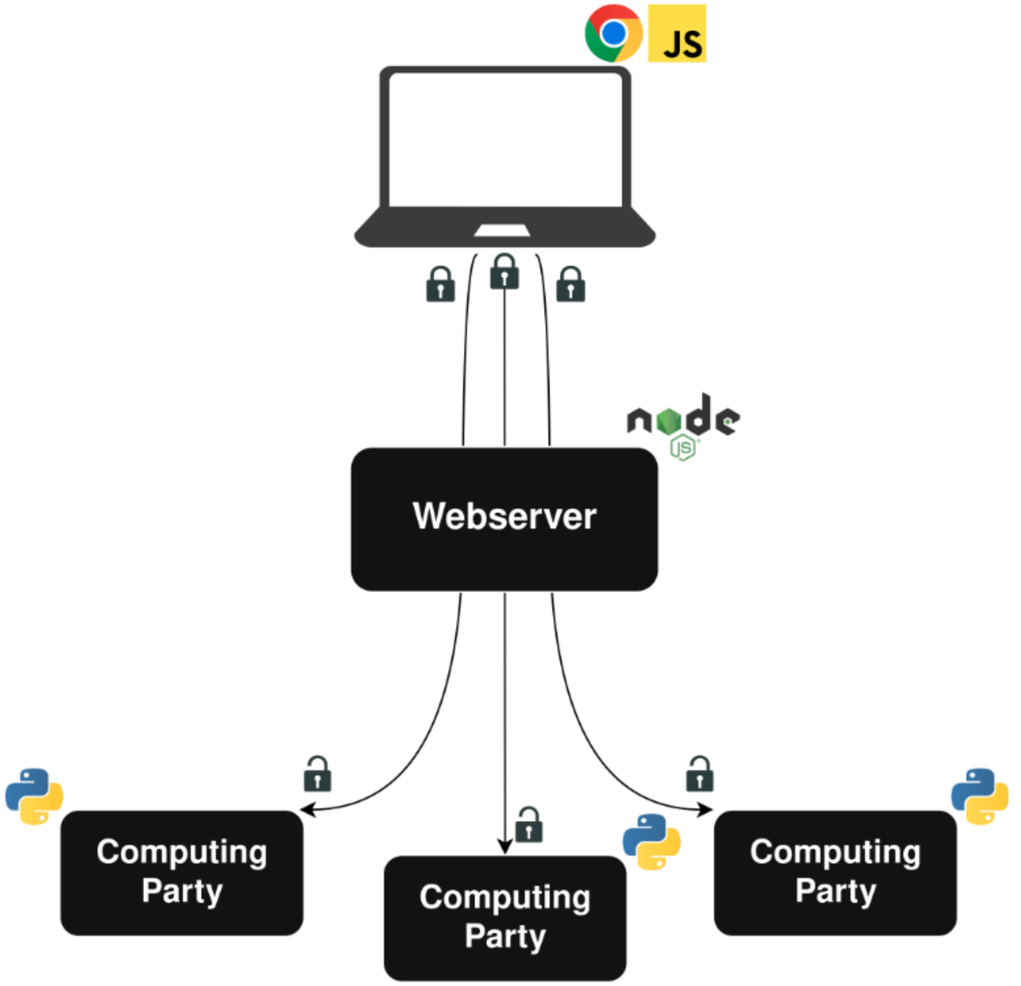

Client Plugin

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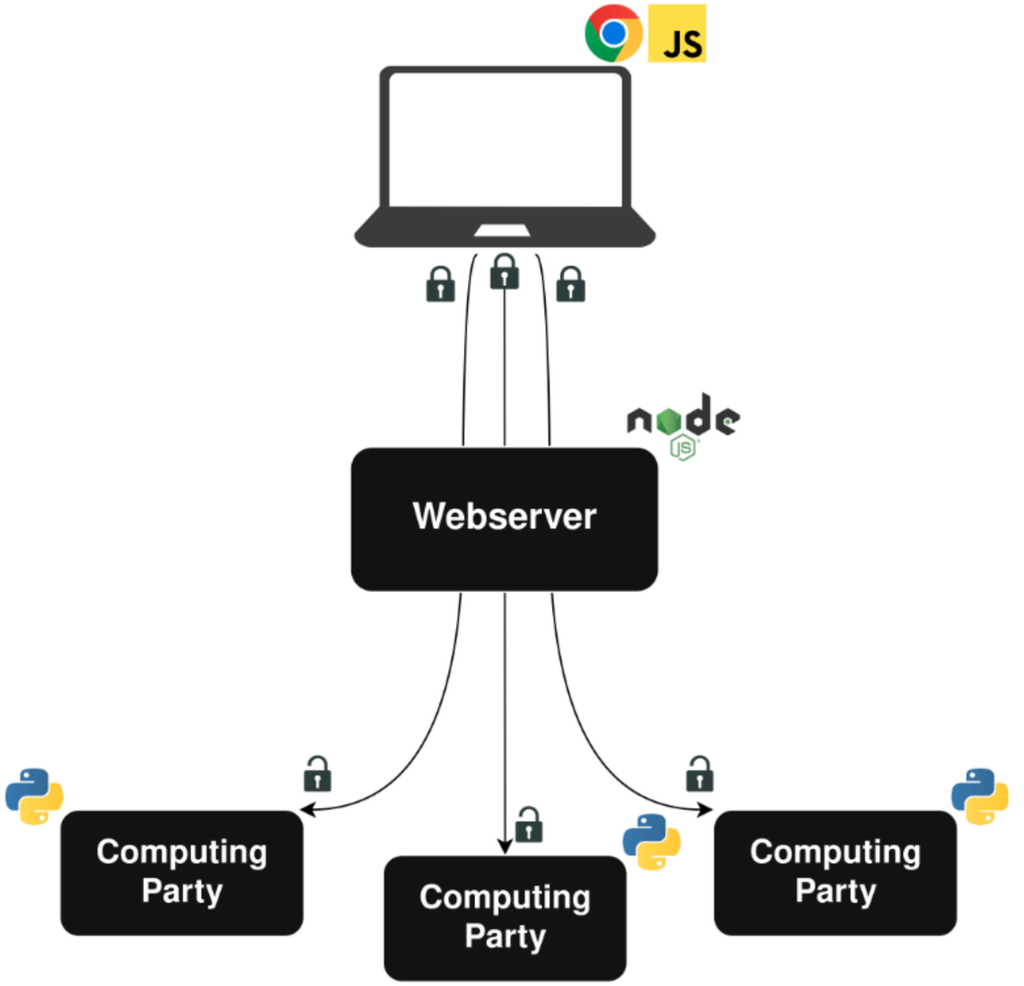

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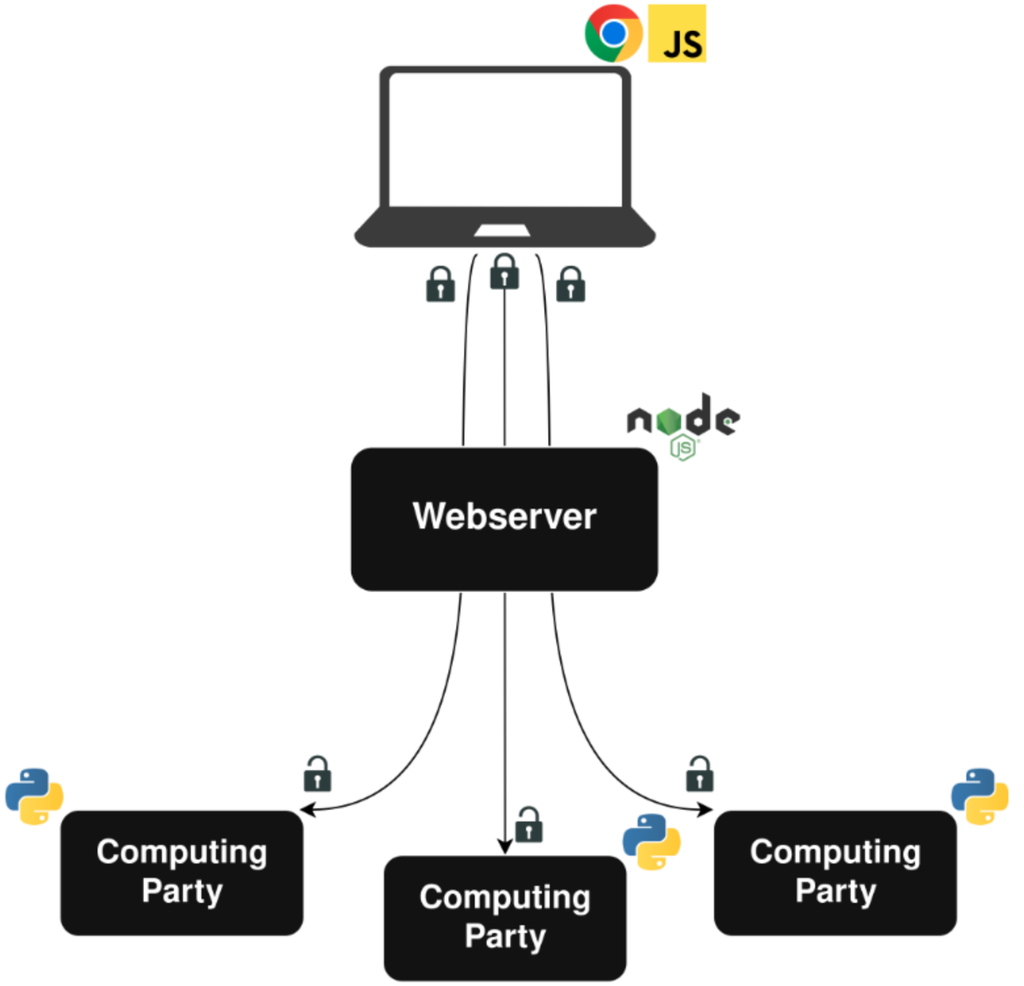
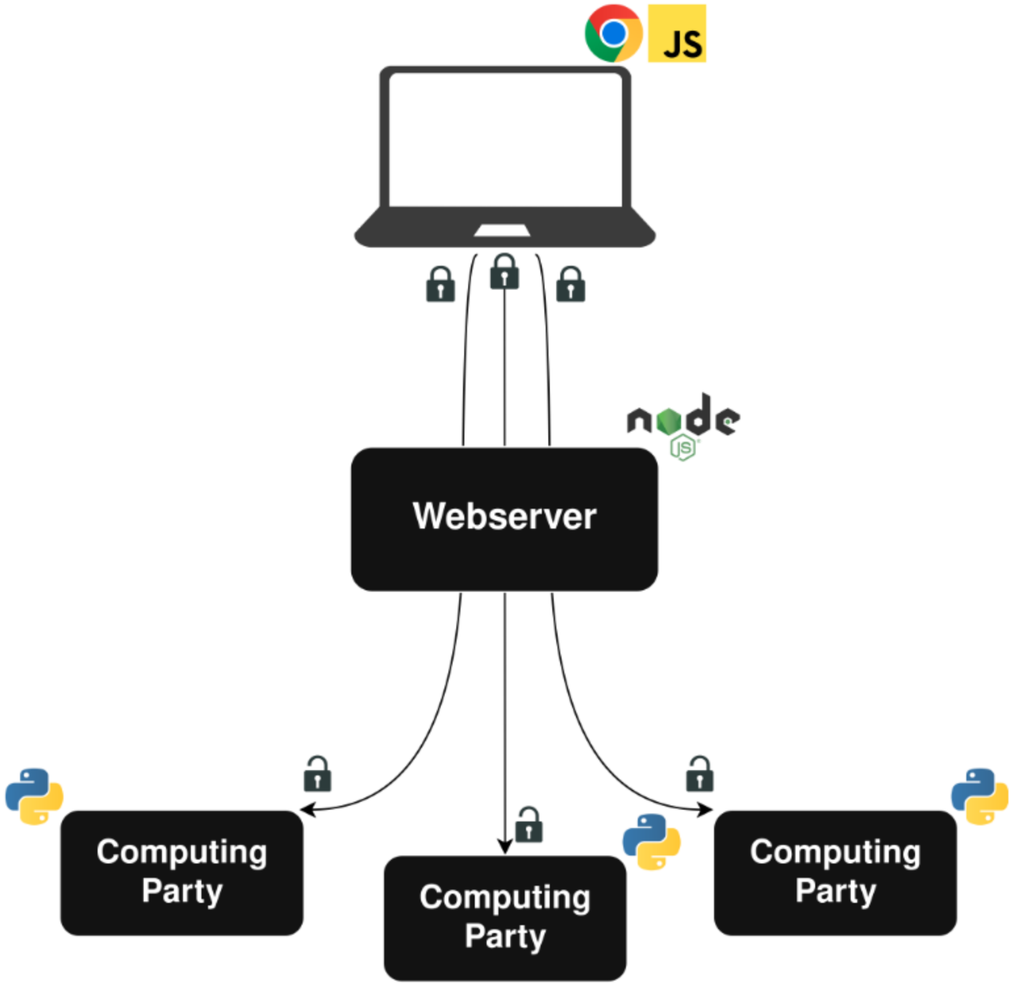
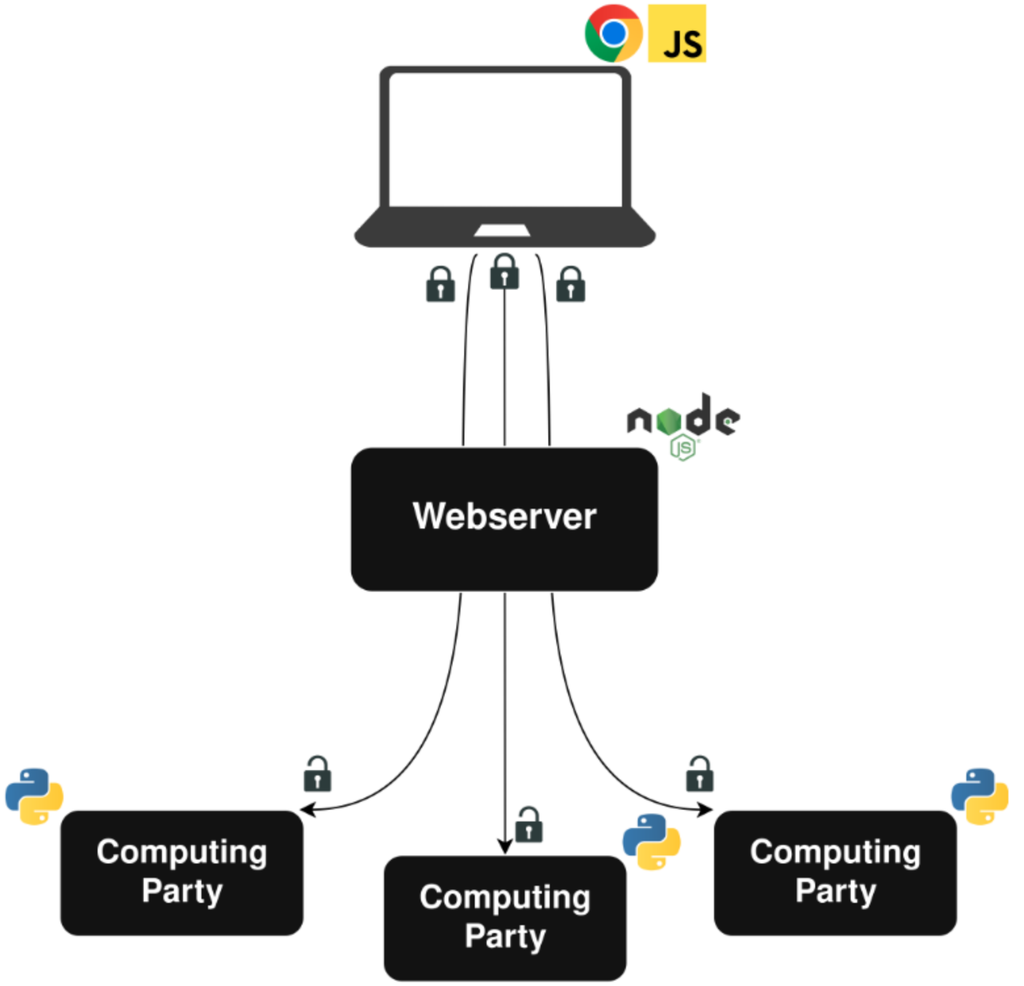


Client Plugin

- Custom-built Chrome plugin to monitor browsing
- Daily data uploads of secret-shared histograms
- Client-side secret sharing and encryption



Webserver


JS
node
Webserver
Computing Party
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Webserver

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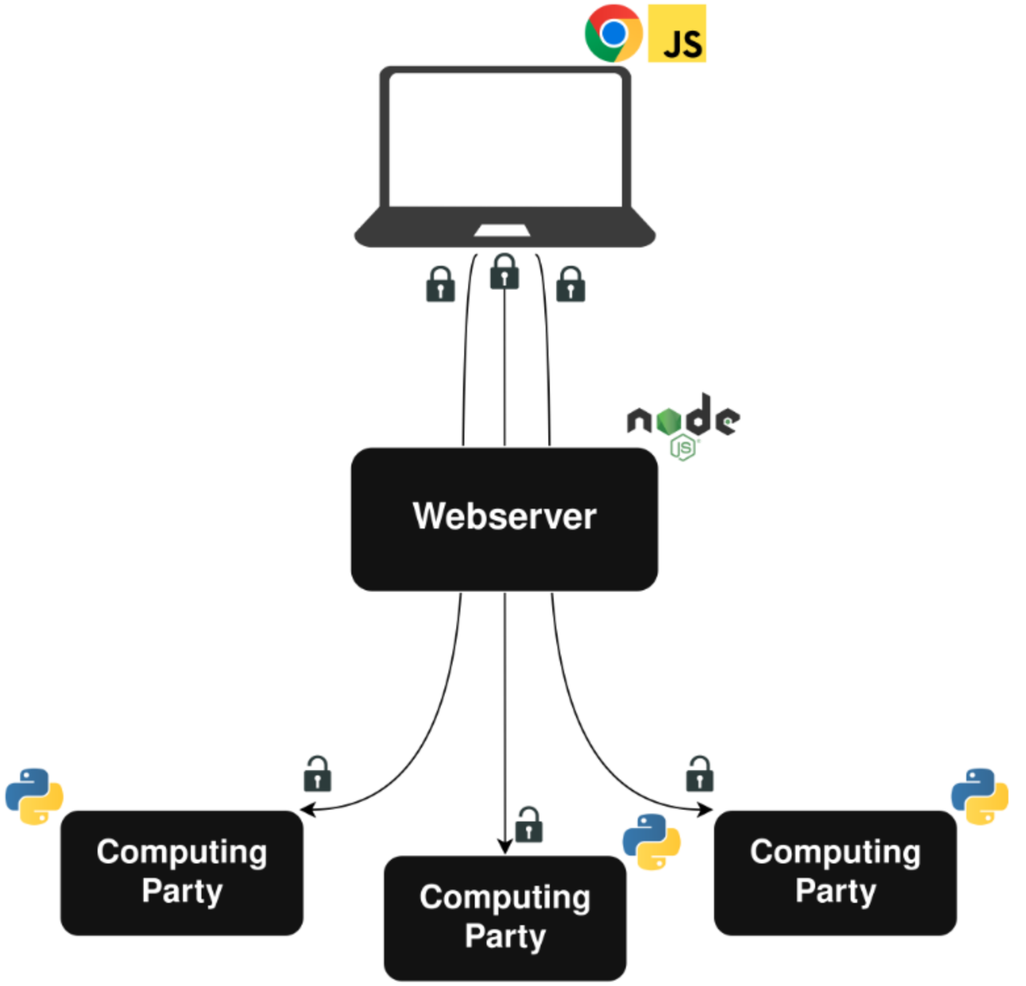

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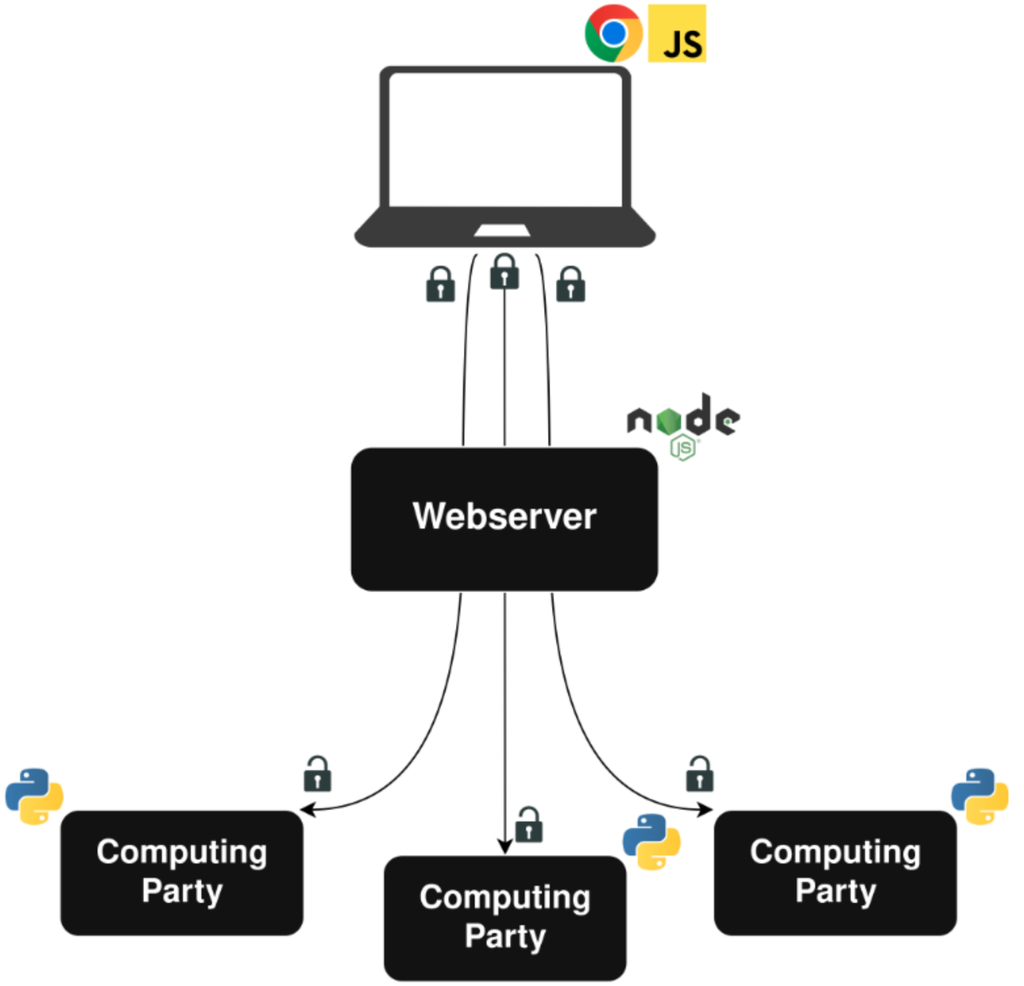
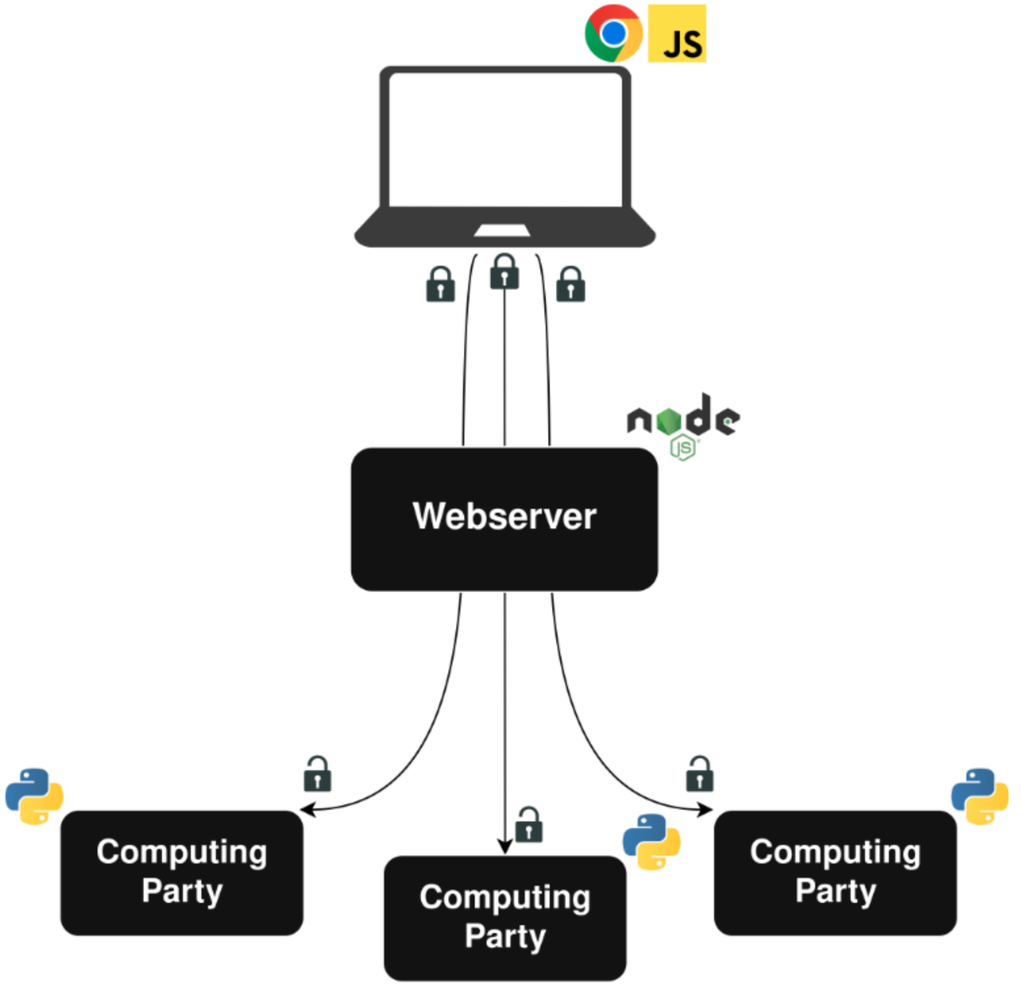
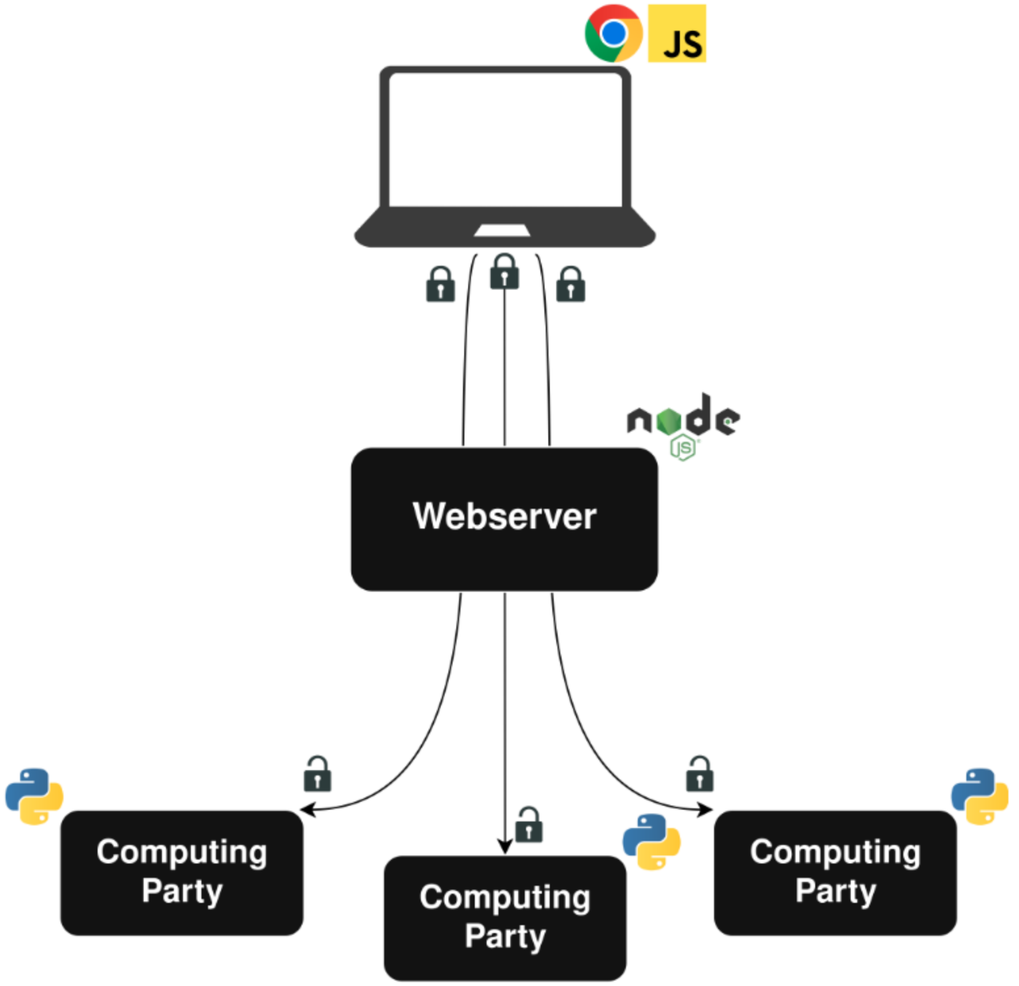


Webserver

- Simplifies interaction with clients
- Collects basic metadata
- Never sees any private data



MPC Backend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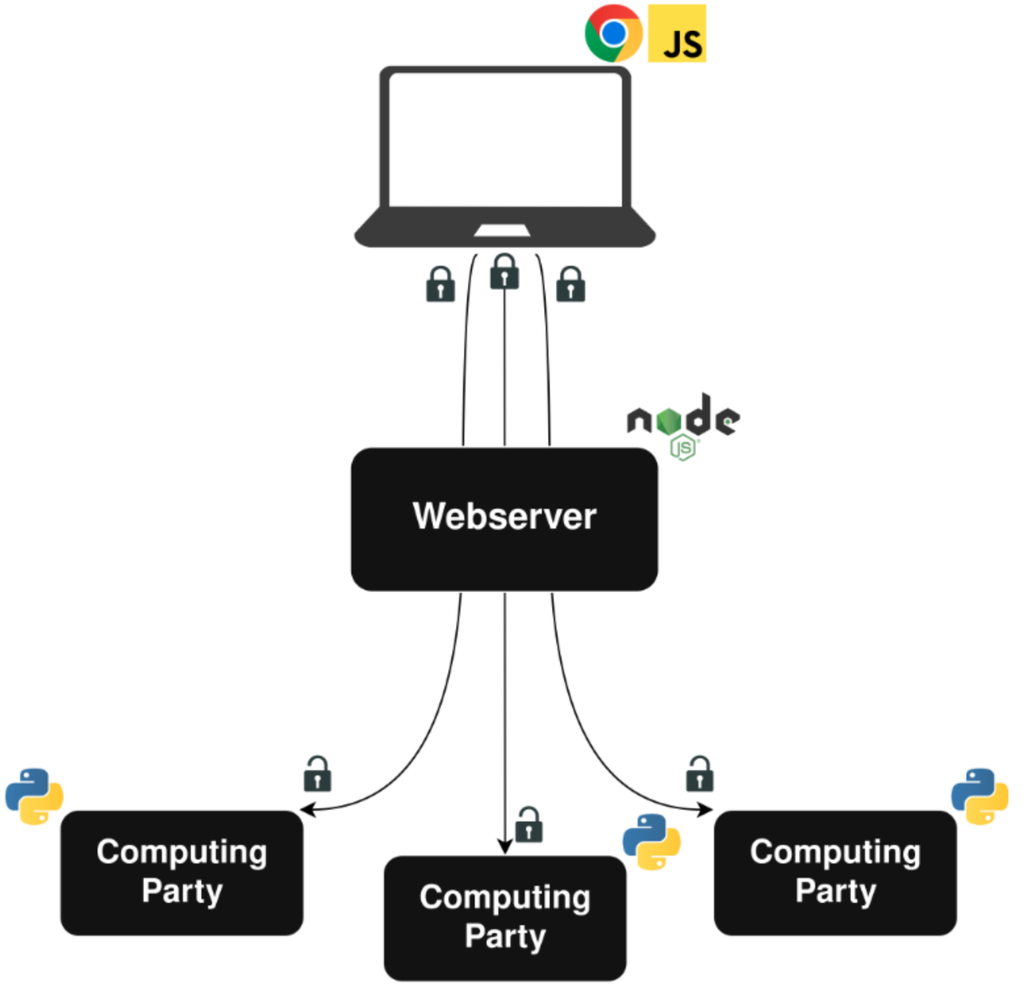
- Three party computation with an honest majority



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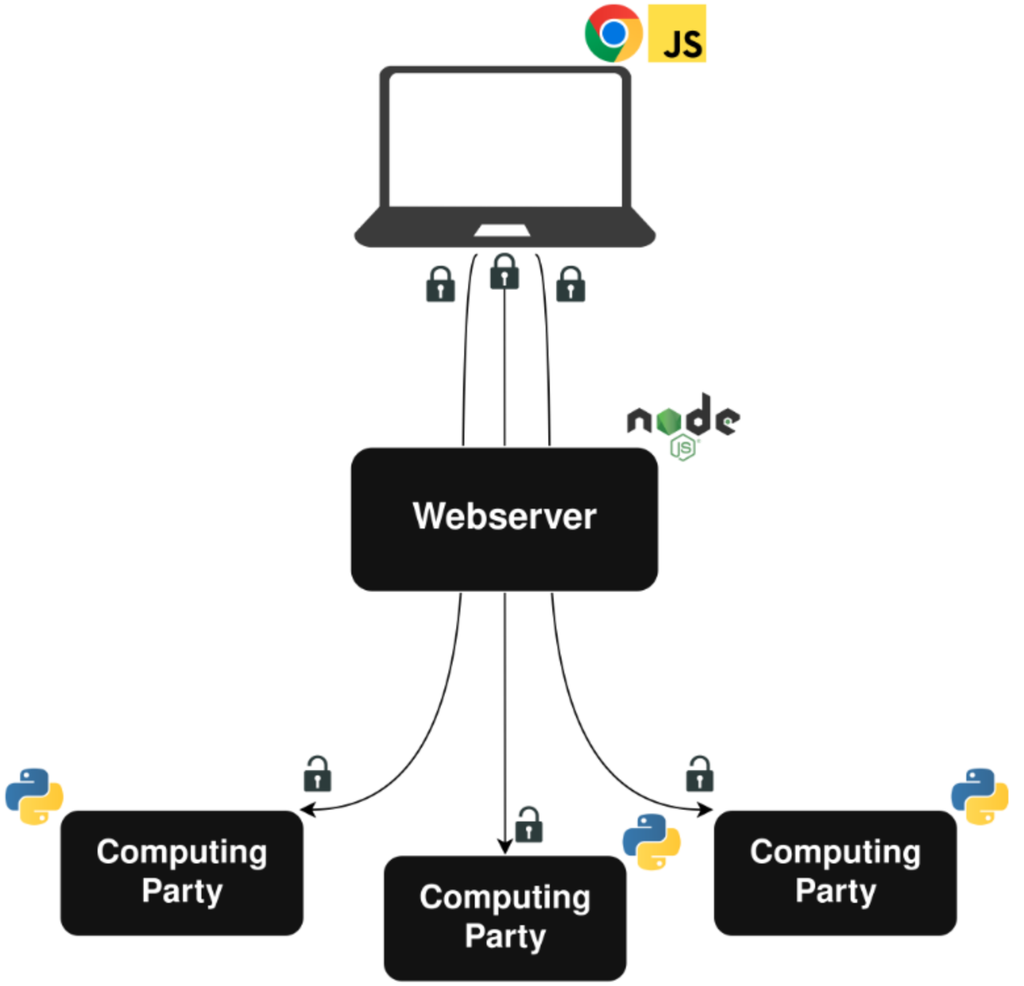

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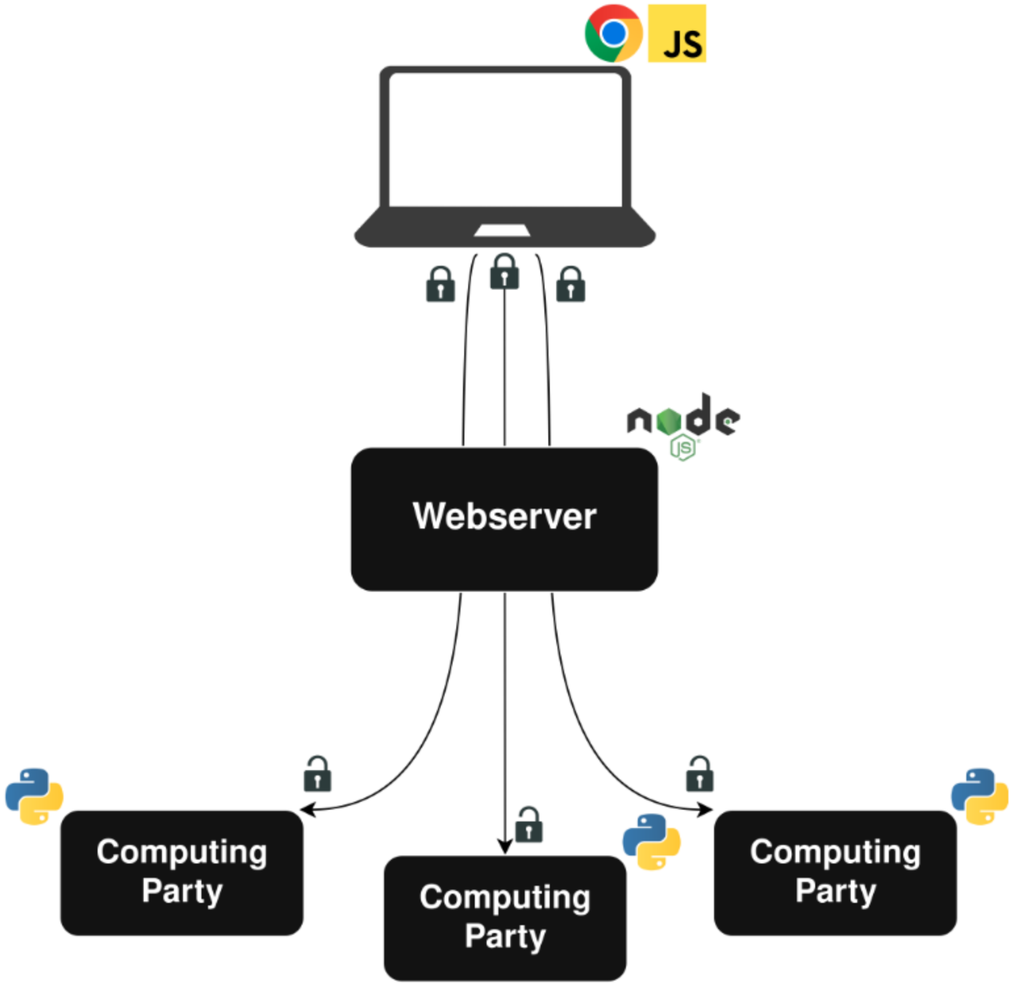

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- The first implementation under MPC of an LLP model
- Mix of out-of-the-box components and custom implementations

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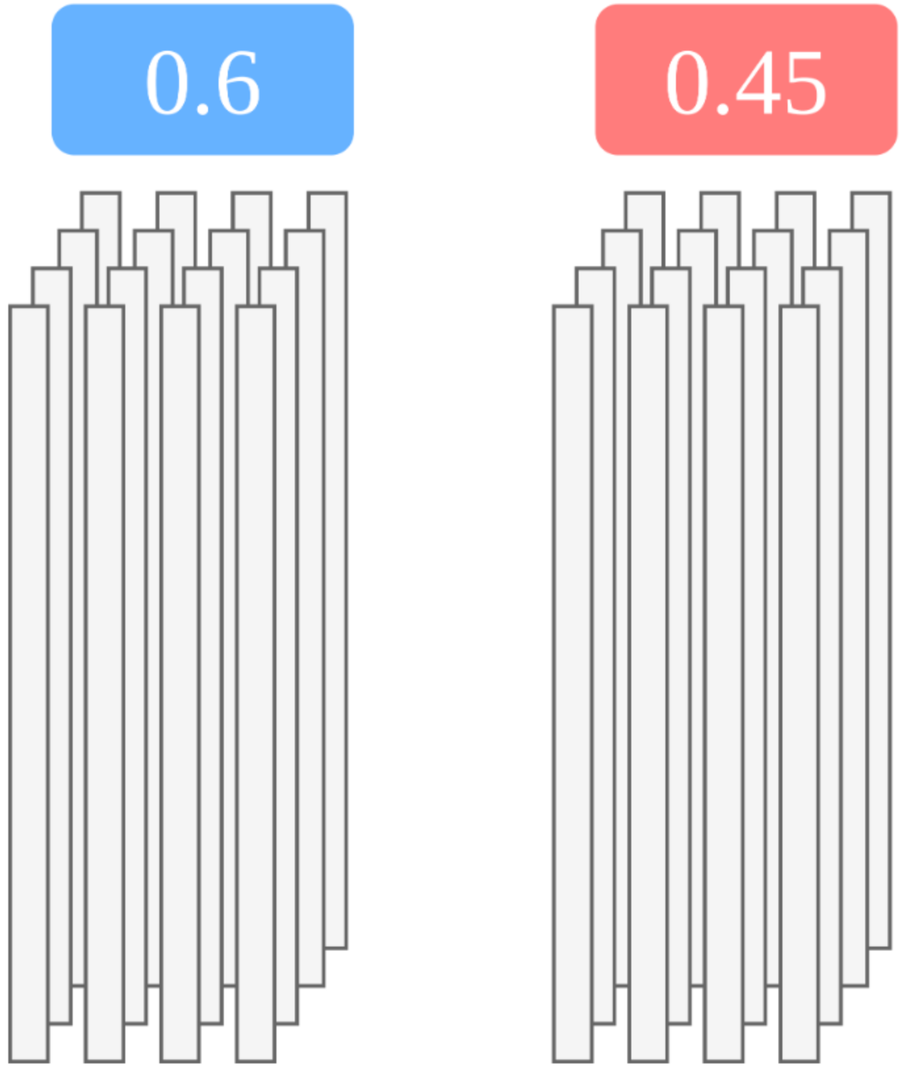

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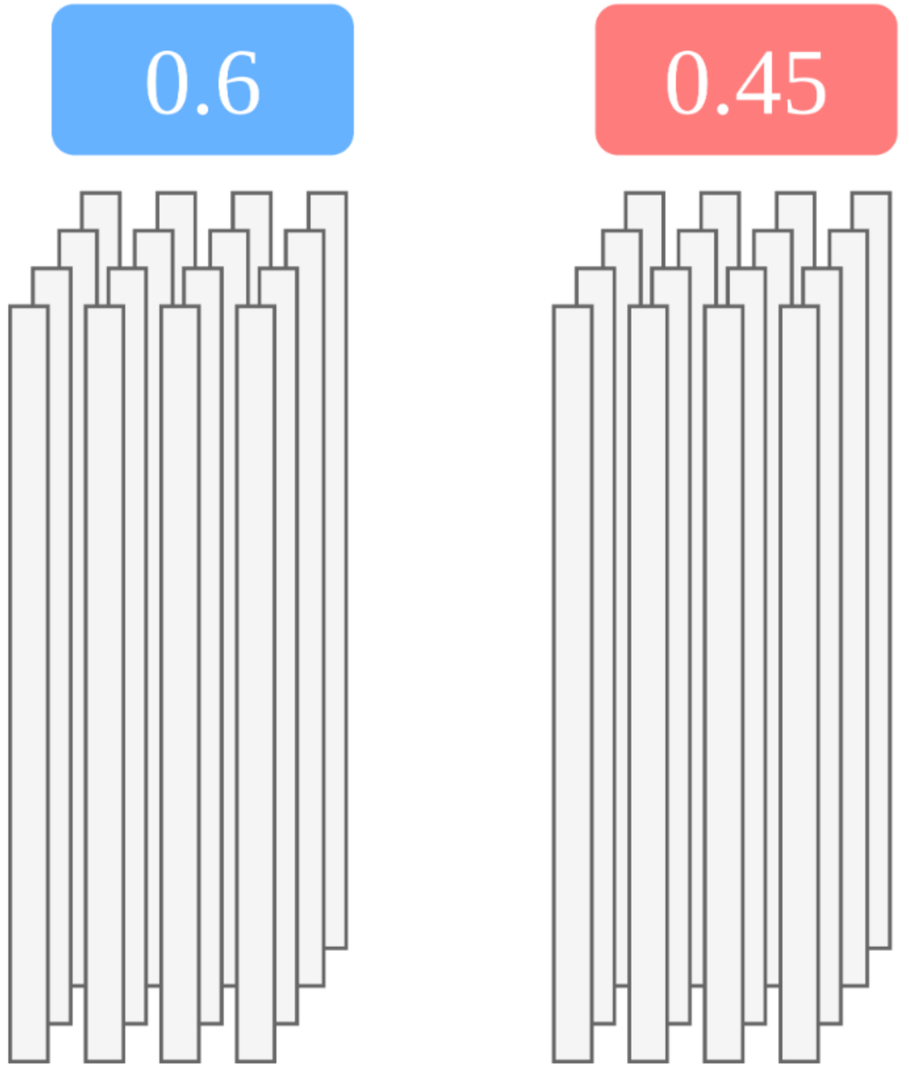
- Train on aggregate ground truth



The Plaintext Algorithm

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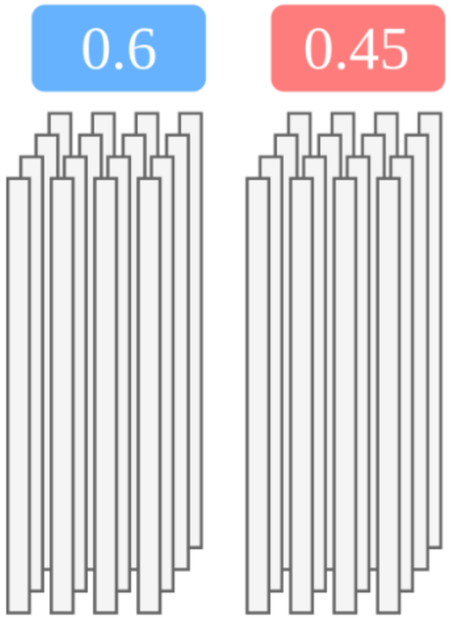

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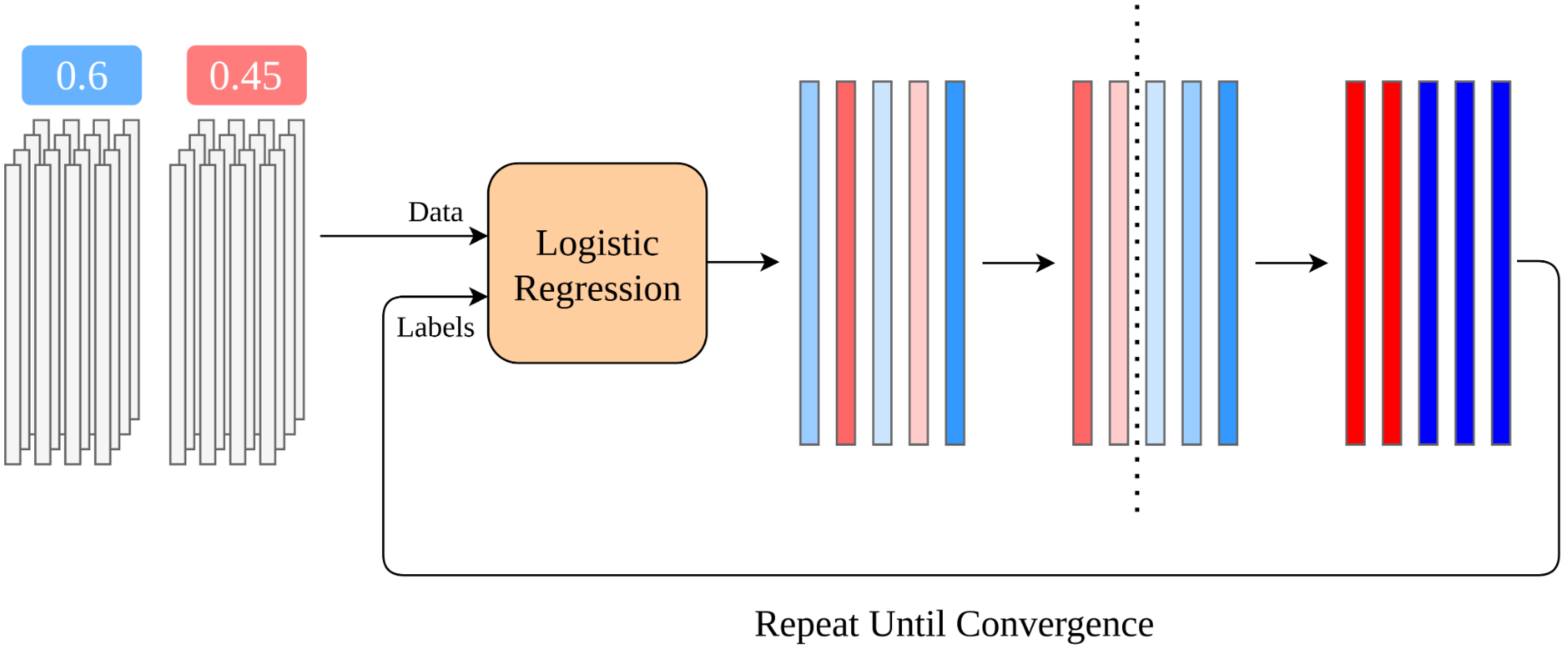

Translating to MPC

```
initialize_labels()

repeat until convergence:
    model = train(data, labels)
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- Sum the comparison results

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Convergence Check

- Vector oblivious comparison
- Sum the comparison results
- Compare sum with convergence threshold

Payment

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- Users are paid and recruited through Mechanical Turk

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Communication by email

Data Partition

Data Partition

Data Partition

Input	Web Browsing Data	Registered IDs

Data Partition

	BOSTON UNIVERSITY	WashU
Input	Web Browsing Data	Registered IDs
Output	Aggregation by ID	ID Set per Day

Calculating Payments

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WashU needed to know which users sent data on which days.

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- BU sends all collected tags to WashU
- WashU computes all combinations of users and dates and checks for matches

Computing Anonymous IDs

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WashU maps IDs to anonymized IDs between 1 and n.

- WashU sends all possible tags for all anonymous IDs
  - All `(date, hash)` combinations
- BU aggregates data from same anonymous ID for each week

Looking Ahead to 2028

Looking Ahead to 2028

- Anonymous payments

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- Anonymous payments
- More fine-grained data

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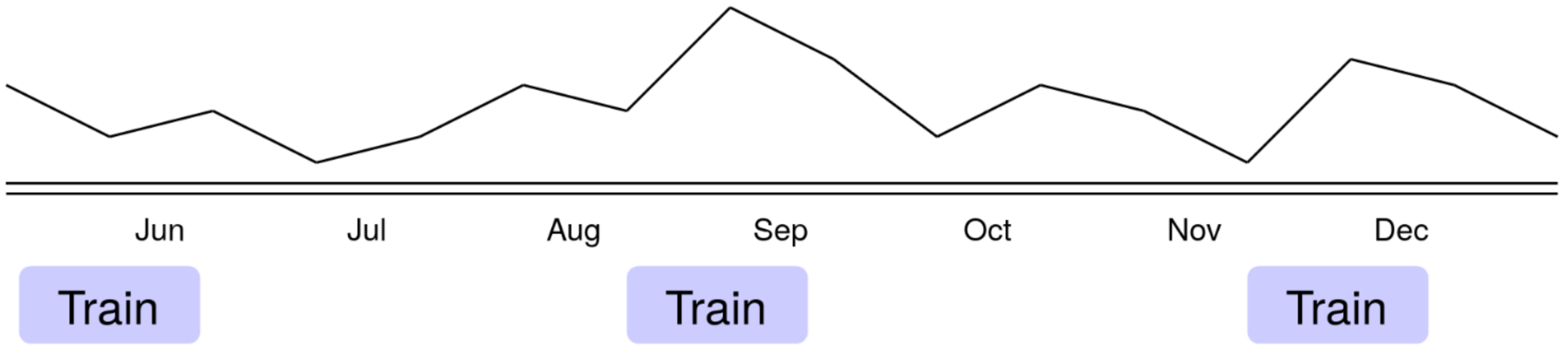

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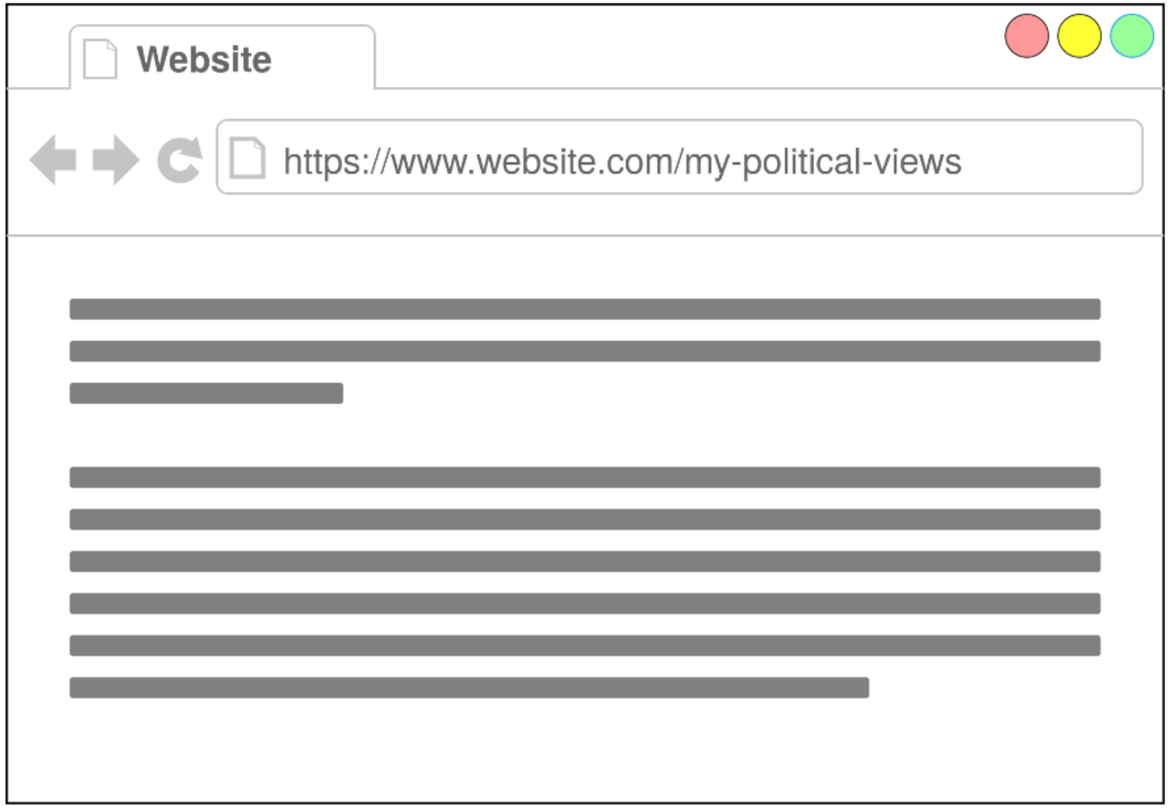

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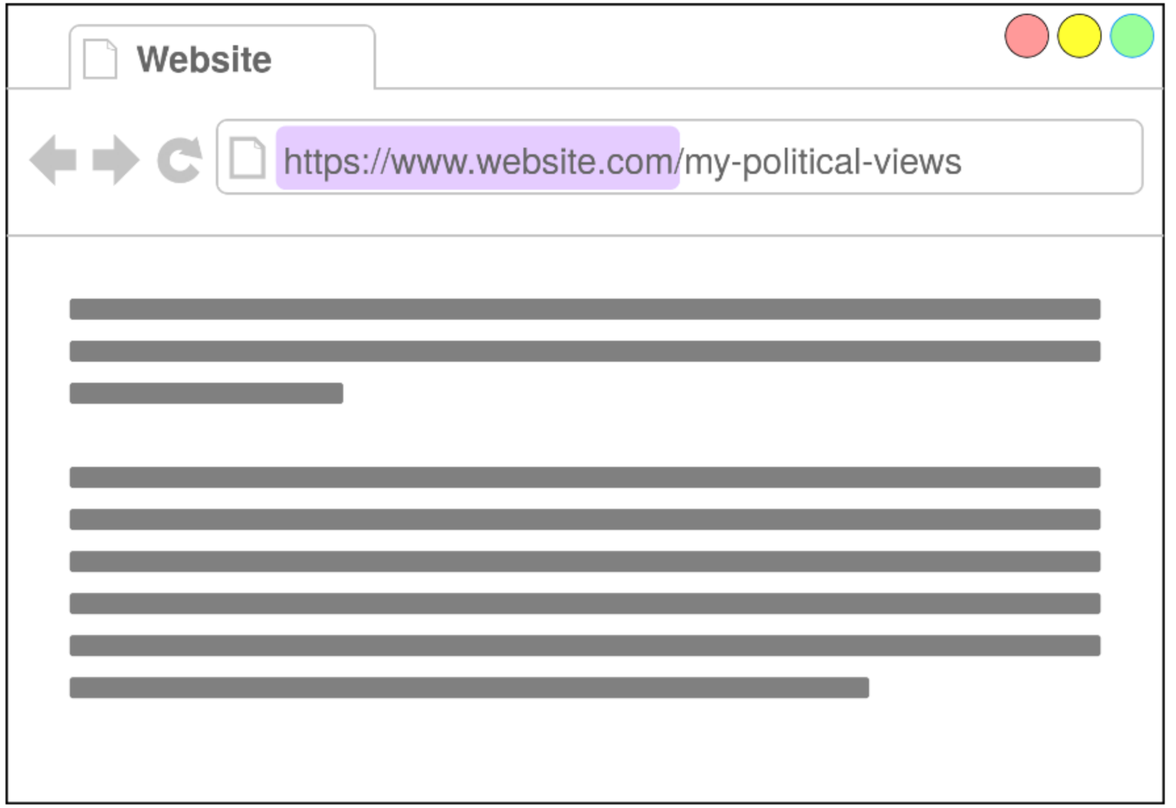

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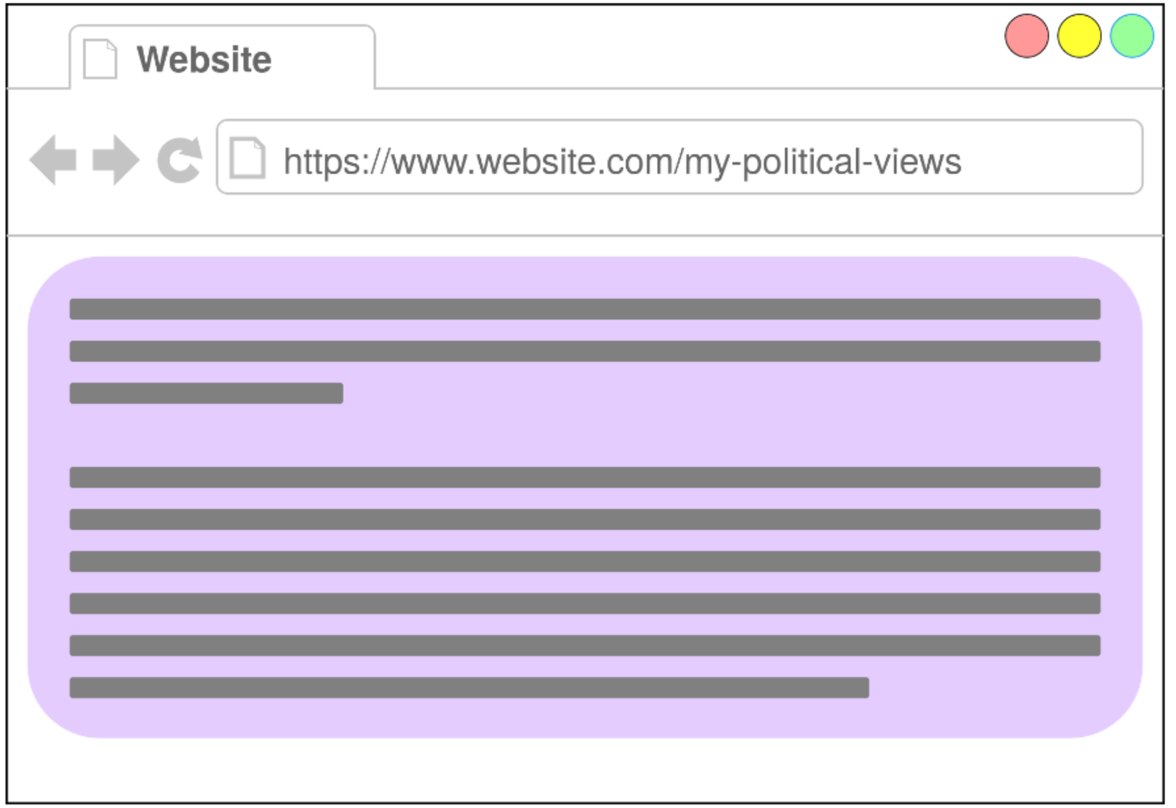
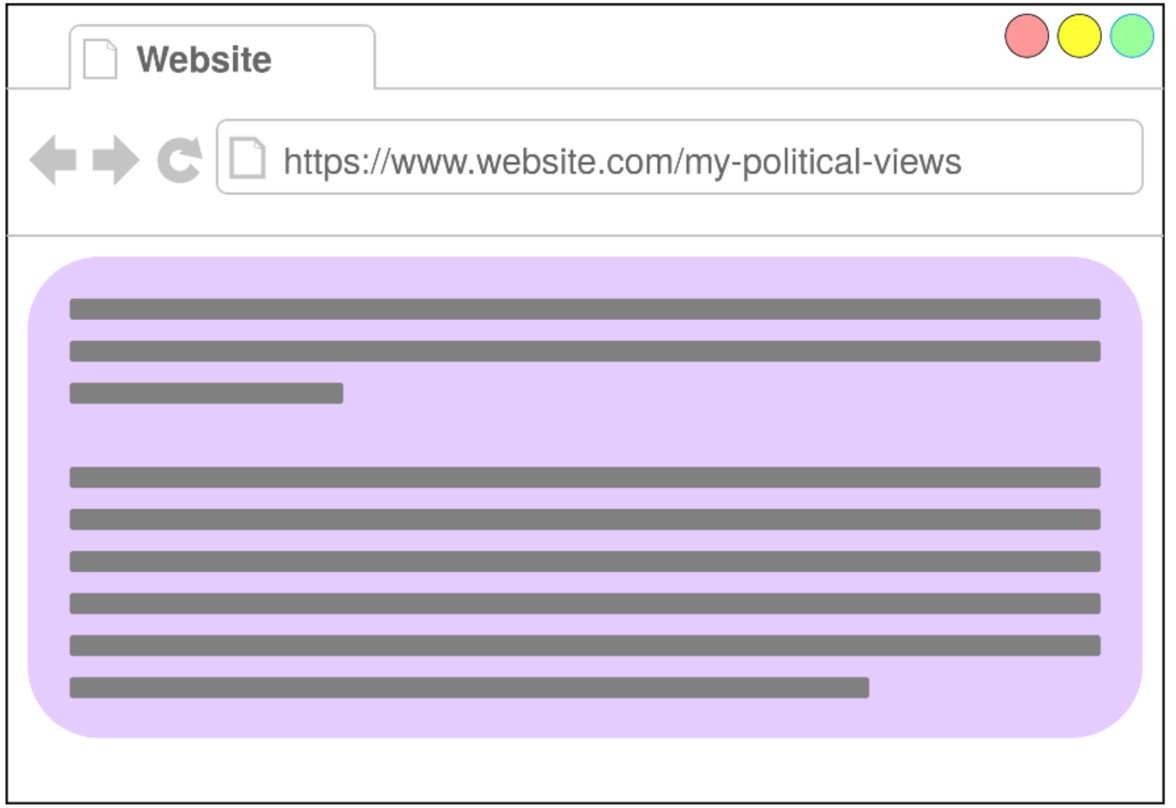


More Fine-Grained Data

- Beyond domains
- LLM in browser to generate embeddings



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Scaling Up Training

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State of the art is $1000\times$ faster!

Oblivious Sorting

Roadmap

~~1. Deployment of MPC for Political Polling~~

2. Oblivious Sorting

3. ORQ

Oblivious Sorting

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Sorting Networks

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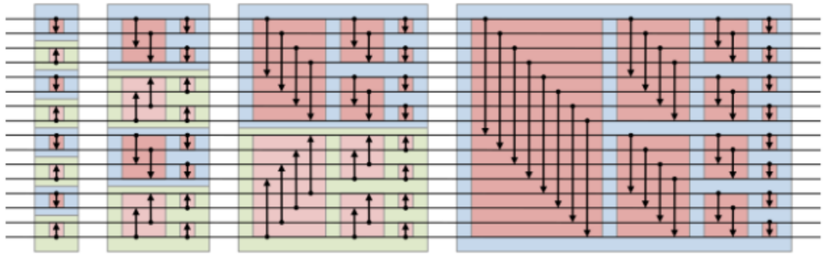

Sorting Networks

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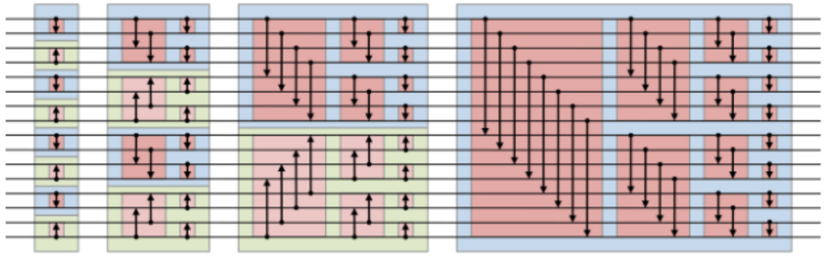

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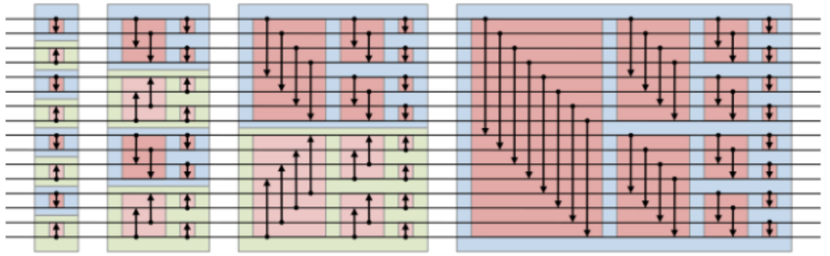

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Goal: Match the $O(n \log n)$ plaintext algorithms.

Our Sorting Contributions

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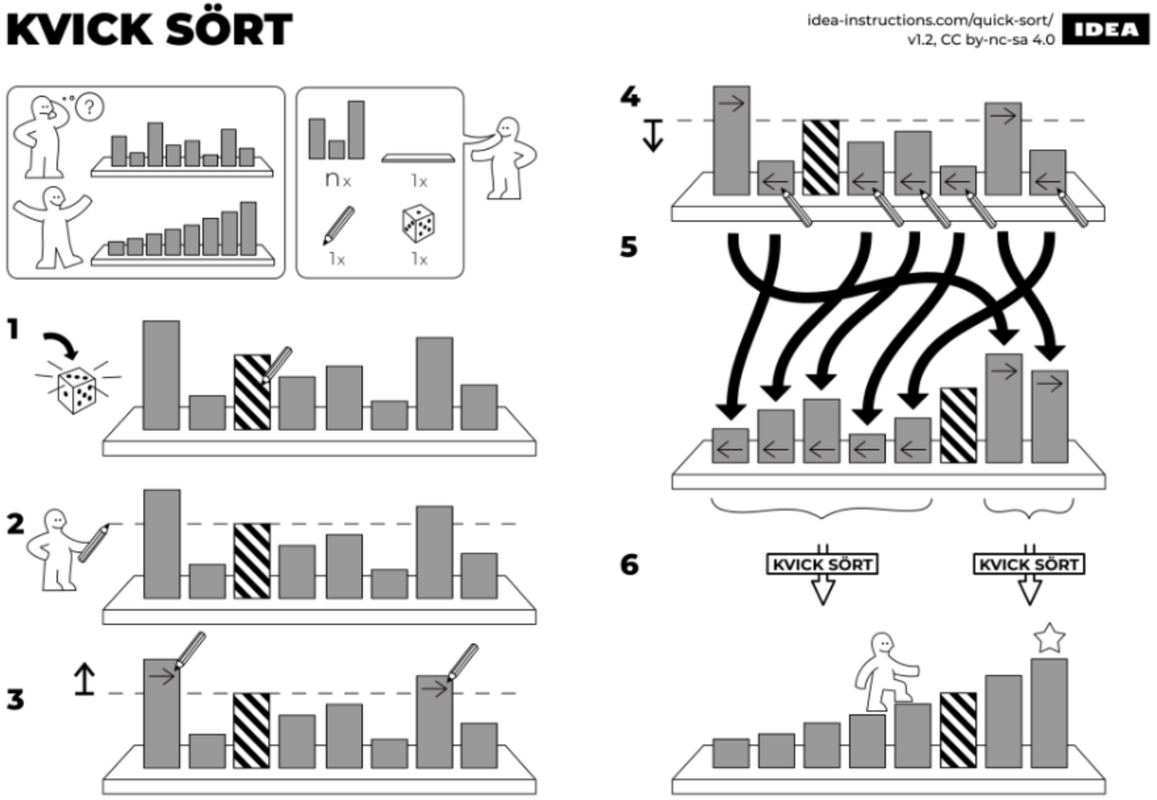
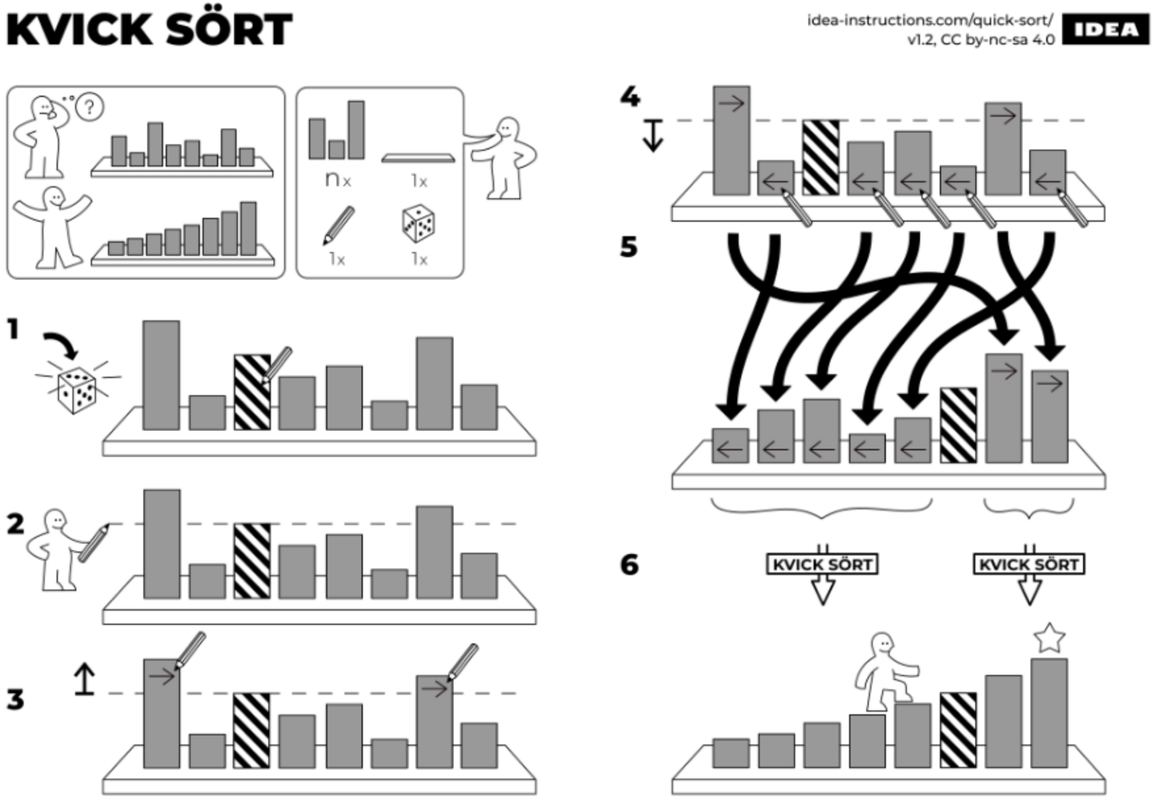


Oblivious Quicksort

- Control flow is non-oblivious



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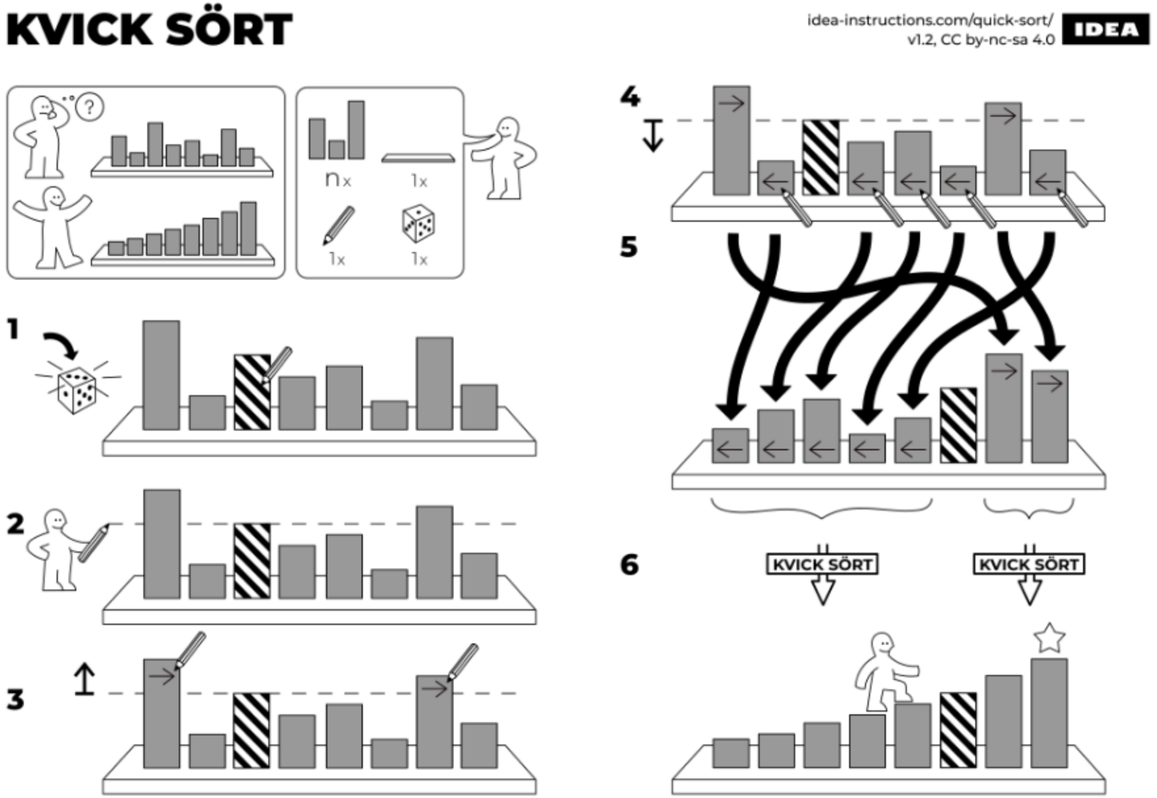

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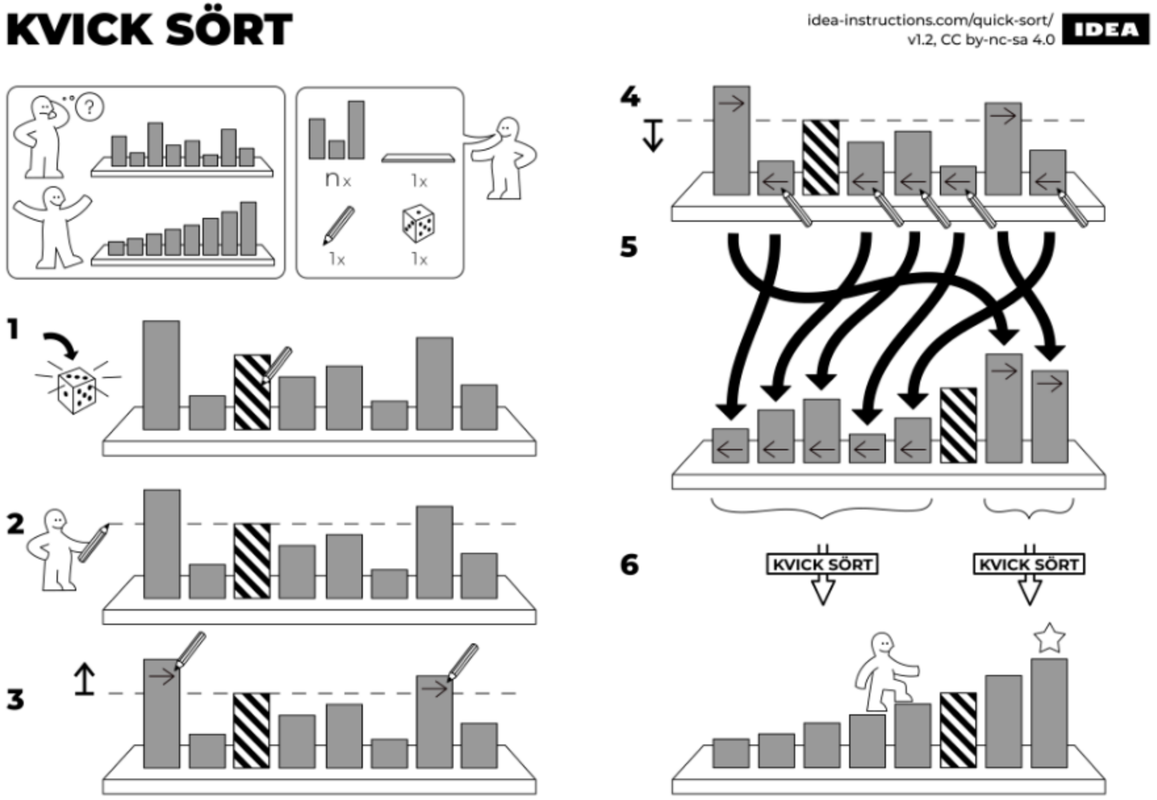

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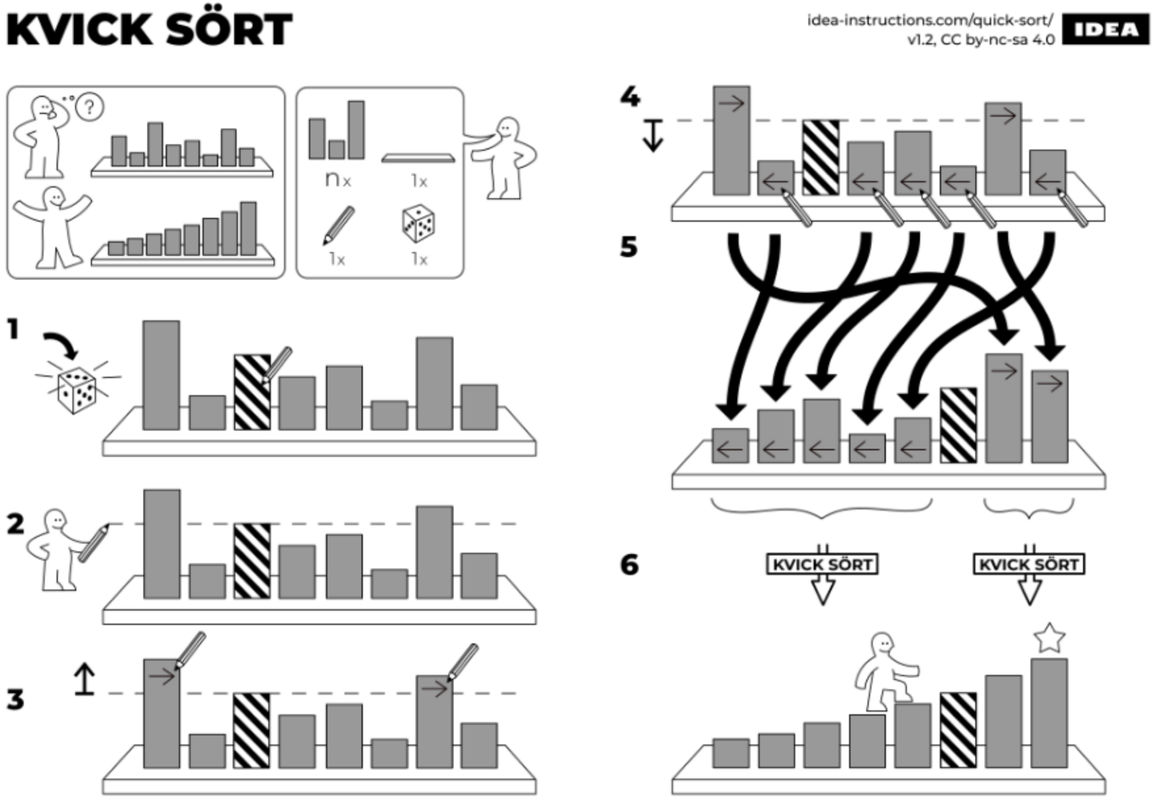

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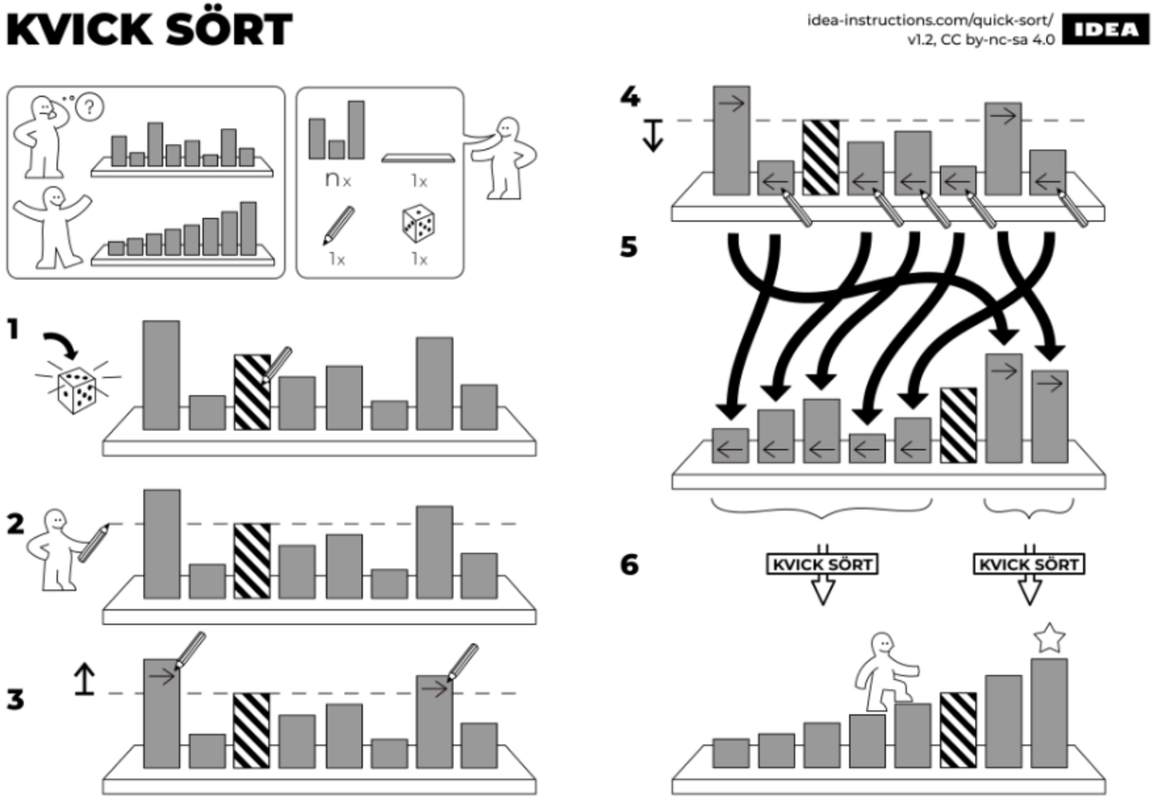
$$[\vec{x}] \to [\pi(\vec{x})] \to [\sigma(\pi(\vec{x}))]$$



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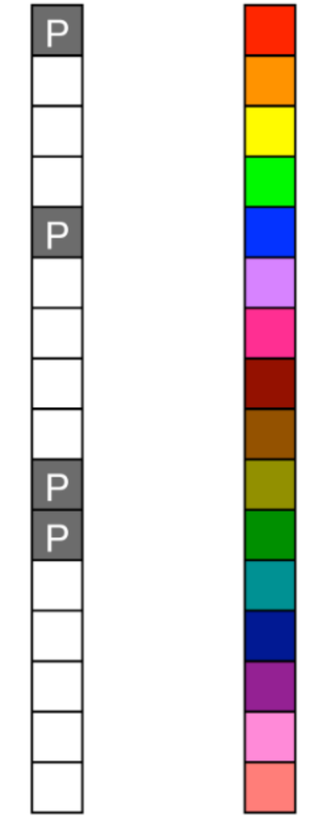

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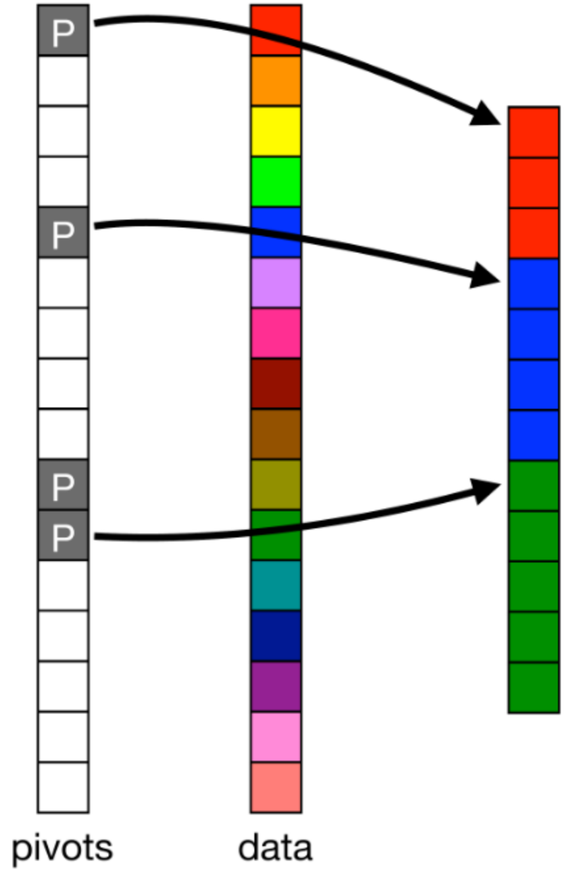

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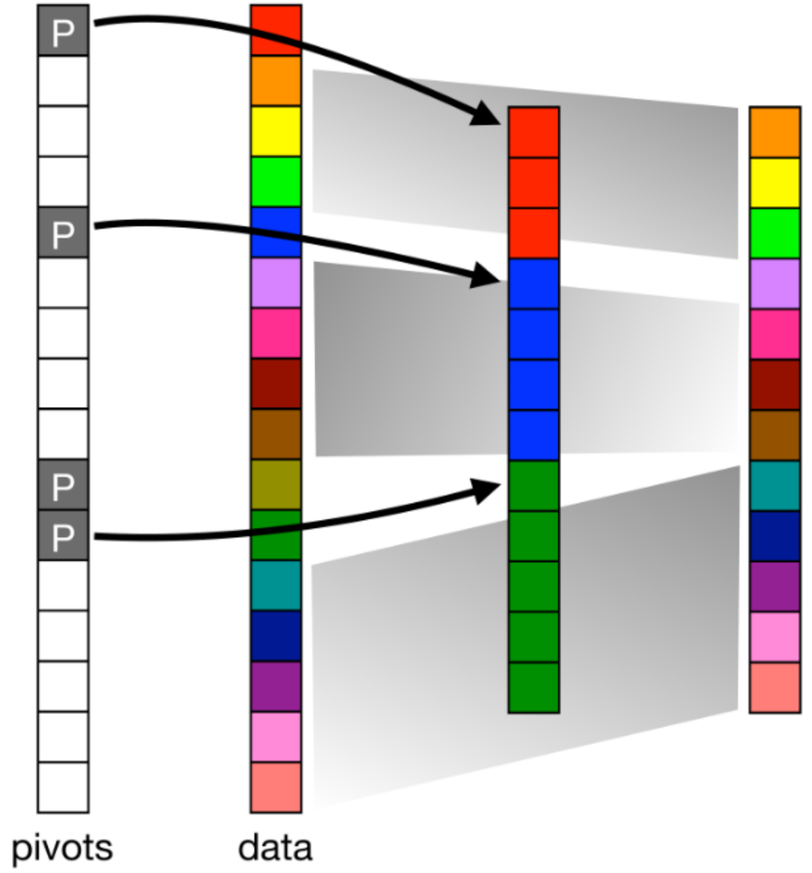

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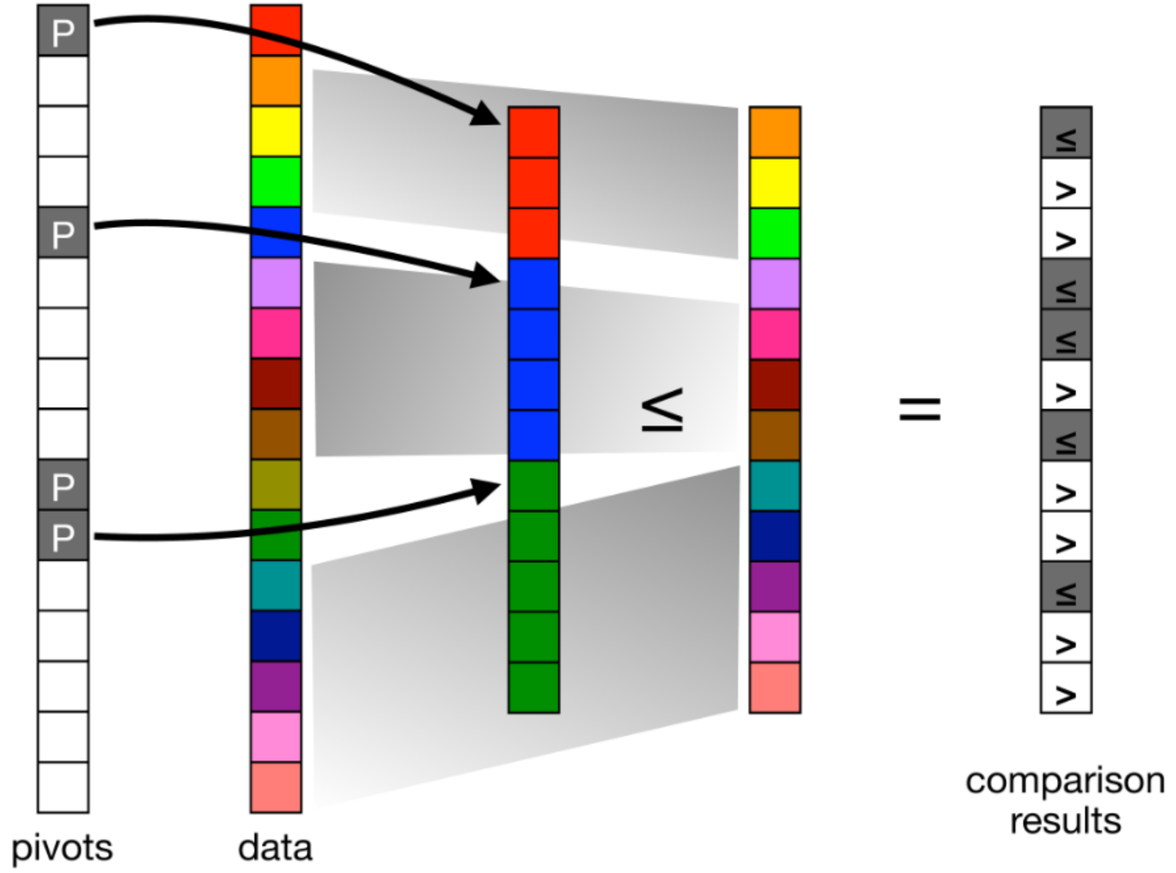

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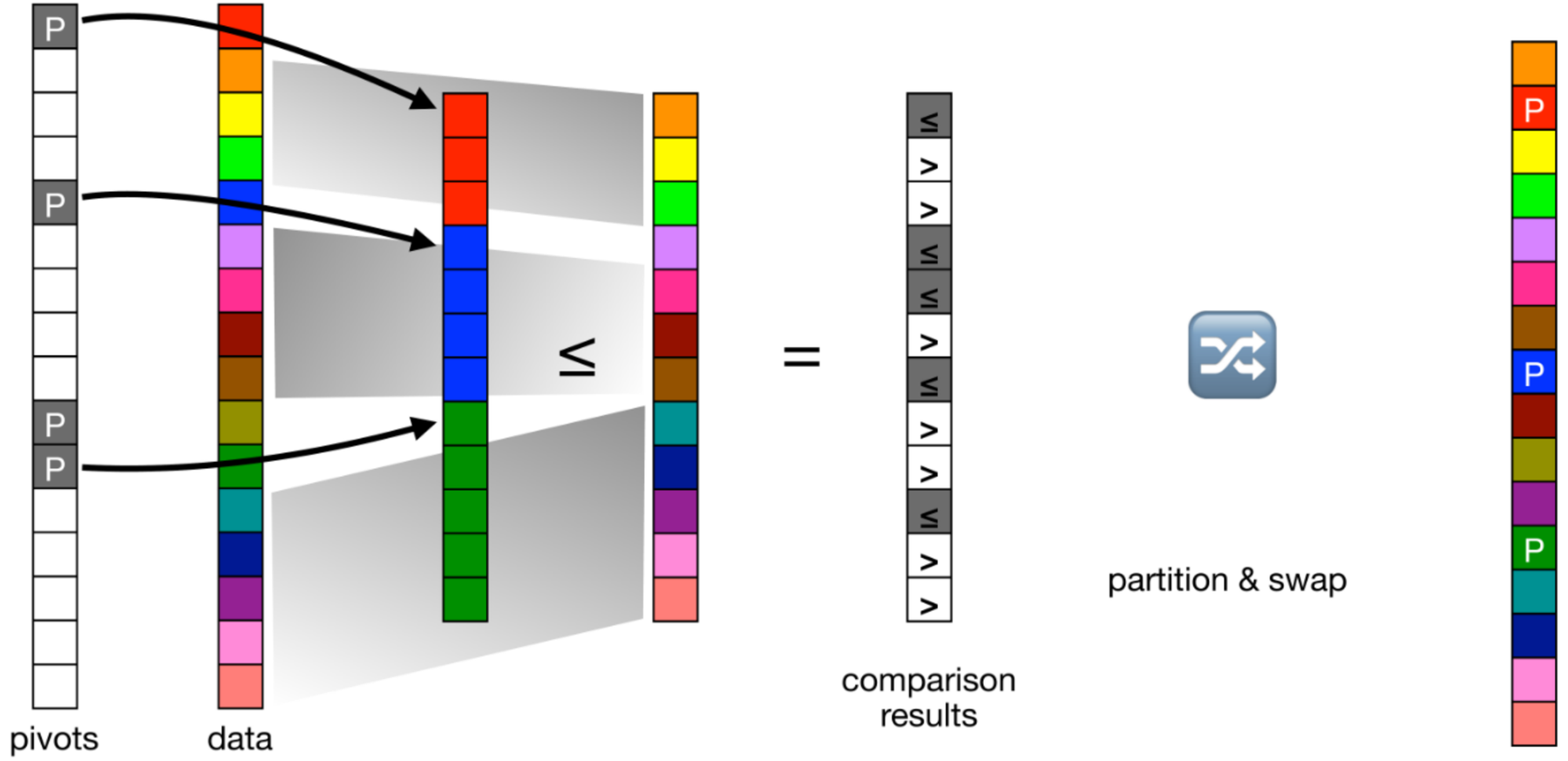

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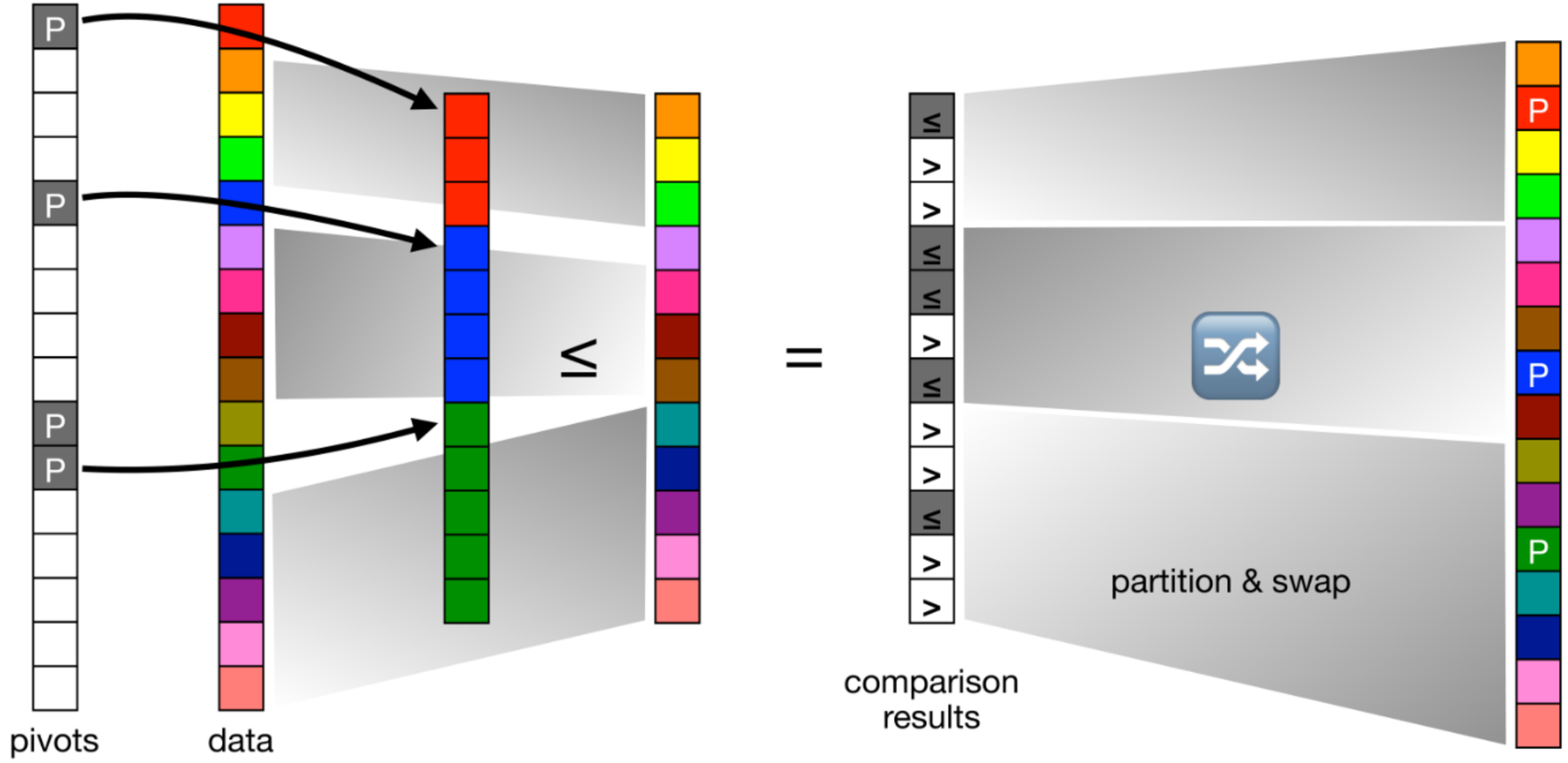

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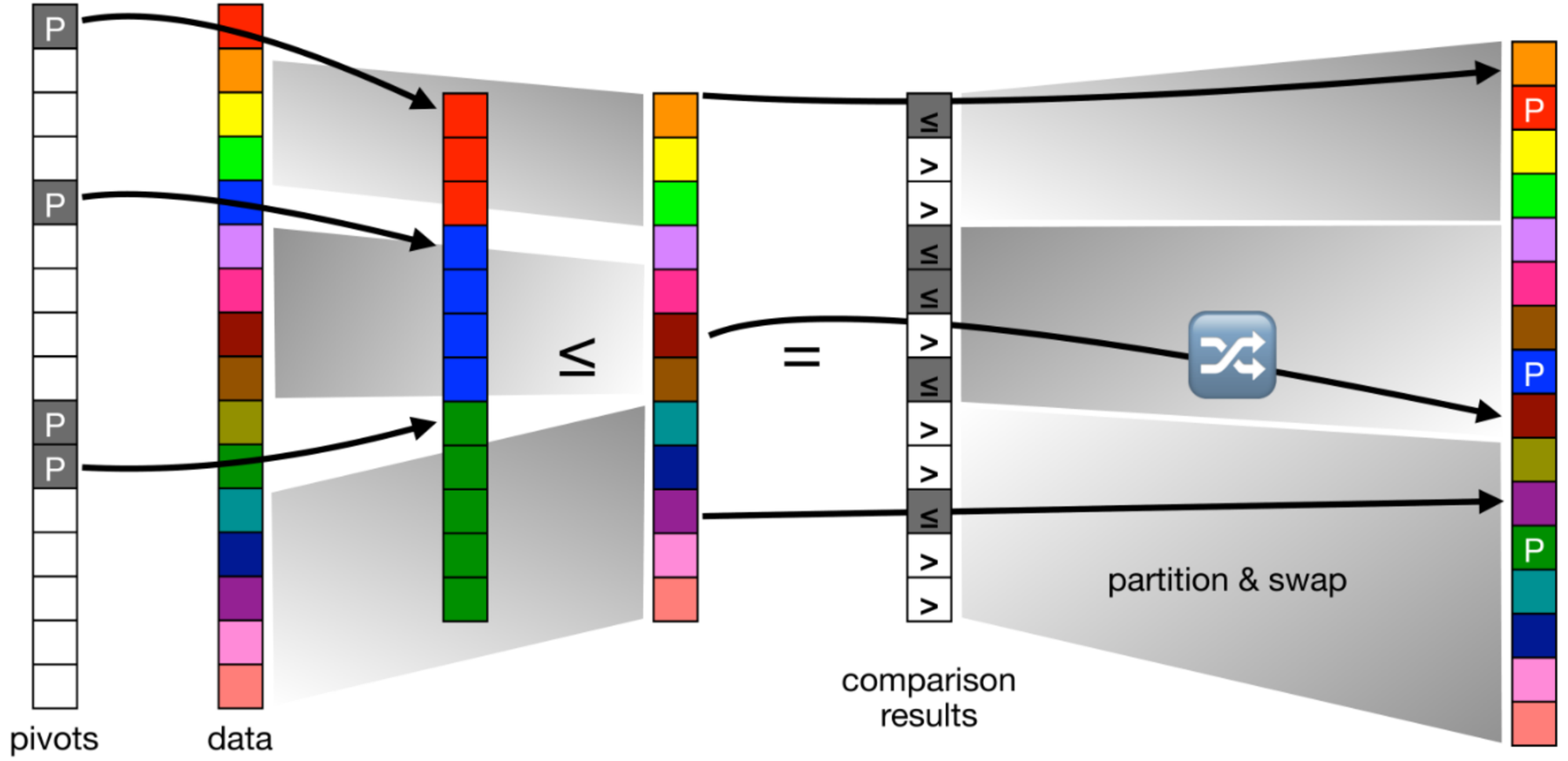
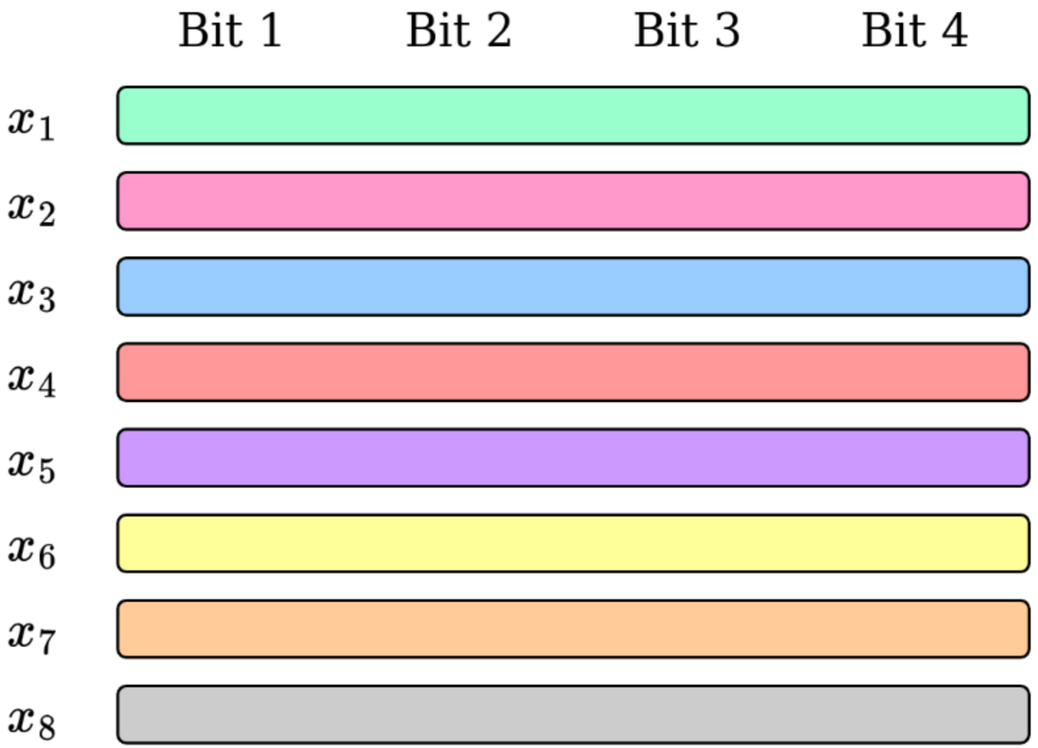


Radix Sort



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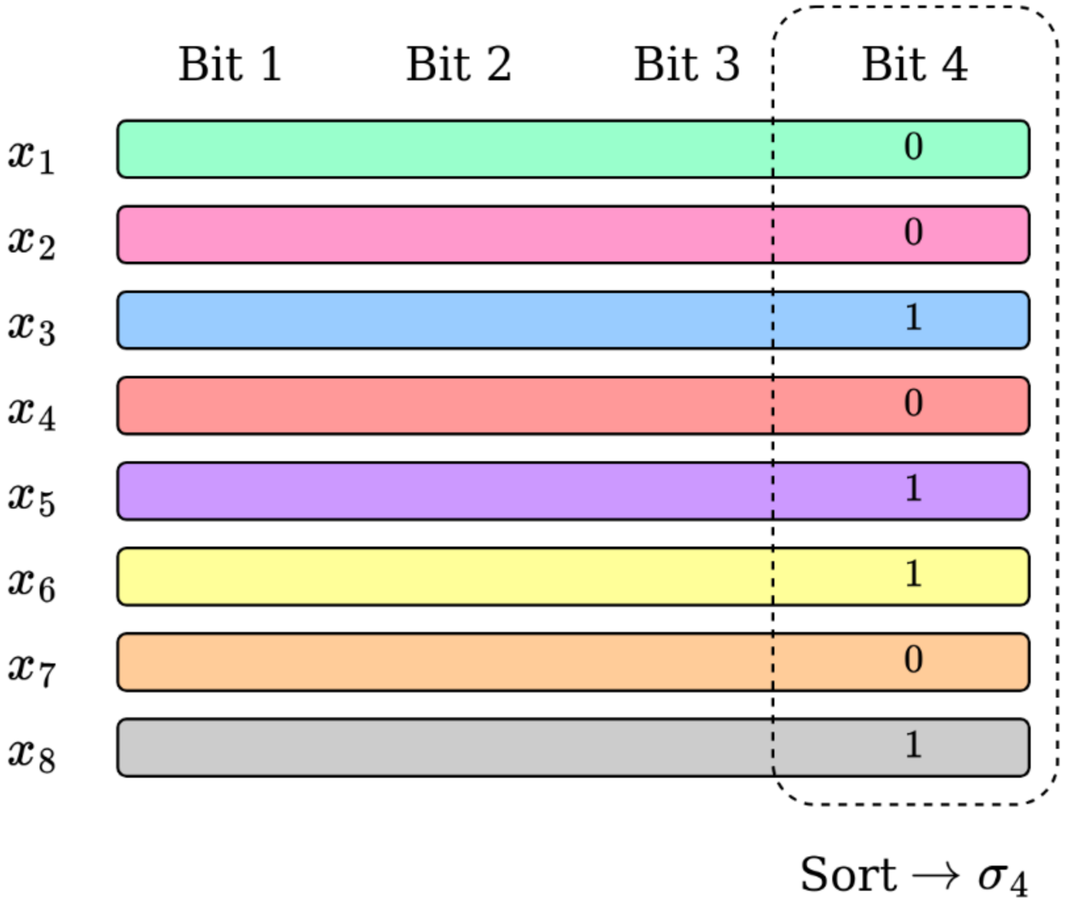
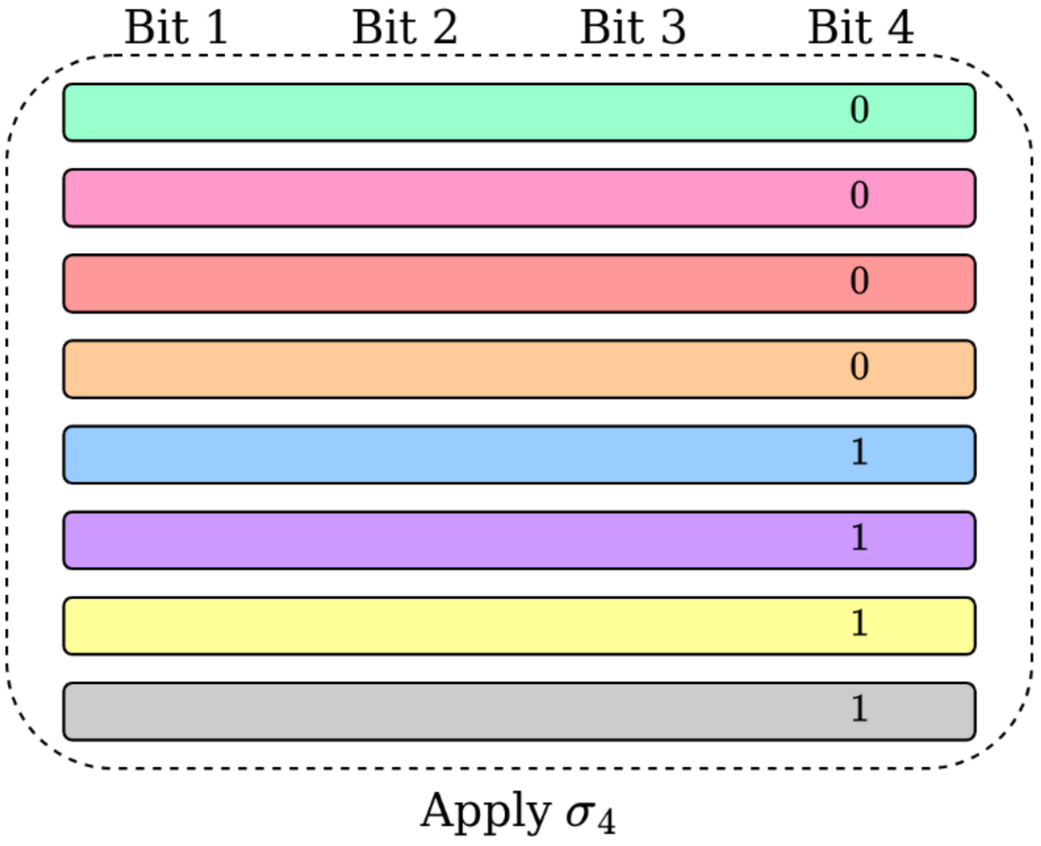
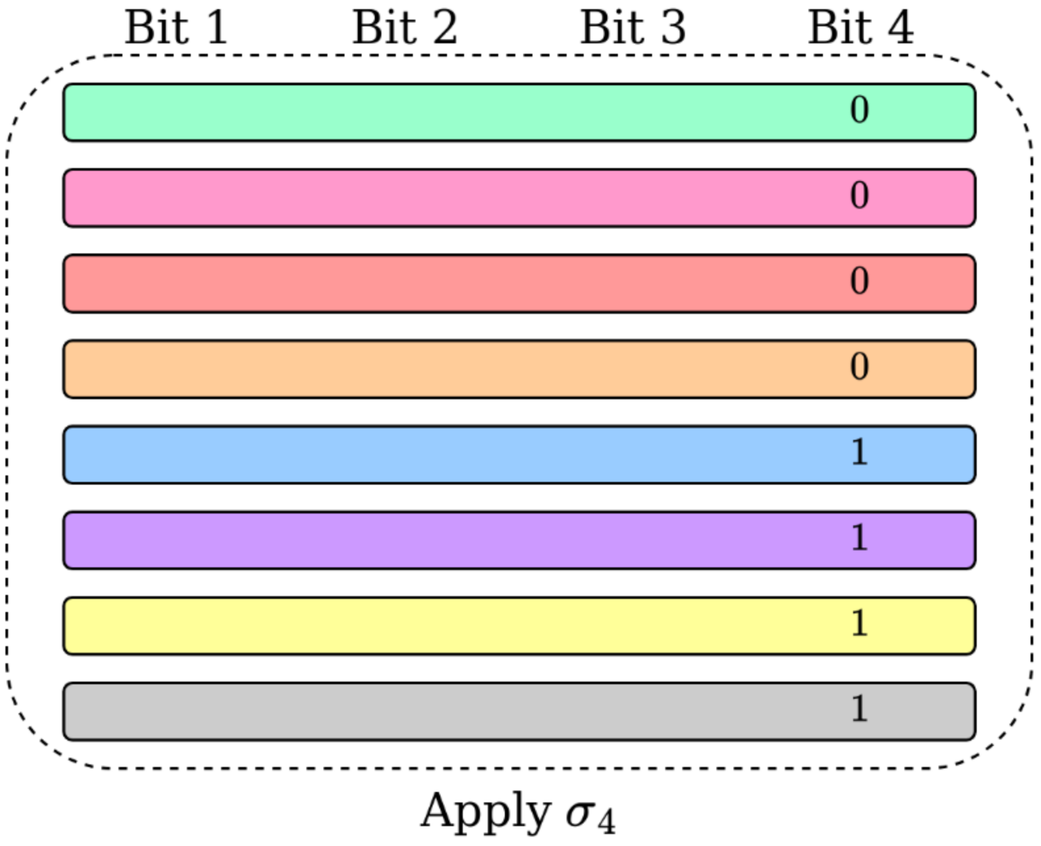

Bit 1
Bit 2
Bit 3
Bit 4
x_1
x_2
x_3
x_4
x_5
x_6
x_7
x_8
0
0
1
0
1
1
0
1
Sort $\rightarrow \sigma_4$

Radix Sort



Radix Sort

Single bit sorting is $O(n)$



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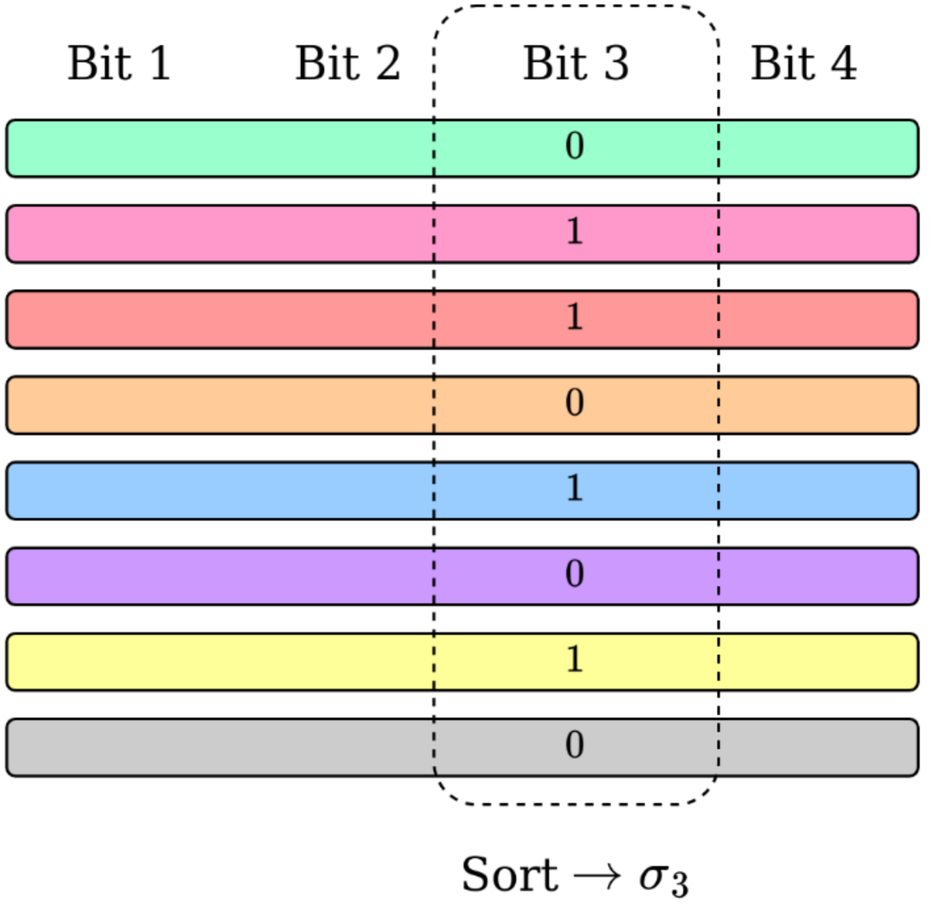

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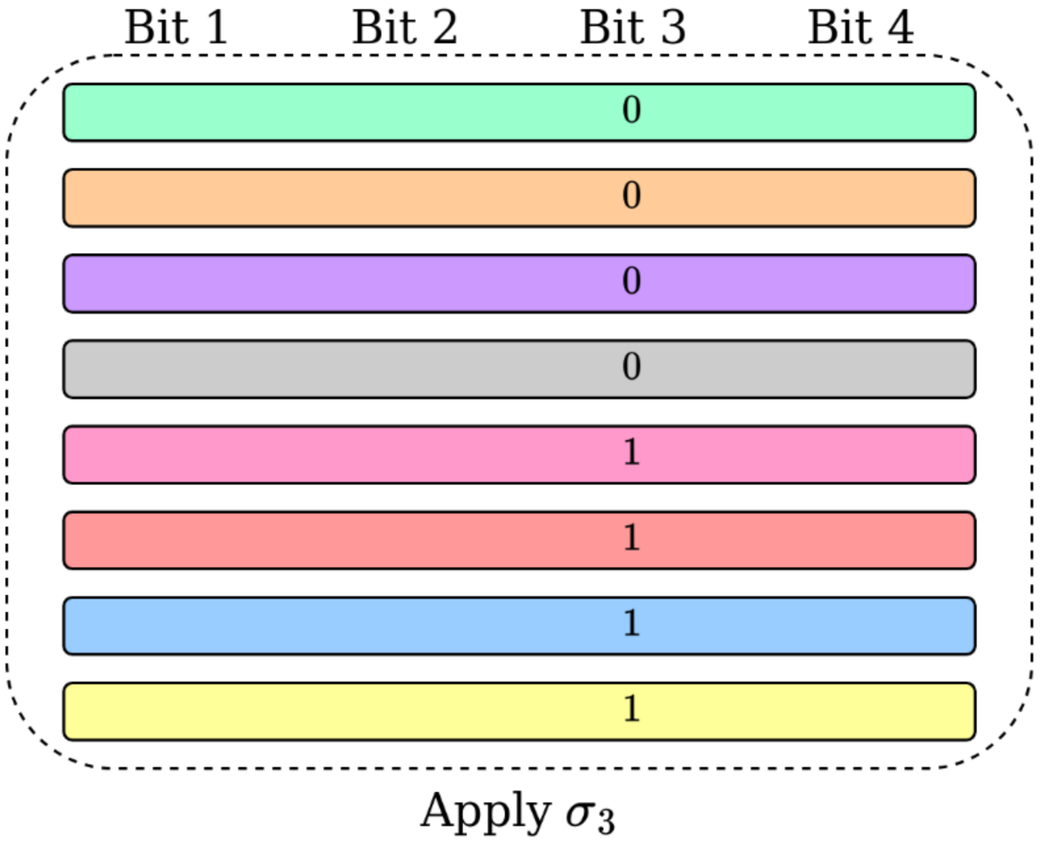

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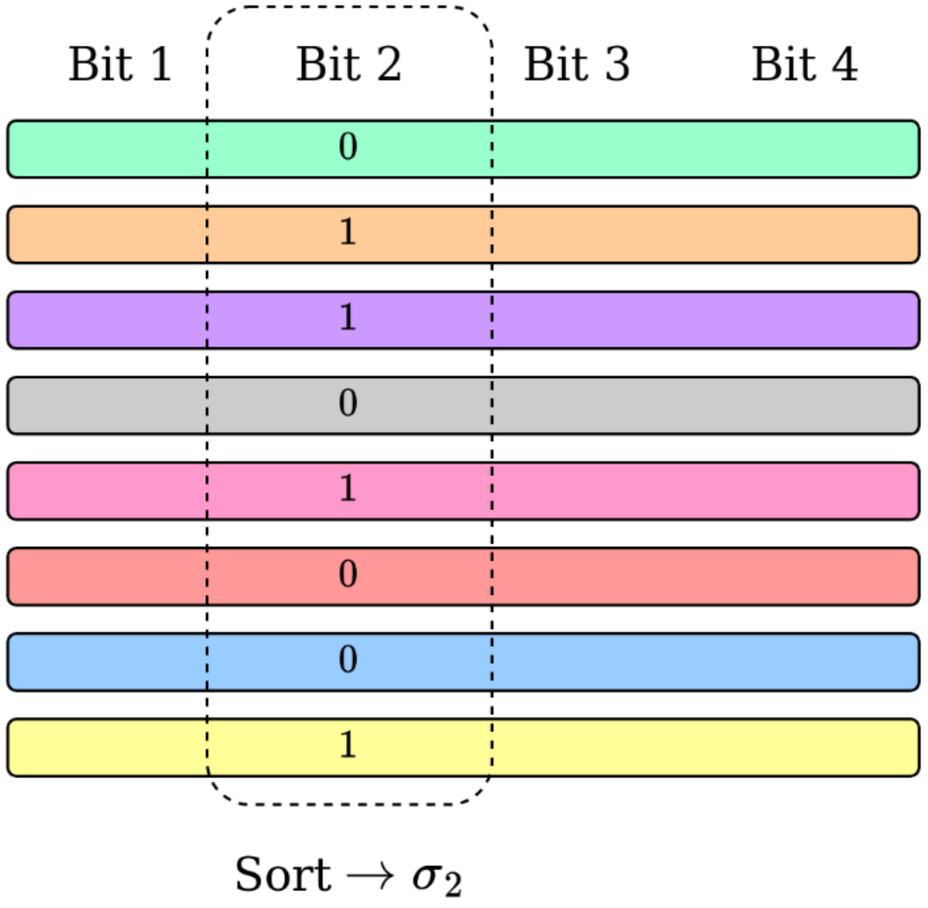

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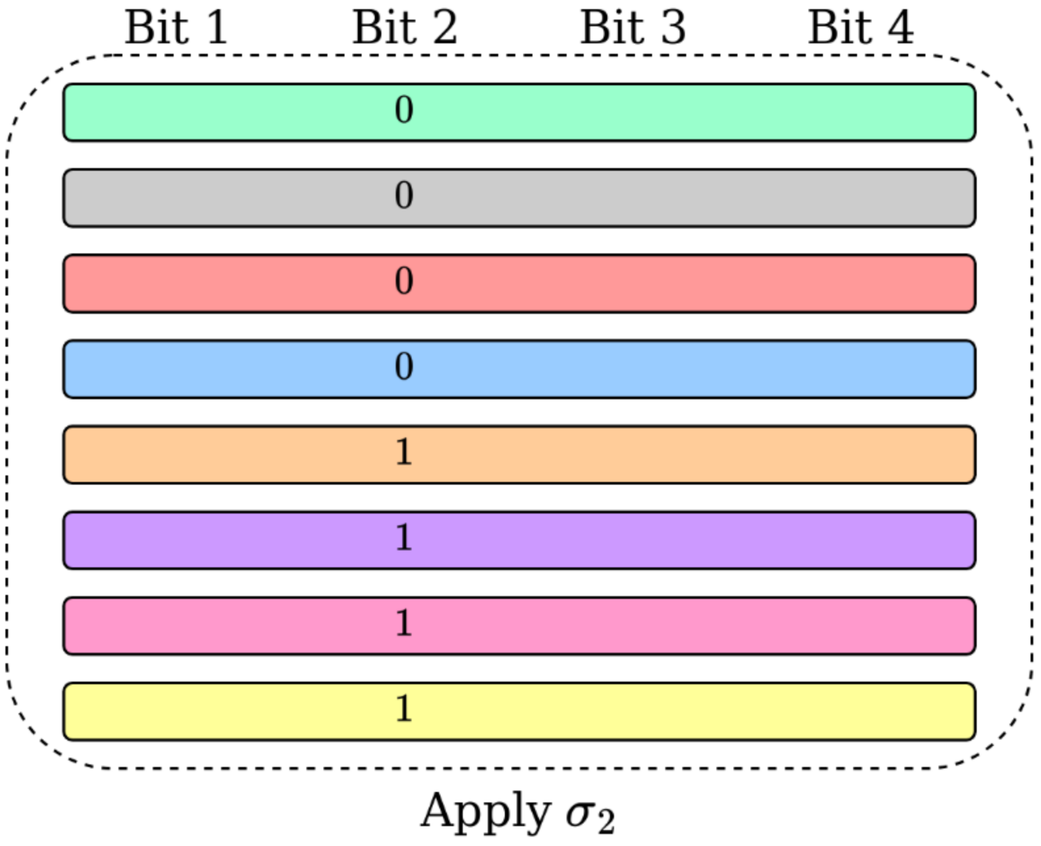

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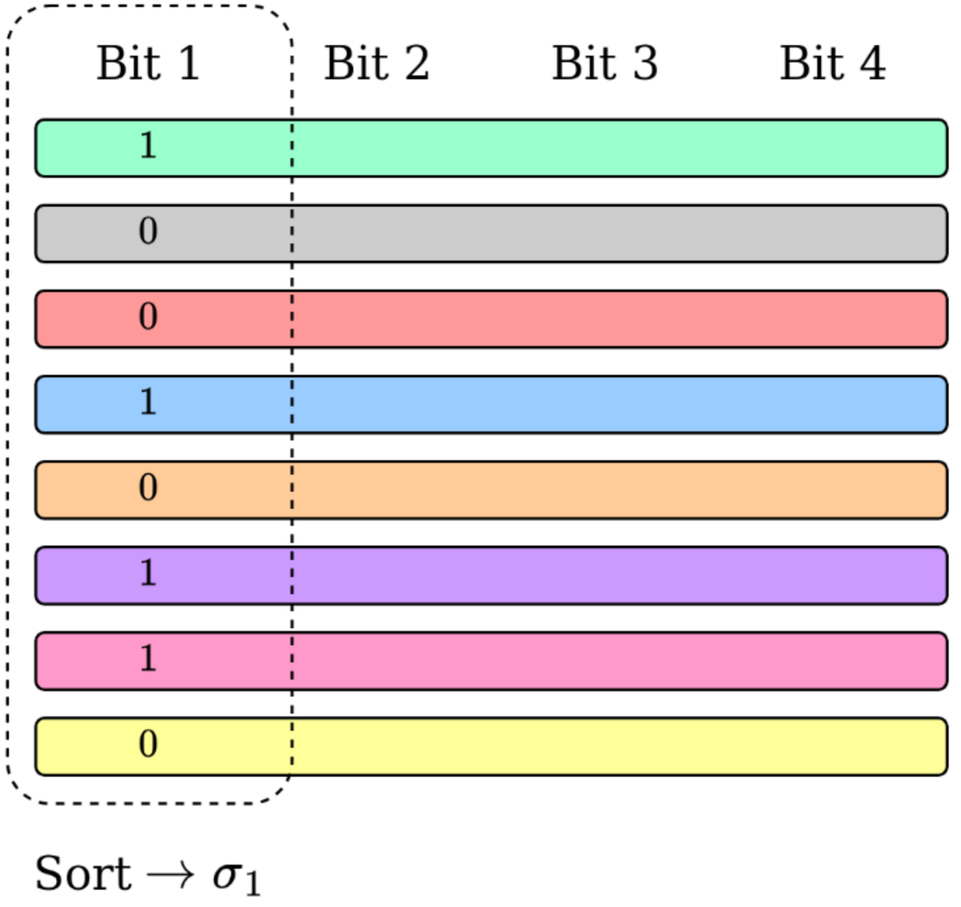

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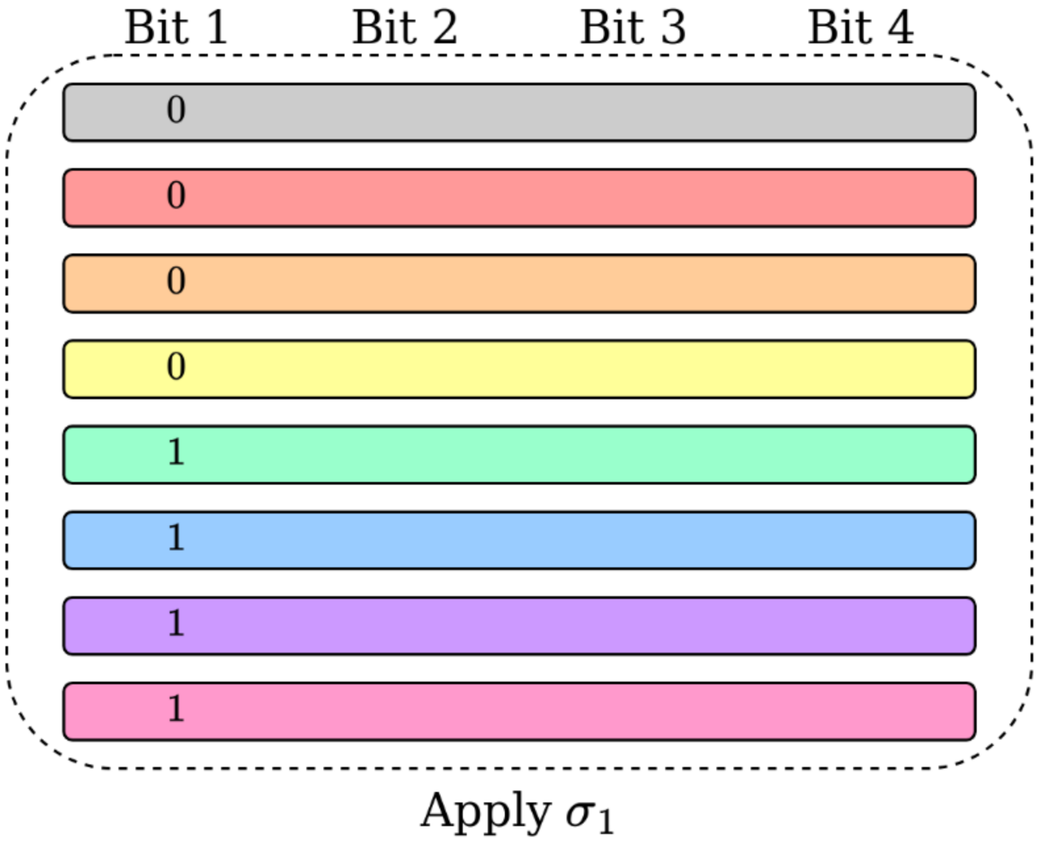

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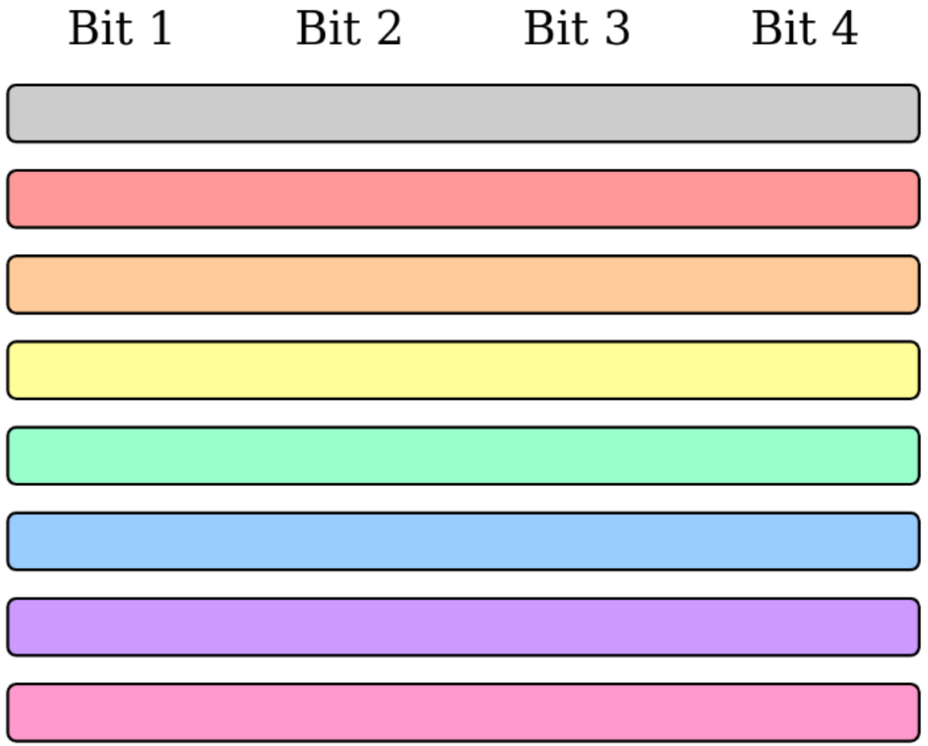
Single bit sorting is $O(n)$



$$\sigma = \sigma_1 \circ \sigma_2 \circ \sigma_3 \circ \sigma_4$$

Analyzing Quicksort and Radixsort

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Quicksort: $O(n \log n)$ operations

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Algorithm	Communication
Quicksort	$O(\ell n \log n)$
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Gold standard of oblivious sorting is $O(n \log^2 n)$ communication.

Extracting Sorting Permutations

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Two Challenges

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- Ensuring stability

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- Extracting the applied permutation

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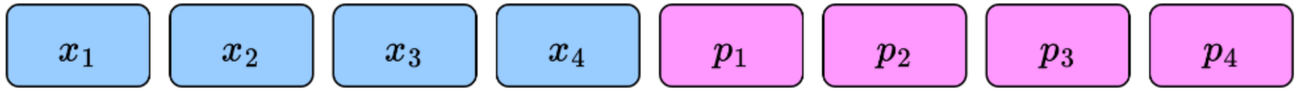

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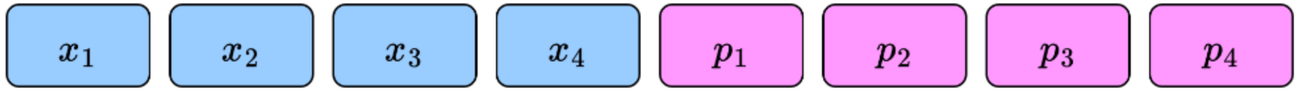


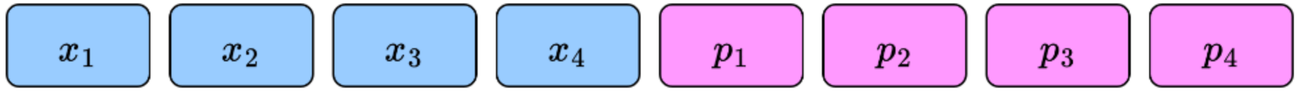
$$x_i \rightarrow x_i||i$$

Extracting Sorting Permutations

Two Challenges

- Ensuring stability
- Extracting the applied permutation

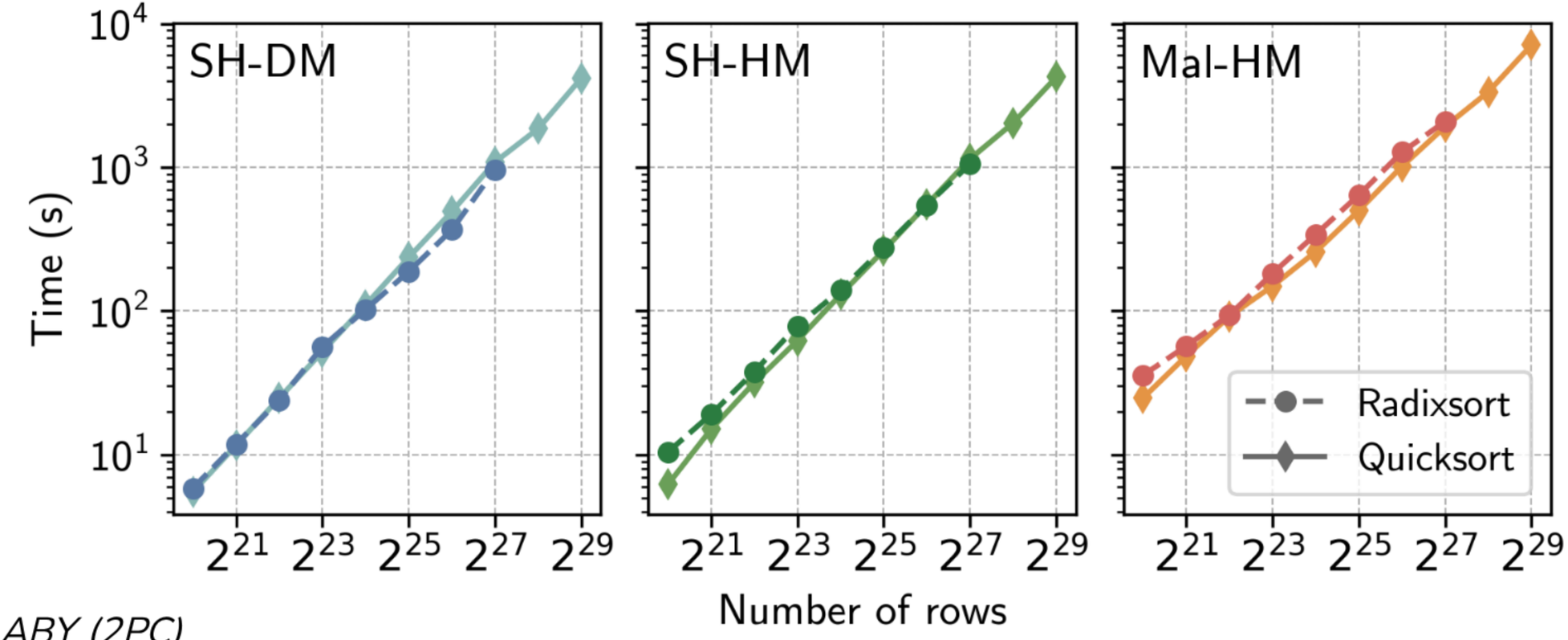
One Solution



- Input padding
- Padded bits represent the applied permutation

$$x_i \rightarrow x_i||i$$

Experimental Results



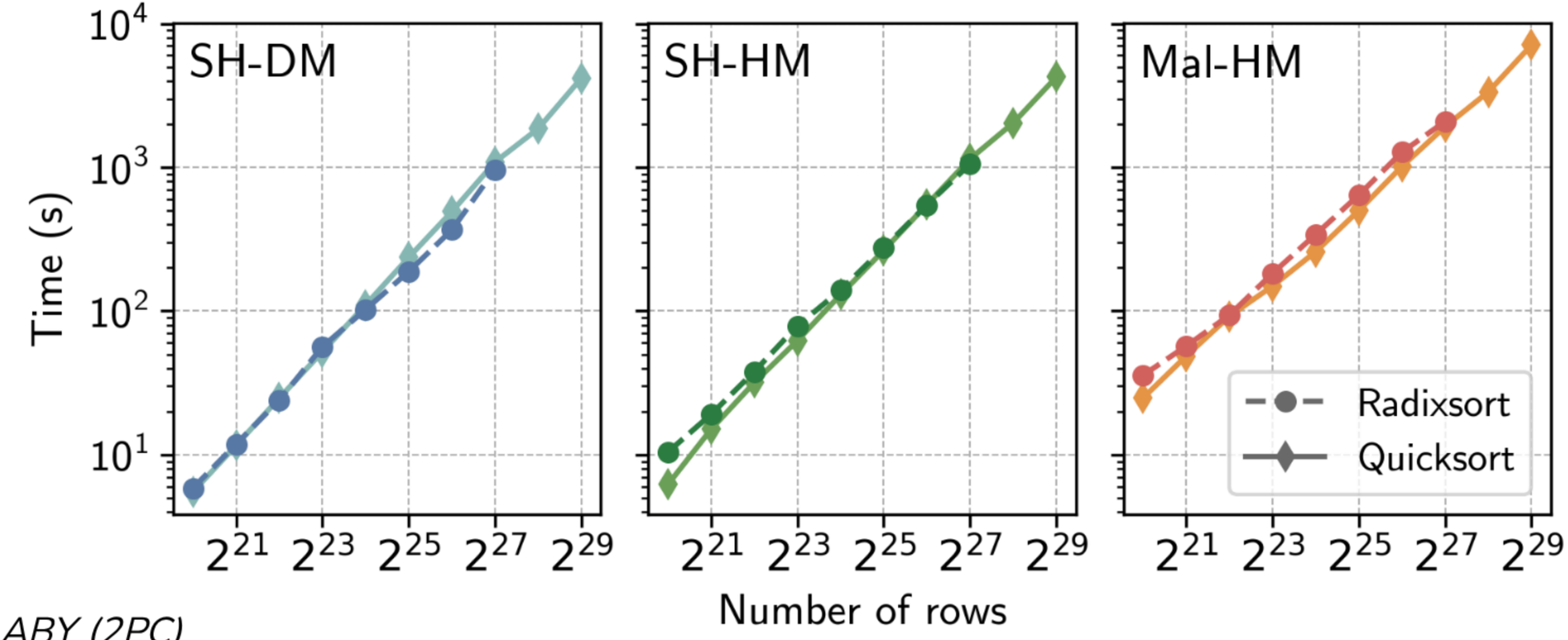
SH-DM: *ABY (2PC)*

SH-HM: *Araki et al. (3PC)*

Mal-HM: *Fantastic Four (4PC)*

Experimental Results

Up to 500m elements!



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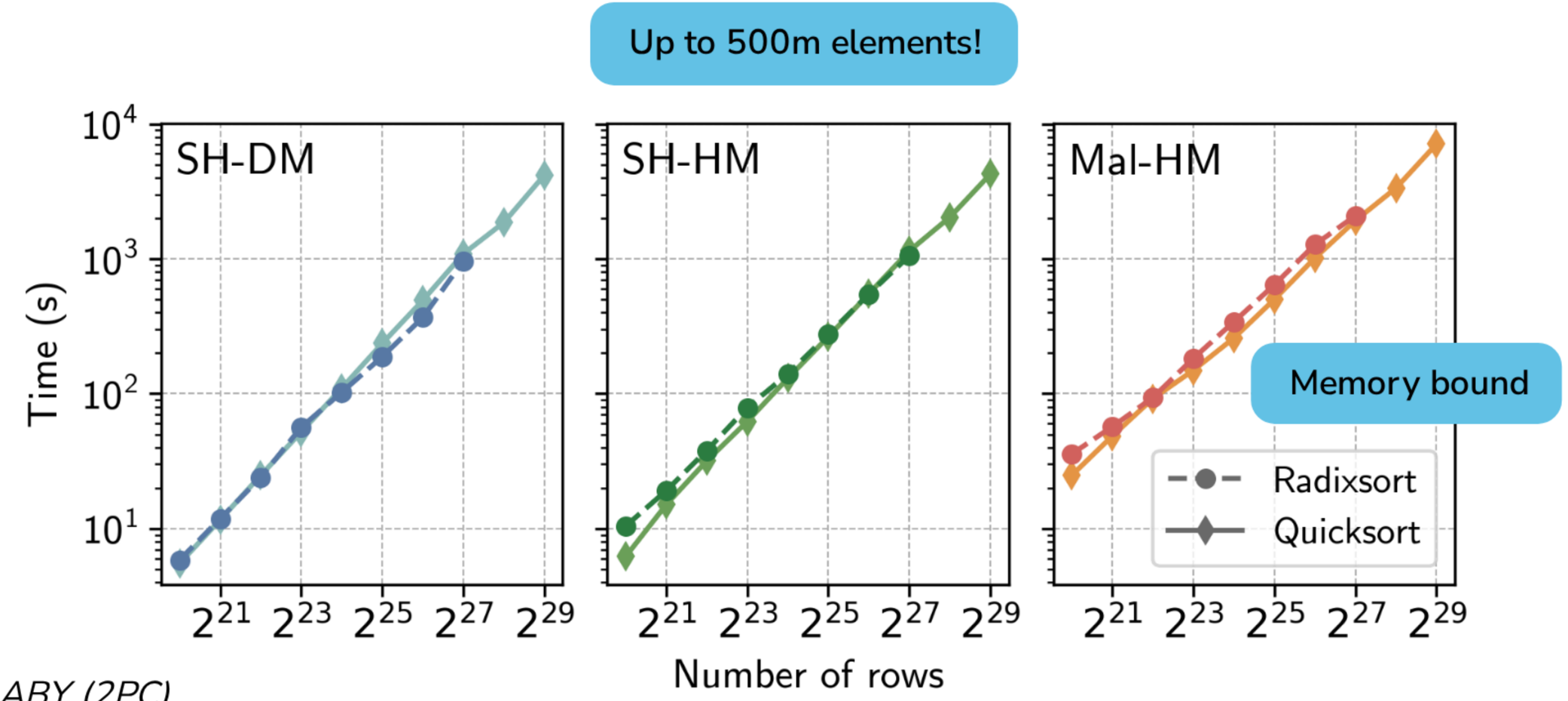

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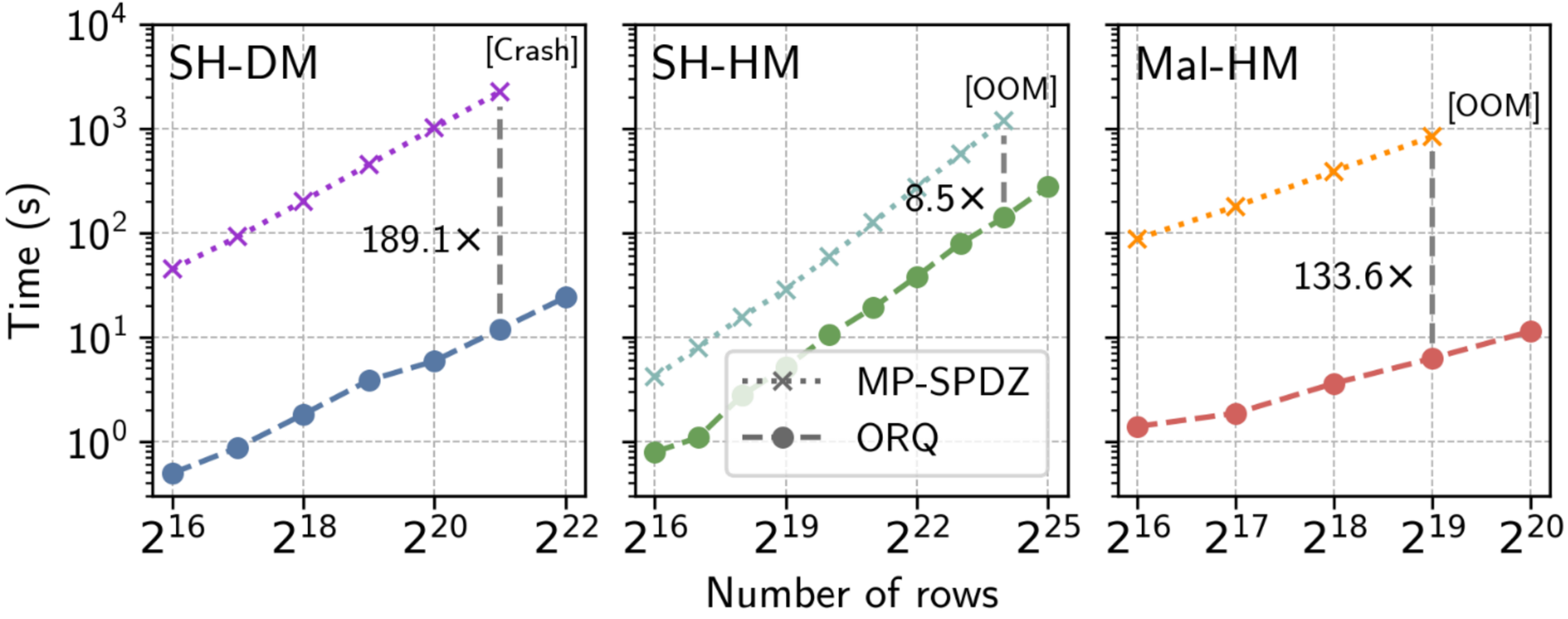
Comparison with MP-SPDZ



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ORQ

Roadmap

~~1. Deployment of MPC for Political Polling~~

~~2. Oblivious Sorting~~

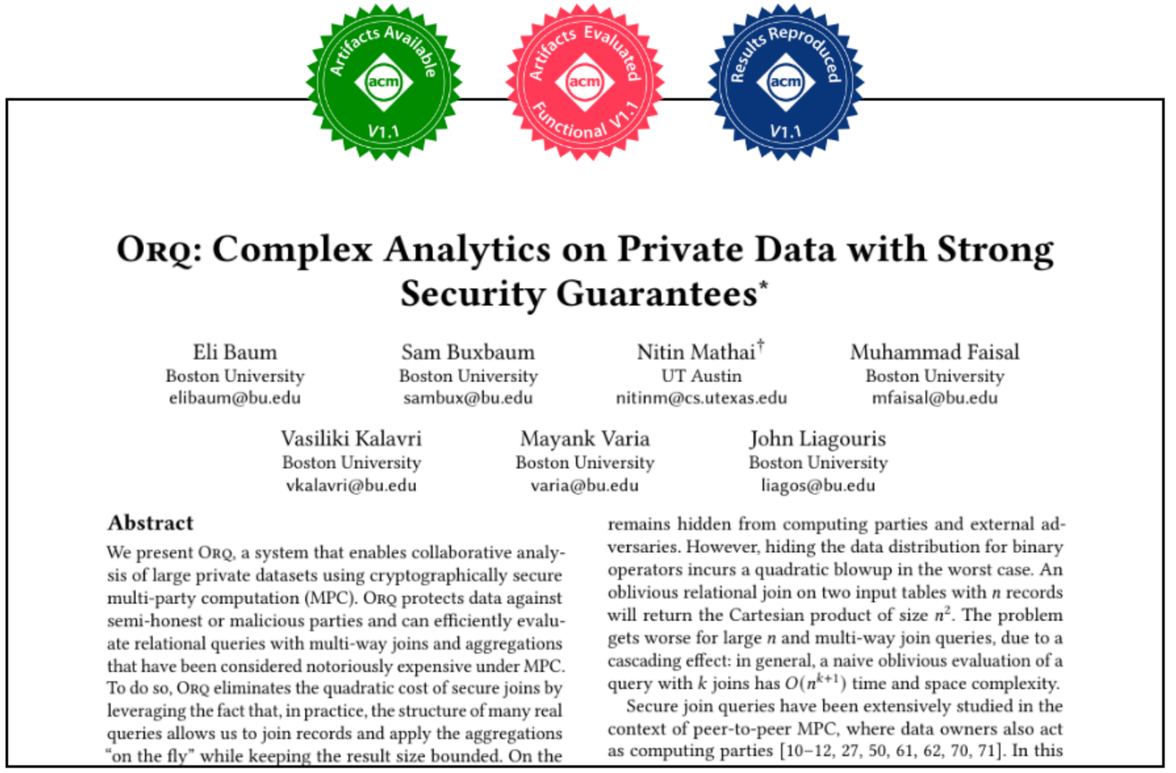
3. ORQ

ORQ: Complex Analytics on Private Data with Strong Security Guarantees

Distinguished Artifact Award at SOSP 25!

Paper

GitHub



ORQ: Complex Analytics on Private Data with Strong Security Guarantees*

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Abstract

We present ORQ, a system that enables collaborative analysis of large private datasets using cryptographically secure multi-party computation (MPC). ORQ protects data against semi-honest or malicious parties and can efficiently evaluate relational queries with multi-way joins and aggregations that have been considered notoriously expensive under MPC. To do so, ORQ eliminates the quadratic cost of secure joins by leveraging the fact that, in practice, the structure of many real queries allows us to join records and apply the aggregations "on the fly" while keeping the result size bounded. On the remains hidden from computing parties and external adversaries. However, hiding the data distribution for binary operators incurs a quadratic blowup in the worst case. An oblivious relational join on two input tables with n records will return the Cartesian product of size n^2. The problem gets worse for large n and multi-way join queries, due to a cascading effect: in general, a naive oblivious evaluation of a query with k joins has $O(n^{k+1})$ time and space complexity.

Secure join queries have been extensively studied in the context of peer-to-peer MPC, where data owners also act as computing parties [10–12, 27, 50, 61, 62, 70, 71]. In this

Oblivious Analytics

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Secure computation must be *oblivious*.

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Customer	AcctBalance
🤡	$1000
😈	$2000
🤡	$500
🐸	$60
🥶	$8000
🥶	$700
🐸	-$200

`filter(Customer ≠ 🐸)`

Customer	AcctBalance
▓▓▓	▓▓▓
▓▓▓	▓▓▓
▓▓▓	▓▓▓
▓▓▓	▓▓▓
▓▓▓	▓▓▓
▓▓▓	▓▓▓
▓▓▓	▓▓▓

Oblivious Joins

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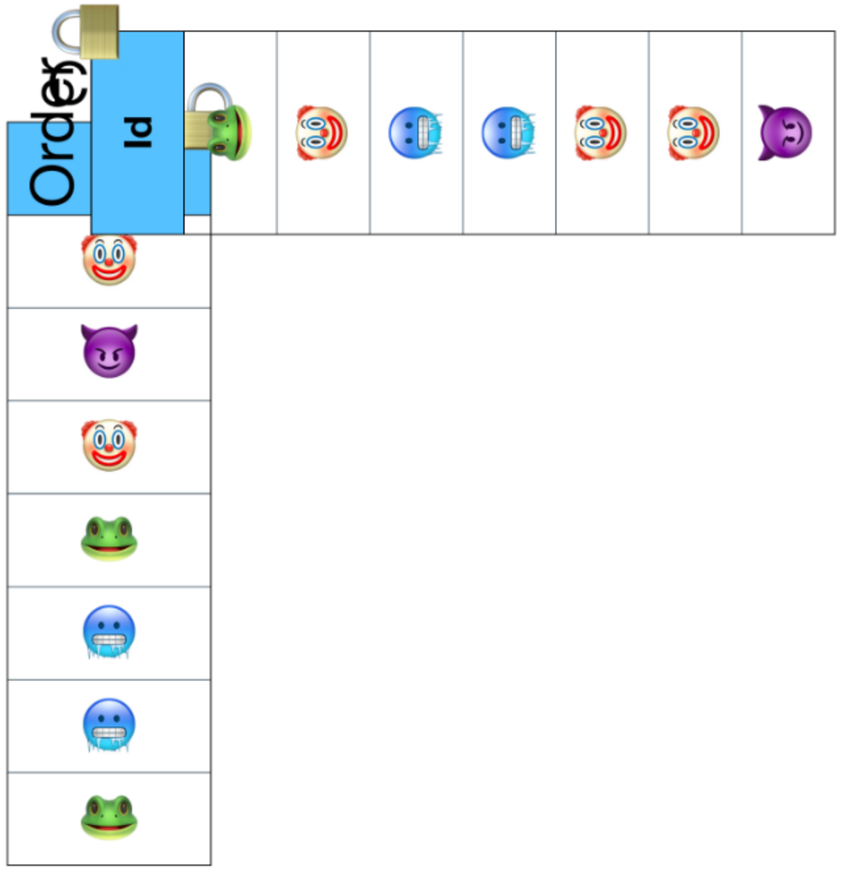

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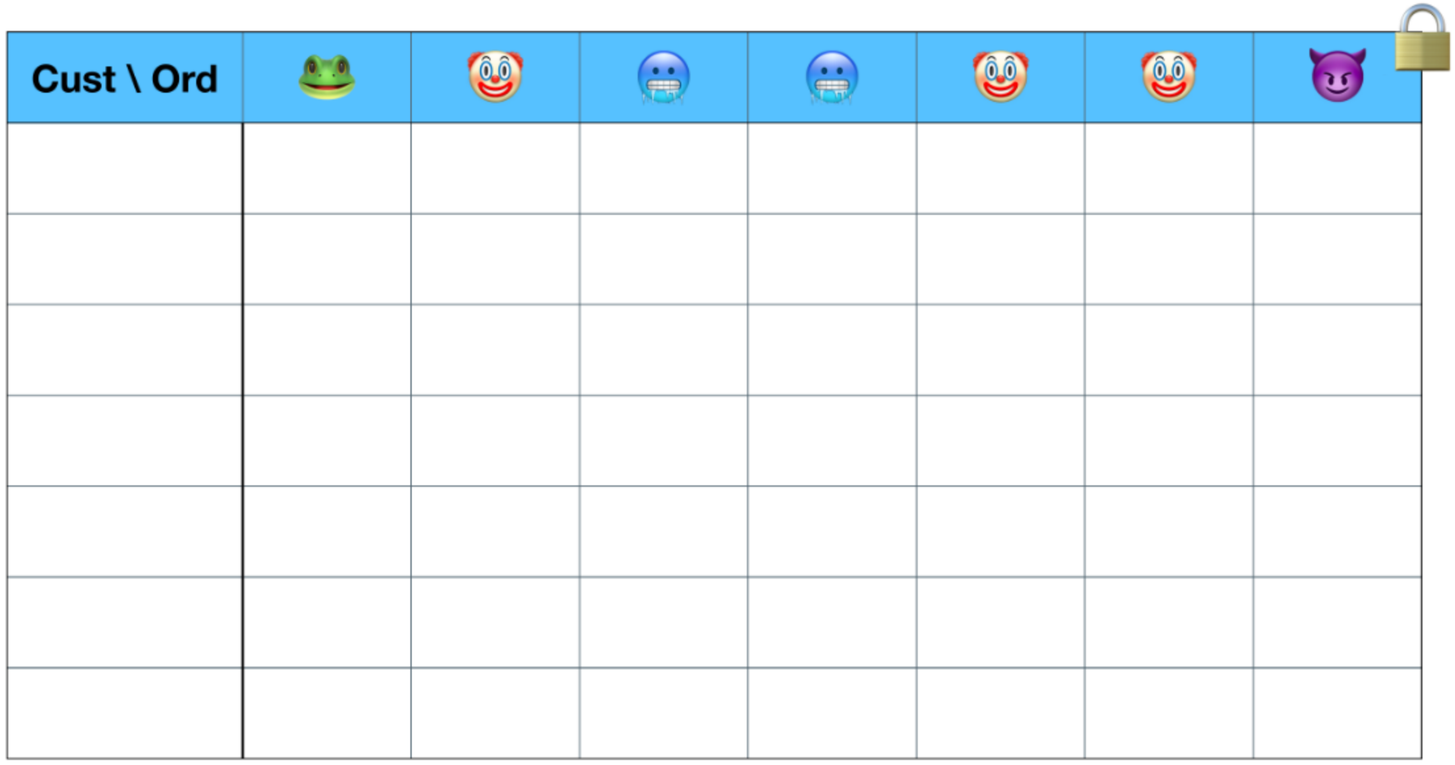


Oblivious Joins

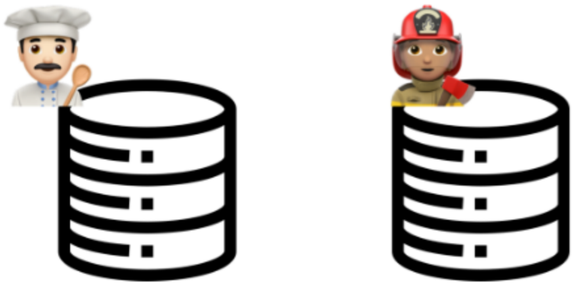
Cust \ Ord	🐸	🤡	🥶	🥶	🤡	🤡	😈 🔒
🤡	❌	✅	❌	❌	✅	✅	❌
😈	❌	❌	❌	❌	❌	❌	✅
🤡	❌	✅	❌	❌	✅	✅	❌
🐸	✅	❌	❌	❌	❌	❌	❌
🥶	❌	❌	✅	✅	❌	❌	❌
🥶	❌	❌	✅	✅	❌	❌	❌
🐸	✅	❌	❌	❌	❌	❌	❌

Oblivious Joins

Cust \ Ord	🐸	🤡	🥶	🥶	🤡	🤡	😈 🔒
🤡	❌	✅	❌	❌	✅	✅	❌
😈	❌	❌	❌	❌	❌	❌	✅
🤡	❌	✅	❌	❌	✅	✅	❌
🐸	✅	❌	❌	❌	❌	❌	❌
🥶	❌	❌	✅	✅	❌	❌	❌
🥶	❌	❌	✅	✅	❌	❌	❌
🐸	✅	❌	❌	❌	❌	❌	❌

$O(n^2)$ time and space

The Problem



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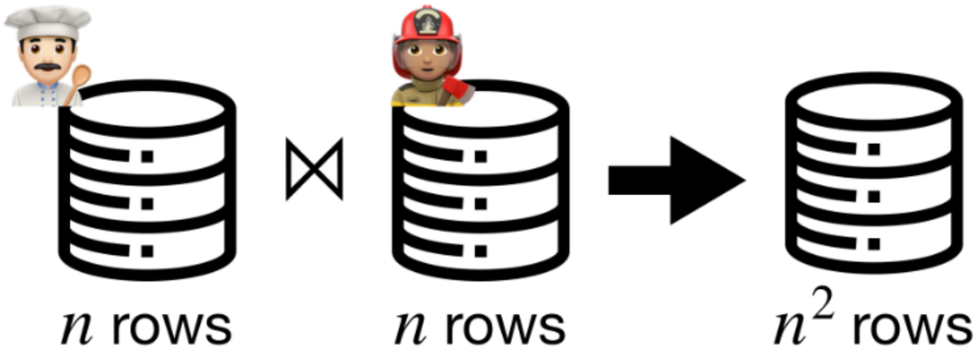

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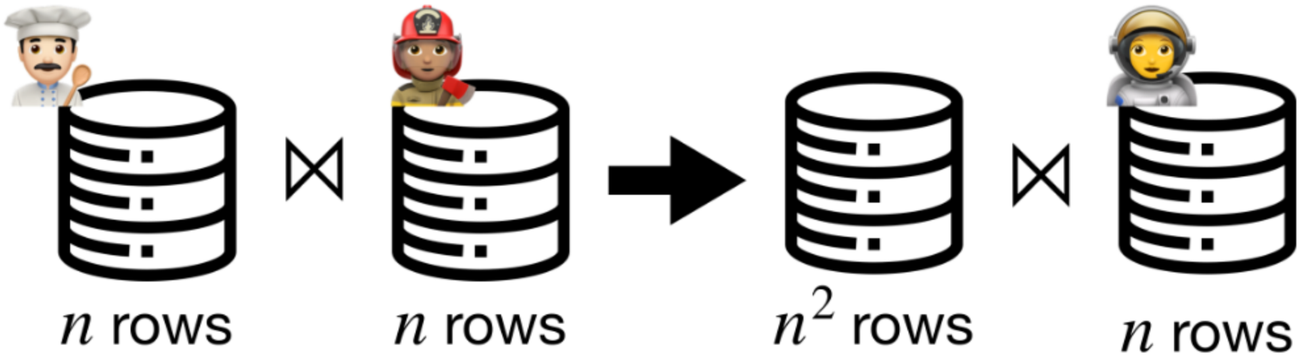

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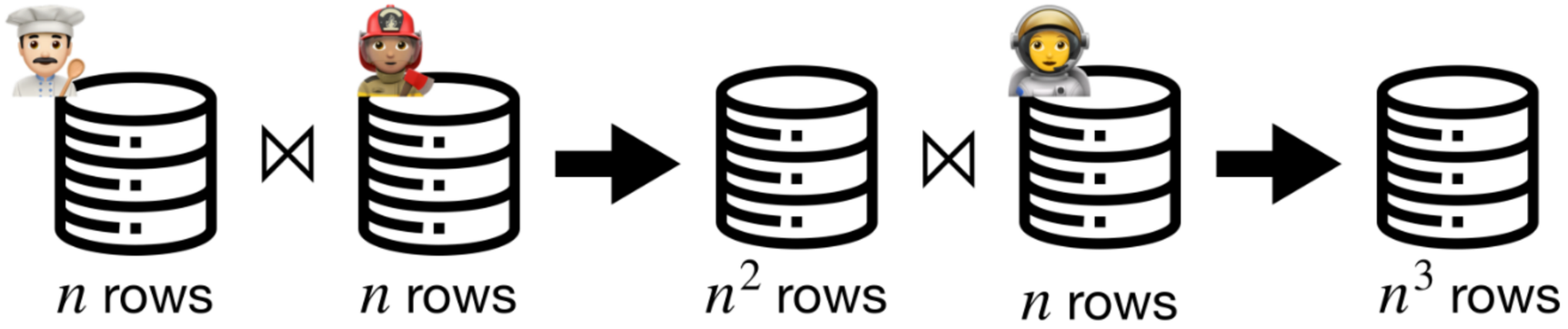

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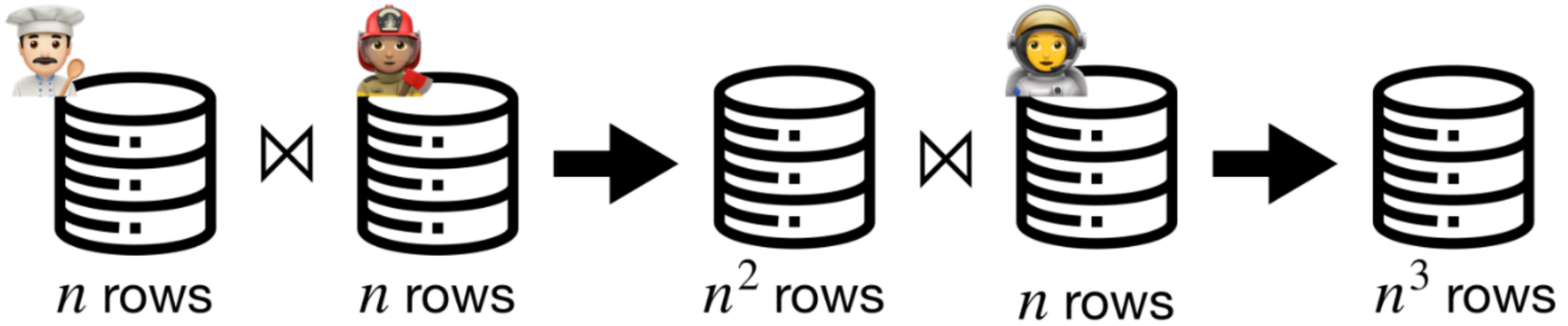
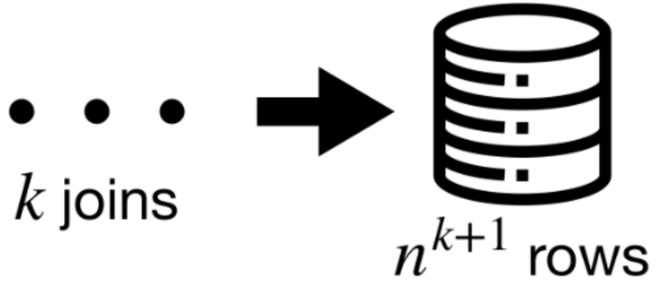

n rows
$\bowtie$
n rows
n^2 rows
$\bowtie$
n rows
n^3 rows


k joins
n^{k+1} rows

Our Insight

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Joins with incremental aggregations can avoid the quadratic blowup.

$$f(X, Y) = f(f(X), f(Y))$$

Join-Aggregation Operator

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Starting point: one-to-many joins

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```
func join(L, R, keys):
   T := concat*(L, R)
   T.sort(keys)
   T.prefix_aggregate(…)
   return T
```

Join-Aggregation Operator

Starting point: one-to-many joins

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func join(L, R, keys):
   T := concat*(L, R)
   T.sort(keys)
   T.prefix_aggregate(…)
   return T
```

Reduces to sorting and aggregation!

Supporting Many-to-Many Joins

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Customer	C.Total	O.Cost
∅	∅	
🤡	$1500	
😈	$2000	
∅	∅	
🐸	-$140	
∅	∅	
🥶	$8700	
🐸		$30
🤡		$20
🥶		$55
🥶		$18
🤡		$70
🤡		$12
😈		$8

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- The aggregation must be decomposable

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We can transform a many-to-many join into a one-to-many join.

Pre-aggregate one of the tables and mark duplicate keys as invalid.

- The aggregation must be decomposable
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These constraints hold for all 31 queries from prior work.

Customer	C.Total	O.Cost
∅	∅	
🤡	$1500	
😈	$2000	
∅	∅	
🐸	-$140	
∅	∅	
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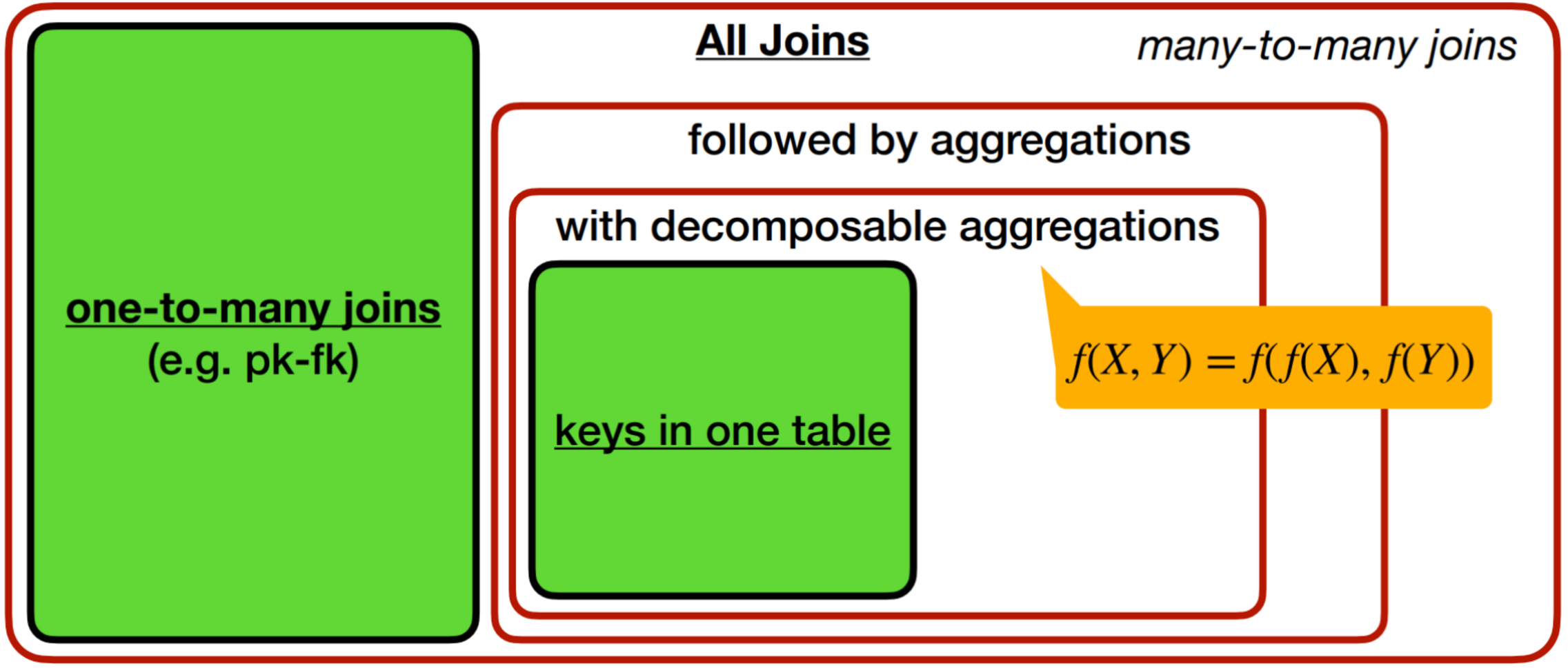
What Can We Support?



Connection to Oblivious Sorting

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- 12 sorts for TPC-H Q21

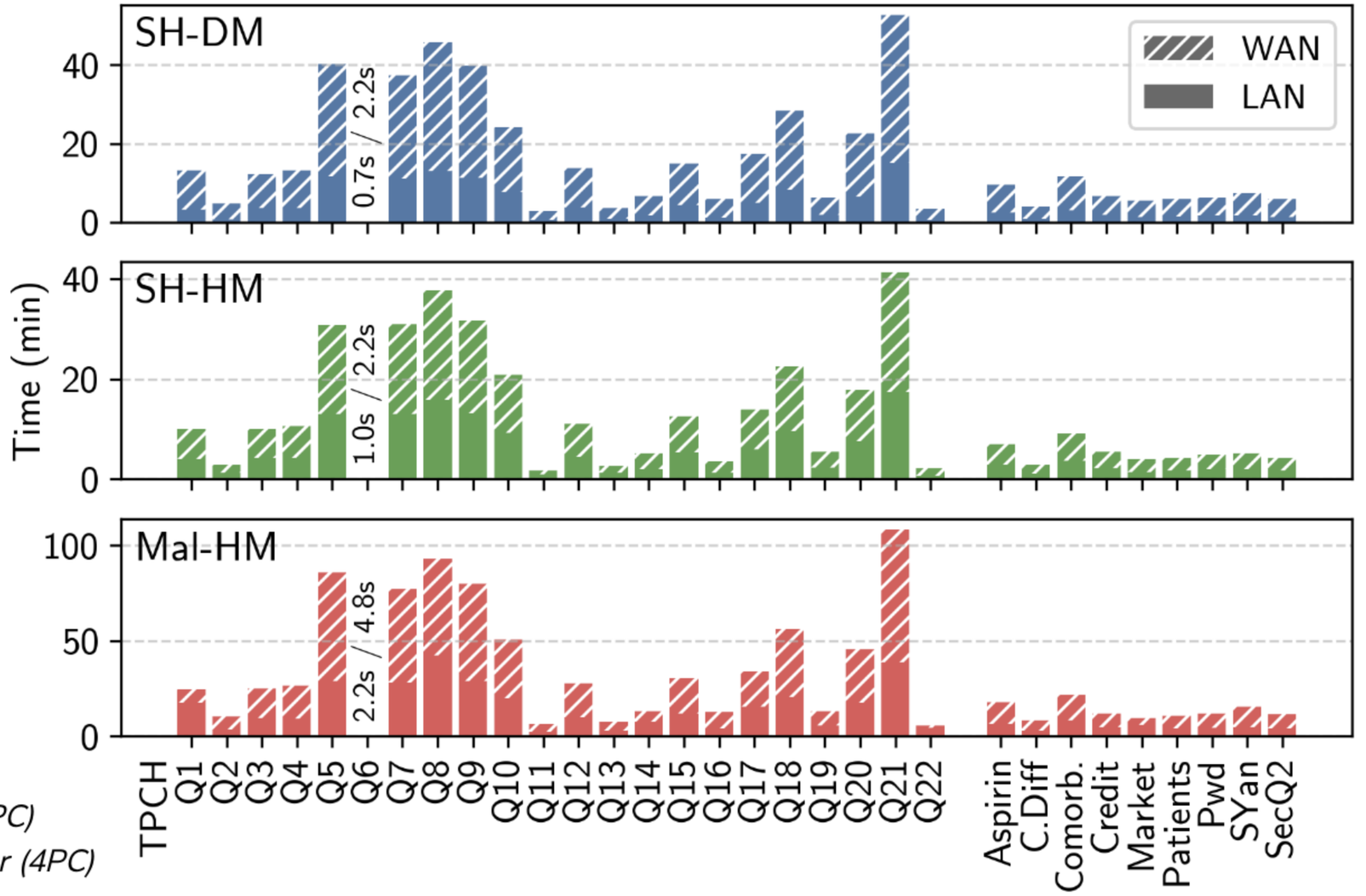
Connection to Oblivious Sorting

Queries consist of joins, joins consist of (many) sorts.

- 12 sorts for TPC-H Q21

Sorting is no longer the main bottleneck!

Query Benchmarks



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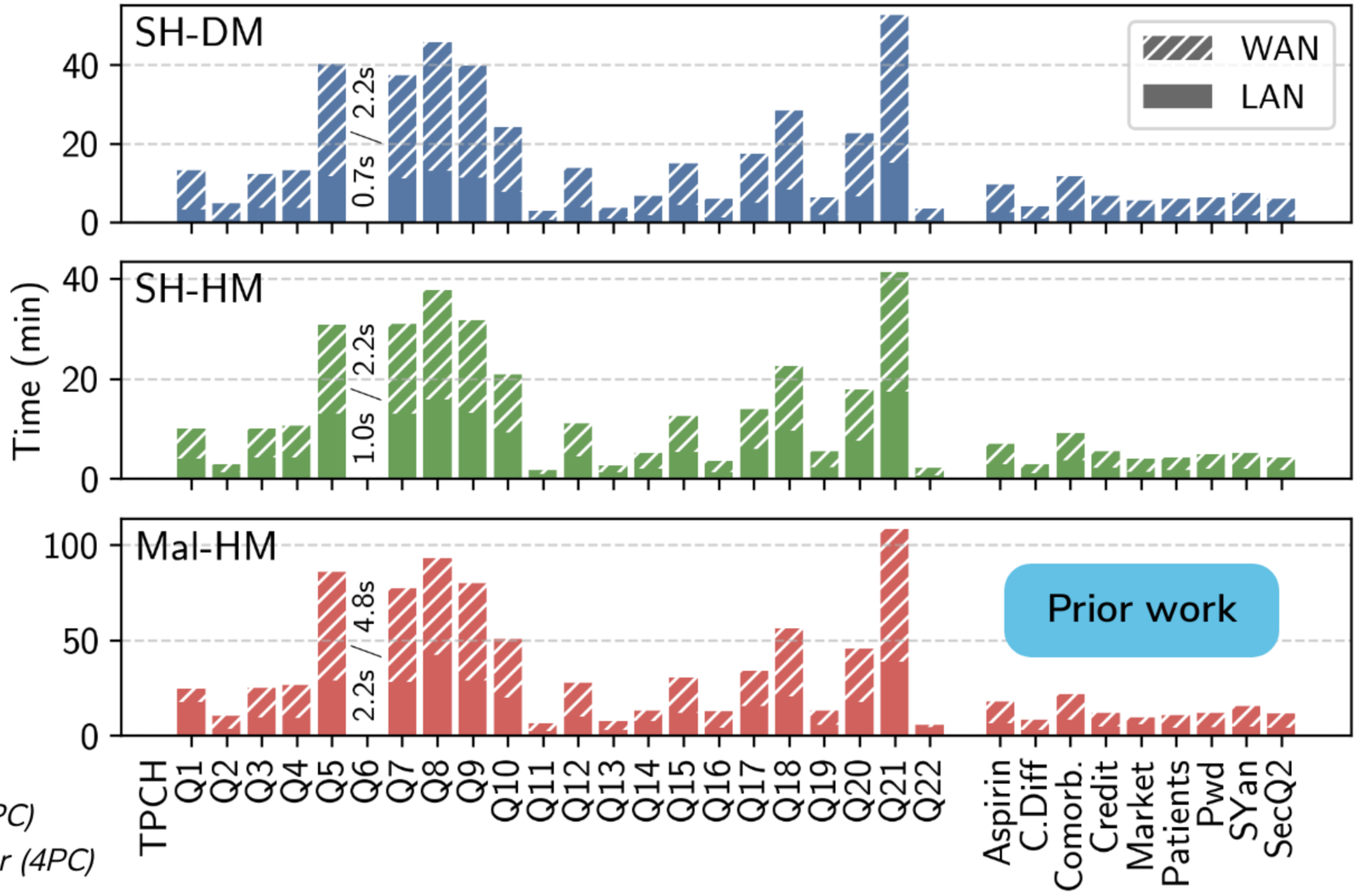

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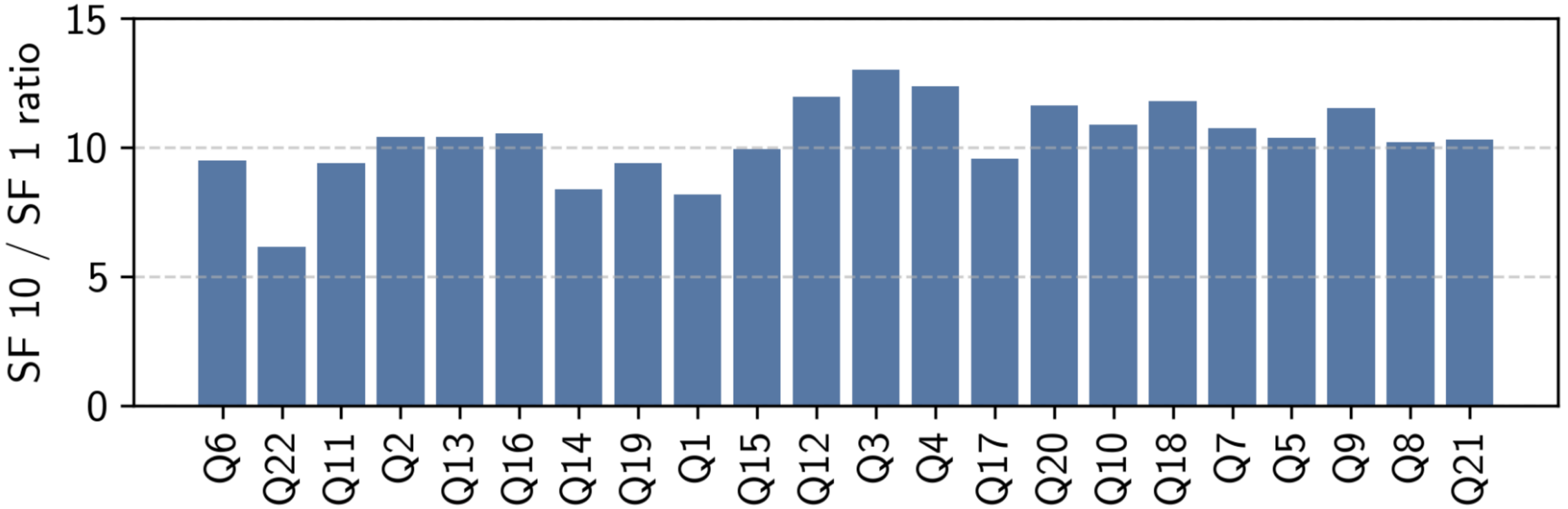
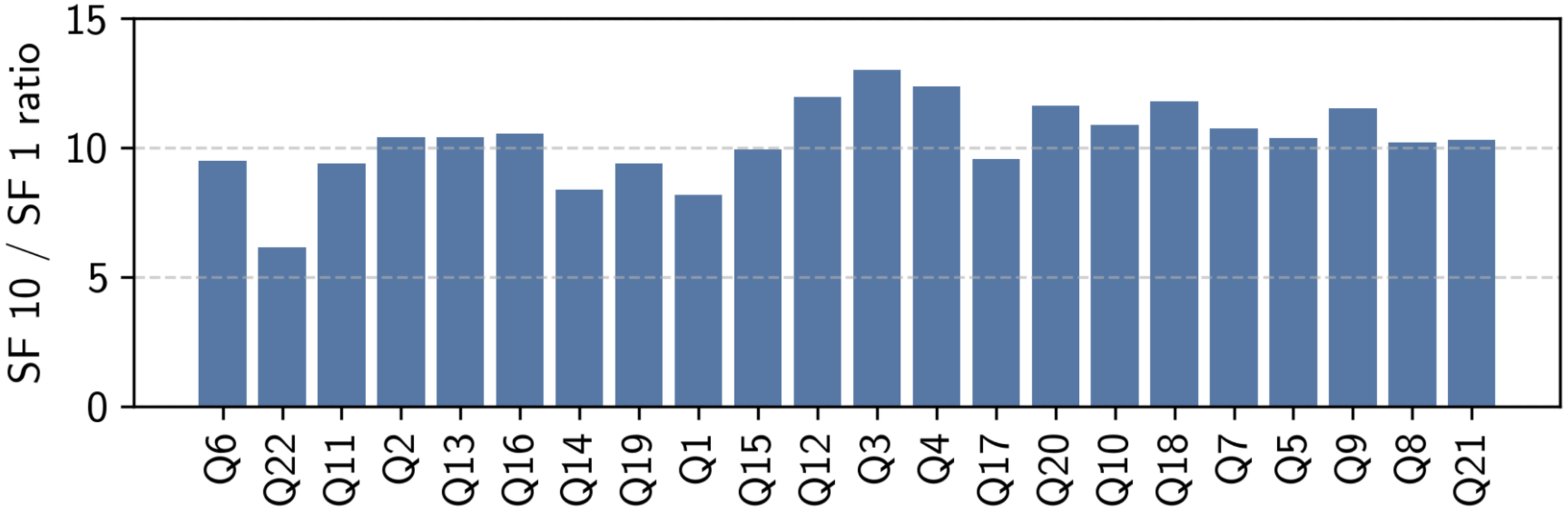
Scalability



Scalability

Theoretical overhead $\approx 11\times$



The ORQ System

The ORQ System

- Familiar API

```
auto C = db.getCustomersTable();
auto O = db.getOrdersTable();

O.filter(O["[Comment]"] ≠ 0);

auto T = C.left_outer_join(O,
    {"[CustKey]"},
    {{"Count", "Count", count}});

T.convert_a2b("Count", “[Count]");

auto F = T.aggregate(
    {"[Count]"},
    {{"CustDist", "CustDist", count}});

F.convert_a2b("CustDist", "[CustDist]");
F.sort({"[CustDist]", "[Count]"}, DESC);
```

The ORQ System

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- Protocol-agnostic interface

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The ORQ System

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  - Support for 2PC, 3PC, 4PC
- Data-parallel runtime
- Custom TCP Communicator

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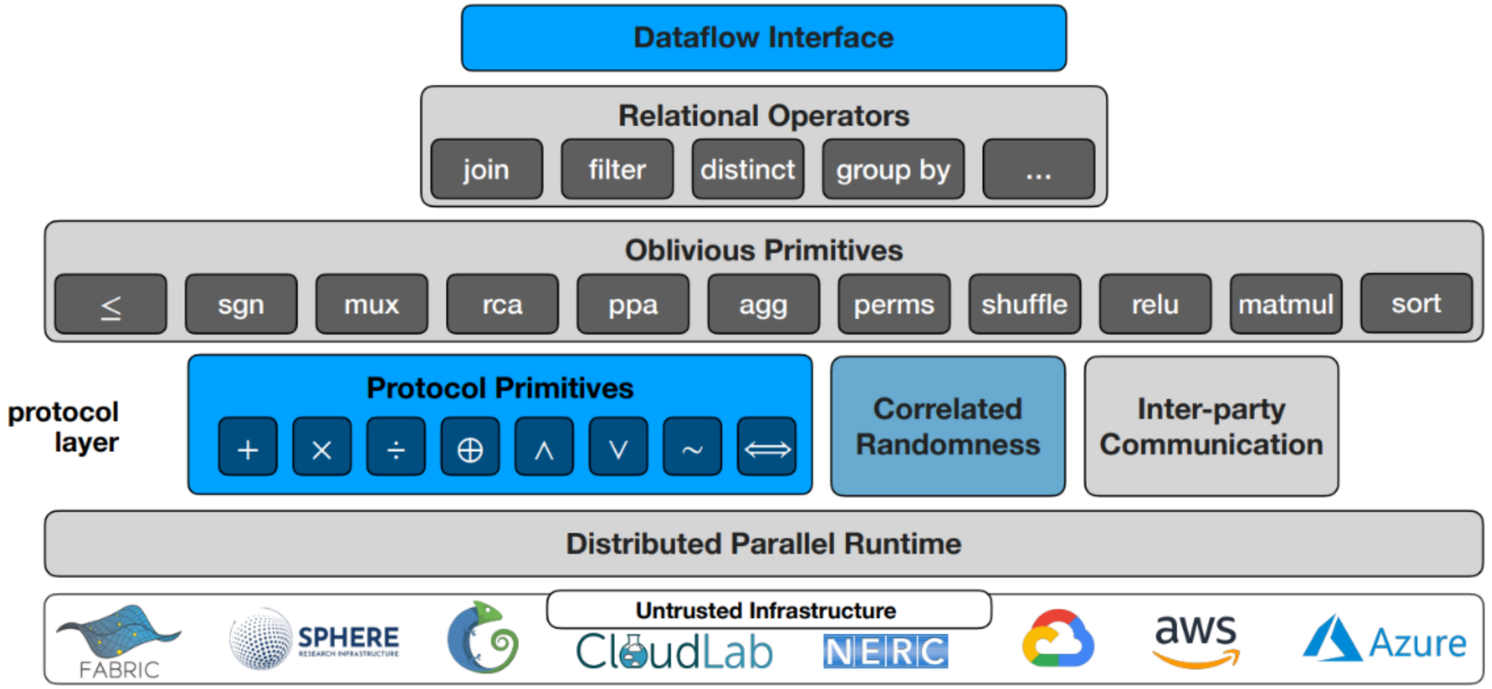
The ORQ System

- Familiar API
- Protocol-agnostic interface
  - Support for 2PC, 3PC, 4PC
- Data-parallel runtime
- Custom TCP Communicator



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  - Beaver triples over $\mathbb{Z}_{2^k}$
  - Oblivious shuffling

Thank You!

Polling Paper

Polling GitHub

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